



डॉ. श्यामा प्रसाद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान - नया रायपुर  
Dr. Shyama Prasad Mukherjee  
International Institute of Information Technology - Naya Raipur

## ANNUAL REPORT : 2022-23





**Fostering All Round Development of Students**



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**Dr. Shyama Prasad Mukherjee**

**International Institute of Information Technology - Naya Raipur**

Sector-24, Naya Raipur, Atal Nagar - 493661 (Chhattisgarh)

Website : [www.iiitnr.edu.in](http://www.iiitnr.edu.in)



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Dr. Shyama Prasad Mukherjee

International Institute of Information Technology - Naya Raipur

## Annual Report : 2022-23

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## 01

## The Chancellor



Hon'ble Governor of Chhattisgarh, Shri Biswabhusan Harichandan, is the Chancellor of Dr. Shyama Prasad Mukherjee International Institute of Technology - Naya Raipur (IIIT-NR). He assumed the office of the Governor on February 23, 2023. Before serving as the Governor of Chhattisgarh, he was the Governor of the State of Andhra Pradesh.

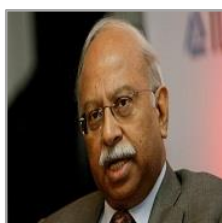
Shri Harichandan is an ardent democrat in all respect who fights for human rights, democracy, democratic values, and citizens' rights, especially rights of down trodden people and against corruption. He has an enriching political career. He was elected to the Odisha State Legislative Assembly five times. He has served people in various capacities including Revenue Minister, Law Minister, Industry Minister in the then Odisha Government. He also served as Cabinet Minister in 1977 (during emergency), and as the Chairman of the Cabinet Sub-Committee on Rehabilitation and Re-settlement of the displaced persons which was then the best Rehabilitation and Re-settlement Policy of the Country.

Shri Harichandan is also a reputed columnist who has written several articles on contemporary political issues, historical, social and cultural matters, which have been published in all leading newspapers of Odisha and some English weeklies of Delhi. He is an inspiring author and has significant literary contributions to his name, including his autobiography- '*Battle Not Yet Over*', which is focussed on his struggle in political, administrative, social, cultural and other fields during his long public life.



## 02

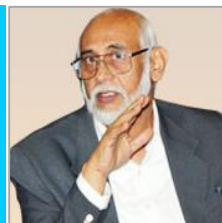
# The Board of Governors



**Shri Sourabh Srivastava**

Co-founder and Former Chairman of NASSCOM

**Chairperson**



**Dr. B. K. Gairola**

Former Mission Director (e-Governance), Dept. of Electronics & Information Technology, Government of India.

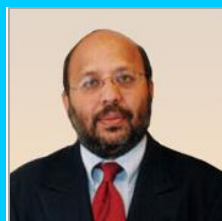
**Member**



**Shri Sujit Bakshi**

Former President, Emerging Market at Tech Mahindra

**Member**



**Shri Steven A. Pinto**

Former Lead Director, Emirates Investment Bank, UAE

**Member**



**Professor Uday B. Desai**

Founding Director and Professor Emeritus, IIT Hyderabad

**Member**



**Dr. Alok Shukla**

Principal Secretary, Directorate of Technical Education, Employment and Training, Government of Chhattisgarh

**Member**



**Shri Subrat Sahoo**

Additional Chief Secretary, Electronics and Information Technology Department, Government of Chhattisgarh

**Member**



**Shri Awanish Kumar Sharan**

Director, Directorate of Technical Education, Employment and Training, Government of Chhattisgarh.

**Member**



**Shri Ashwini Kumar Tripathi**

Regional Executive Director WR-II & USSC, NTPC Limited

**Member**



**Shri Diwakar Kaushik**

Executive Director Lara Project, Raigarh NTPC Limited

**Member**



**Ms. Rachna Singh Bhal**

General Manager Power Management Institute (PMI), NTPC Limited

**Member**



**Dr. Pradeep Kumar Sinha**

Vice Chancellor & Director International Institute of Information Technology - Naya Raipur (IIIT-NR)

**Member Secretary**

## 03

## Director's Note

It's indeed a privilege to present the Annual Report 2022-23 of this rapidly growing IIIT-Naya Raipur.

About 8 years ago, IIIT-NR embarked upon its journey to realize the State Government's vision to create a premier technical institute of the country in Chhattisgarh. I am proud to report that "Team IIIT-NR" has successfully converted this dream into reality. With an annual intake of only 80 students in two B. Tech. programmes in 2015, today IIIT-NR offers three B. Tech. programmes, four M. Tech. disciplines, multi-disciplinary M. S. (by Research) and Ph. D. programmes, with a total annual intake of 230 students. These programmes cover a wide range of state-of-the-art technology disciplines.



Our industry-oriented teaching-learning pedagogy helped achieve 100% placements in the first four consecutive years, with an average package higher than parallel institutes. Some of our key campus recruiters include Microsoft Research, American Express, Zeiss, Apra Labs, Big Basket, Cognizant, Lumiq, Infosys, KPMG, Capgemini, PharmEasy, Ericsson, Deloitte etc.. In addition, students interested in higher studies secured admissions in renowned academic institutes both in India and abroad. These include IIT Bombay, IIT Delhi, University of Massachusetts Amherst (USA), University of Illinois Urbana-Champaign (USA), Cornell University (USA), and Carnegie Mellon University (USA).

Team IIIT-NR firmly believes that both education and research are equally important for a premier Higher Education Institute (HEI). Hence, just like innovative academic programmes, IIIT-NR took a unique approach to carve out its research landscape. It decided to make IIIT-NR "*A Living Research Lab for Agriculture, Forestry and Rural Development*", by focusing on applications of ICET (Information, Communication and Electronics Technologies) for agriculture, forestry and rural development. Being situated in Chhattisgarh, a state rich in agriculture and forestry, and dominated by rural population, we chose to focus on these areas so that we can effectively contribute to the inclusive development of the state. I am proud to report that IIIT-NR is currently pursuing several funded research projects in these areas which have resulted in the publication of 242 journal papers, 18 book chapters, 4 books, and 7 patents in the year 2022-23. Our faculty members also conducted several Faculty Development Programs (funded by AICTE) in futuristic technology domains like Robotics, Artificial Intelligence, Data Science, etc.

With its firm belief in advancing society through multidisciplinary collaborative research, IIIT-NR has signed MoUs with several premier academic and research institutes. These include All India Institute of Medical Sciences (AIIMS) Raipur, DRDO, C-DAC, Microsoft Research Lab, Samsung R&D, Louisiana State University (USA), Wright State University (USA), Indira Gandhi Krishi Vishwavidyalaya (IGKV), SECQAI Ltd. (UK) etc. For increased interactions with researchers across the globe, IIIT-NR started its own international conference named TRIBES (Technology, Research, and Innovation for BETterment of Society). To encourage its members to actively participate in professional societies globally, IIIT-NR has established IEEE Student Branch and IEEE Women in Engineering Affinity Group. In December 2022, it also hosted the IEEE International Women in Engineering Conference. With these initiatives, the institute and the State of Chhattisgarh are gaining increased popularity nationally and internationally as a research hub.

Through its entrepreneurship initiatives, IIIT-NR also motivates and nurtures its students to become job creators as an entrepreneur. The institute received the '*Start Me Up Award*' by Wadhvani Foundation for its entrepreneurship initiatives and active participation in global entrepreneurship competitions. Recognizing the new wave where startups will be the dominant job creators of the future, the institute is also setting up an incubation facility in the campus. The institute got 3.5-Star rating by the Institution Innovation Council (IIC), Ministry of Education (MoE) and '*Peformers Band*' in Atal Rankings of Institutions on Innovation Achievements (ARIIA).

To develop the sense of social responsibility in its members, IIIT-NR has established ISR (Institute Social Responsibility) Cell and NSS (National Social Service) unit in its campus. Through these units, IIIT-NR members help less privileged sections of our society in various ways. For example, the institute has adopted two nearby villages (Rakhi and Uparwara) where it conducts special classes for village school children; conducts cleanliness and awareness programmes for the villagers; undertakes projects for tribal and rural development; conducts invited talks for members of other technical institutes to share the best practices of IIIT-NR; conducts summer camps for school children; conducts an annual event called '*SCI<sub>n</sub>TFIC*' (Science and Technology over Fiction) to motivate school children towards career in Science and Technology etc.

In addition to our students excelling globally in various ways as mentioned above, several faculty members received international recognitions like selection/ nomination in executive positions in professional societies, elevation to IEEE Senior Member Grade, Gold Faculty Award by Wadhvani Foundation, etc. One of our faculty, Dr. Ruhul Amin, got listed for two consecutive years, among the top 2% most-cited scientists globally in the list released by Stanford University, USA.

To sum up, 2022-23 has been a year of success and accelerated growth of IIIT-NR. I acknowledge that the success and the brand that IIIT-NR has earned in this short span would not have been possible without the wholehearted support of Govt. of Chhattisgarh and NTPC and our visionary Board members for their guidance and support in all matters. I extend my sincere thanks to all stakeholders including the brilliant efforts of talented students, academic accomplishments of our faculty members and committed staff.

Dr. Pradeep Kumar Sinha  
Vice Chancellor and Director  
Dr. SPM IIIT-Naya Raipur

## 04

## About the Institute

Dr. Shyama Prasad Mukherjee International Institute of Information Technology – Naya Raipur (IIIT-NR) is a joint venture by Govt. of Chhattisgarh and National Thermal Power Corporation (NTPC). The institute is established by the International Institute of Information Technology University Act, 2013 of the Govt. of Chhattisgarh “to promote and facilitate studies, research, incubation and extension work in Information Technology and in its application domain and related areas of knowledge and to achieve excellence in these fields and matters connected therewith or incidental thereto.”



As enshrined in the Act, IIIT-Naya Raipur is committed to pursue excellence in higher education, research and development in Information Technology (IT) and associated disciplines. It firmly believes in bestowing knowledge dissemination and imparting entrepreneurial skills to the students to enable them to address real world problems. To empower this vision, IIIT-NR fosters state-of-the-art research and product development laboratories to carry out inter-disciplinary research and product development endeavours. The institute is mentored by a highly eminent board comprising of industry leaders, erudite academicians and administrators.

## IIIT-NR Campus and Location

IIIT-NR's fifty-acre fully residential campus is located in the smart city of Nava Raipur, Atal Nagar approximately 23 Kms from Raipur - the capital city of Chhattisgarh state. Raipur is one of the unique cities where various Institutions of National Importance (INIs) and premier academic institutions like IIT, IIM, AIIMS, NIT, HNLU, PRSU, IGKV and IIIT-NR are located in close proximity. Availability of these premier higher educational and research organisations at one place makes the city a perfect educational hub. It gives students and scholars a perfect platform with plenty of opportunities and appropriate ambience and ecosystem for teaching, learning and research on wide spectrum of subjects.

The city is well connected by air, rail and road to all parts of the country.

The vibrant lush green campus of IIIT-NR is enabled with Wi-Fi connectivity, CCTV surveillance, hi-tech security measures and excellent amenities for sports and recreation. The modern architecture of the buildings and the campus's serene environment fosters an ambience for creativity and innovation. The classrooms and labs are equipped with state-of-the-art technologies for teaching, learning and research.



**IIIT-NR Campus**



## IIIT-NR Logo

### Description of Logo

The logo of Dr. Shyama Prasad Mukherjee IIIT-Naya Raipur is designed to capture the Institute's vision towards its social responsibility along with its name and location. The logo indicates that our institute is located in the State of Chhattisgarh and provides an environment to foster knowledge enrichment and development of humane qualities to enable individuals to emerge as enlightened citizens, who can lead a peaceful and contented life. The various components of the logo signify the following:



**Buddha:** “Buddha” signifies enlightenment (an ideal state of intellectual and ethical perfection) through gain of knowledge, discipline, dedication and concentration. This object in the logo indicates that DSPM-IIIT Naya Raipur nurtures such qualities in its students and other members.



**Circle Around Buddha:** This artistically designed circle has both the letters ‘C’ and ‘G’ embedded, signifying Chhattisgarh. This object in the logo indicates that DSPM-IIIT Naya Raipur is located in and is supported by the State of Chhattisgarh.

## ज्ञानं लब्ध्वा परं शान्तिम्

**Sanskrit Shloka:** Taken from Bhagavad Gita (4:39), it means that ultimate wisdom and peace in life can be achieved only by

acquiring knowledge. This object in the logo indicates that DSPM-IIIT Naya Raipur strives hard to impart as much knowledge as possible to its students and other members to help them lead a peaceful and contented life.



**IIIT**  
NAYA RAIPUR

**Institute Name in Long and Short Forms:** The logo also contains the institute's name in both long and short forms.

## IIIT-NR Vision and Mission

### Vision

To be a leading technology institute that produces high quality IT professionals and fosters technological innovation for social upliftment.

### Mission

To create an Institute of Excellence in IT Education and Research for imparting knowledge through education, advancing knowledge through research, and providing service to the region, nation and mankind to contribute towards the creation of a better society.

### Vision 2.0

The first five-year focus of the institute was to properly establish its education and culture components. The next five year vision (IIIT-NR 2.0) focuses on strengthening the research component of the institute with a motto to transform the institute into “*A Living Research Lab for Agriculture, Forestry and Rural Development*” by focusing on the application of ICET (Information, Communication and Electronics Technologies) for agriculture, forestry, and rural development.



## 05

## Administrative Structure

## ✿ FINANCE COMMITTEE ✿

|                                                                                                   |                  |
|---------------------------------------------------------------------------------------------------|------------------|
| Dr. Pradeep Kumar Sinha<br>Director, IIIT-Naya Raipur                                             | Chairperson      |
| Shri Dinesh Bahuguna<br>GM (Finance), WR-II, NTPC                                                 | Member           |
| Shri Steven A Pinto<br>Representative of IIIT-NR Board                                            | Member           |
| Shri A. K. Singh<br>Joint Secretary, Dept. of Finance, Govt. of Chhattisgarh                      | Member           |
| Shri H. R. Chandrawanshi<br>Joint Secretary (Finance), Dept. of Technical Education, Chhattisgarh | Member           |
| Lt. Col. Rajesh Kumar Mishra,<br>Registrar, IIIT-Naya Raipur                                      | Member           |
| Shri Gopal K. Padhy,<br>Chief Finance and Accounts Officer, IIIT-Naya Raipur                      | Member Secretary |

## ✿ BUILDING AND WORKS COMMITTEE ✿

|                                                                |                  |
|----------------------------------------------------------------|------------------|
| Dr. Pradeep Kumar Sinha, Director, IIIT-Naya Raipur            | Chairperson      |
| Shri Sanjay Madan, Executive Director, Lara Project, NTPC      | Member           |
| Shri S. P. Sharma, AGM (PE&M), WR-II, NTPC                     | Member           |
| Shri S. R. Shrivastava, Engineer-in-Chief, NRANVP, Raipur (CG) | Member           |
| Prof. Shrish Verma, Professor, NIT Raipur                      | Member           |
| Shri Sandeep Kumar, Chief Engineer (Civil), BSNL               | Member           |
| Shri Ajay Kumar Garg, Joint Director, DTE, Chhattisgarh        | Member           |
| Lt. Col. Rajesh Kumar Mishra, Registrar, IIIT-Naya Raipur      | Member           |
| Shri Kailash Agrawala, Estate Officer (I/c), IIIT-Naya Raipur  | Member Secretary |

### **SENATE**

|                                                           |                  |
|-----------------------------------------------------------|------------------|
| Dr. Pradeep Kumar Sinha, Director, IIIT-Naya Raipur       | Chairperson      |
| Dr. Debabrata Das, Director, IIIT Bangalore               | Member           |
| Prof. G. Sivakumar, Professor, IIT Bombay                 | Member           |
| Dr. Raghu Krishnapuram, Professor, IISc. Bangalore        | Member           |
| Prof. Srinivasa K.G., Dean (R&I), IIIT-Naya Raipur        | Member           |
| Lt. Col. Rajesh Kumar Mishra, Registrar, IIIT-Naya Raipur | Member           |
| Prof. Rajesh B. Ingle, Dean (Academics), IIIT-Naya Raipur | Member Secretary |

### **BOARD OF RECRUITMENT**

|                                                                                     |                  |
|-------------------------------------------------------------------------------------|------------------|
| Dr. Pradeep Kumar Sinha<br>Director, IIIT-Naya Raipur                               | Chairperson      |
| Shri Pushpendra Kumar Meena<br>Director, Dept. of Technical Education, Chhattisgarh | Member           |
| Shri Gopal K. Padhy<br>Chief Finance and Accounts Officer, IIIT-Naya Raipur         | Member           |
| Shri Rajiva Kumar Sinha<br>General Manager, PMI, NTPC, Noida                        | Member           |
| Prof. Sumit Gupta<br>Professor, IIM Raipur                                          | Member           |
| Prof. Rajesh B. Ingle<br>Dean (Academics), IIIT-Naya Raipur                         | Member           |
| Shri Ajay Kumar Garg<br>Joint Director, Dept. of Technical Education, Chhattisgarh  | Member           |
| Lt. Col. Rajesh Kumar Mishra<br>Registrar, IIIT-Naya Raipur                         | Member Secretary |

### **ACADEMIC AND GENERAL ADMINISTRATION**

|                              |                                  |
|------------------------------|----------------------------------|
| Prof. Rajesh B. Ingle        | Dean (Academics)                 |
| Prof. Srinivasa K.G.         | Dean (Research & Innovation)     |
| Lt. Col. Rajesh Kumar Mishra | Registrar                        |
| Shri Gopal K. Padhy          | Chief Finance & Accounts Officer |



## 06

## Faculty Members

## Faculty Members at IIIT-NR as on 31.03.2023



**Dr. Abhishek Sharma**  
Ph.D. (IIT Guwahati)

**Department : ECE**

**Research Area:** Bio-metrics, Online Signature verification, Write Identification, Image Processing, Pattern Recognition and Machine Learning.



**Dr. Anirban Bhowal**  
Ph.D. (IIT Guwahati)  
Post-doc. (INRS Montreal)

**Department : ECE**

**Research Area:** Reconfigurable Intelligent Surfaces, Short Packet Communication, Optical Wireless Communication, Spatial Modulation



**Dr. Avantika Singh**  
Ph.D. (IIT Mandi)

**Department : DSAI**

**Research Area:** Biometrics, Machine Learning, Deep Learning, Reinforcement Learning, Indexing and Hashing



**Dr. Debanjan Das**  
Ph.D. (IIT Kharagpur)

**Department : ECE**

**Research Area:** IoT & Sensors, Agricultural & Medical IoT, Signal Processing, Sensors & Instruments, Control & Robotics, Energy Harvester, BioMEMS & Microfluidics.



**Dr. Kavita Jaiswal**  
Ph.D. (NIT Raipur)

**Department : CSE**

**Research Area:** Internet of Things, Wireless Sensor Network, Cloud Computing and Fog Computing.



**Dr. Amit Agrawal**  
Ph.D. (IIT Roorkee)

**Department : SHM (Management)**

**Research Area:** Customer Value Co-creation; Technological Adoption; Services; Consumer Behavior; Marketing.



**Dr. Anurag Singh**  
Ph.D. (IIT Guwahati)

**Department : ECE**

**Research Area:** Biomedical Signal and Image Processing, WBAN-enabled remote healthcare applications using different physiological signals such as ECG, EEG etc.



**Dr. Bipin Chandra Mandi**  
Ph.D. (IIT Kharagpur)

**Department : ECE**

**Research Area:** Embedded System Design, Digital Power Management Circuits, VLSI Implementation



**Dr. Deepika Gupta**  
Ph.D. (IIT Indore)

**Department : ECE**

**Research Area:** 3D Charge Trap NAND Flash Memory, Advanced Non-Volatile Memories and Semiconductor Devices.



**Dr. Lakhindar Murmu**  
Ph.D. (IIT (ISM) Dhanbad)

**Department : ECE**

**Research Area:** Electromagnetics, Novel Microstrip Circuits Design for RF and Microwave Communication.



**Dr. Maifuz Ali**  
Ph.D. (IIT Kharagpur)

**Department : ECE**

**Research Area:** Numerical and analytical methods in Electromagnetic, Electromagnetic Radiation and Scattering, Ray tracing, Communication channel modeling, EMI sensors, Radar Cross Section (RCS), Millimeter wave technology and Non-reciprocal devices.



**Dr. Mallikharjuna Rao K.**  
Ph.D (JNTU Hyderabad)

**Department : DSAI**

**Research Area:** Artificial Intelligence, Swarm Intelligence, Evolutionary Algorithms, Data Science, Software Reliability.



**Dr. Mithilesh Kumar Chaube**  
Ph.D. (IIT-BHU)

**Department : SHM (Mathematics)**

**Research Area:** Mathematical Modelling, Bio-fluid Mechanics, Biomechanics, Microfluidics, Biomathematics



**Dr. Manoj Kumar Majumder**  
Ph.D (IIT Roorkee)

**Department : ECE**

**Research Area:** Low Power VLSI Circuits and Systems Designing, Modelling of VLSI Device and Circuit Performance Parameters, Hardware Security, Emerging Electronics, Carbon nanotube and Graphene Nanoribbon based VLSI interconnects etc.



**Dr. Pradeep Kumar Sinha**  
Ph.D. (University of Tokyo)

**Department : CSE**

**Research Area:** Distributed Computing, Parallel Processing, High Performance Computing (HPC), Grid Computing and Health Informatics.



**Dr. Punya Prasanna Paltani**  
Ph.D. (BIT MESRA, Ranchi)

**Department : SHM (Physics)**

**Research Area:** Nonlinear Optics, Photonics, Plasmonics and Photonic Crystals.



**Dr. Rajarshi Mahapatra**  
Ph.D. (IIT Kharagpur)

**Department : ECE**

**Research Area:** Cognitive Radio, Signal Processing, Dynamic Spectrum, Green Communication, Heterogeneous Wireless Network, Optical Access Networks, Molecular Communication.



**Dr. Rajesh Ingle**  
Ph.D. (IIT Bombay)

**Department : CSE**

**Research Area:** Computer Networks, Edge Computing, Distributed Computing, Internet of Things, Blockchain and Distributed System Security.



**Dr. Ramakrishna Bandi**  
Ph.D. (IIT Roorkee)  
Post-doc. (IIT Bombay)

**Department : SHM (Mathematics)**

**Research Area:** Error Correcting Codes, Information Theory and Algebra



**Dr. Rohit Chaurasia**  
Ph.D. (IIT Mandi)

**Department : ECE**

**Research Area:** VLSI Design, Hardware-Friendly Algorithm Design, Digital VLSI Architecture Design, ASIC Chip Design and Testing, FPGA Prototyping of Stand-Alone, Cooperative and Wideband Spectrum Sensor for Cognitive Radio Network, VLSI Design of Signal Processing Tasks in Wireless Communication and Computer Arithmetic Domain.



**Dr. Santosh Kumar**  
Ph.D (IIT-BHU)

**Department : CSE**

**Research Area:** Animal Biometrics, Computer Vision, Deep Learning, Pattern Recognition, Wireless Sensors and Internet of Things (IoT).



**Dr. Shailesh Khapre**  
Ph.D. (Pondicherry University)

**Department : DSAI**

**Research Area:** Intelligent Information Retrieval, Machine Learning, Computer Vision, Robot Perceptions, Sensory Fusion, Computer Vision, Autonomous Mobile Robots.



**Dr. Sresha Yadav Ghosh**  
Ph. D (IIT Roorkee)

**Department : SHM (English)**

**Research Area:** ELT, Cognitive Psychology and Criticism, and Professional Communication.



**Dr. Vijaya J.**  
Ph.D. (NIT Trichy)

**Department : DSAI**

**Research Area:** Data Science, Data Analytics, Data Mining and Data Warehousing, Machine Learning, Soft Computing, Fuzzy Set and Rough Set Theory, Optimization Techniques, Deep Learning and Big Data.



**Dr. Venkanna U**  
Ph. D (NIT Trichy)

**Department : CSE**

**Research Area:** IoT, Software Defined Networks, Data Plane Programmability (P4), Self Driving Networks, Networks Security.



**Dr. Satyanarayana Vollala**  
Ph.D. (NIT Trichy)

**Department : CSE**

**Research Area:** Public-Key Cryptography, Hardware Cryptography, Number Theory and Theoretical Computer Science.



**Dr. Shrivishal Tripathi**  
Ph.D (IIT Jodhpur)

**Department : ECE**

**Research Area:** Ultrawideband and MIMO Antenna, Antenna for 5G/IoT, AI in Electromagnetics, Machine/Deep Learning in RF Circuit Design, Energy Harvesting, Radar, Ad-hoc Network.



**Dr. Srinivasa K.G.**  
Ph.D. (Bangalore University)

**Department : DSAI**

**Research Area:** Data Mining, Cloud Computing, Internet of Things, Data Analytics, Cyber Physical Systems, Augmented Reality and Virtual Reality, Learning Analytics, Digital Pedagogy.



**Dr. Srinivas Naik**  
Ph.D. (University of Hyderabad)

**Department : CSE**

**Research Area:** Big Data, Hadoop, Distributed Systems, and Cloud Computing.



**Dr. Vivek Tiwari**  
Ph.D. (NIT Bhopal)

**Department : CSE**

**Research Area:** Machine Learning, Data Mining, Computer Vision (Bioinformatics), Business Analytics, Data Warehousing.



## 07

# Officers and Staff

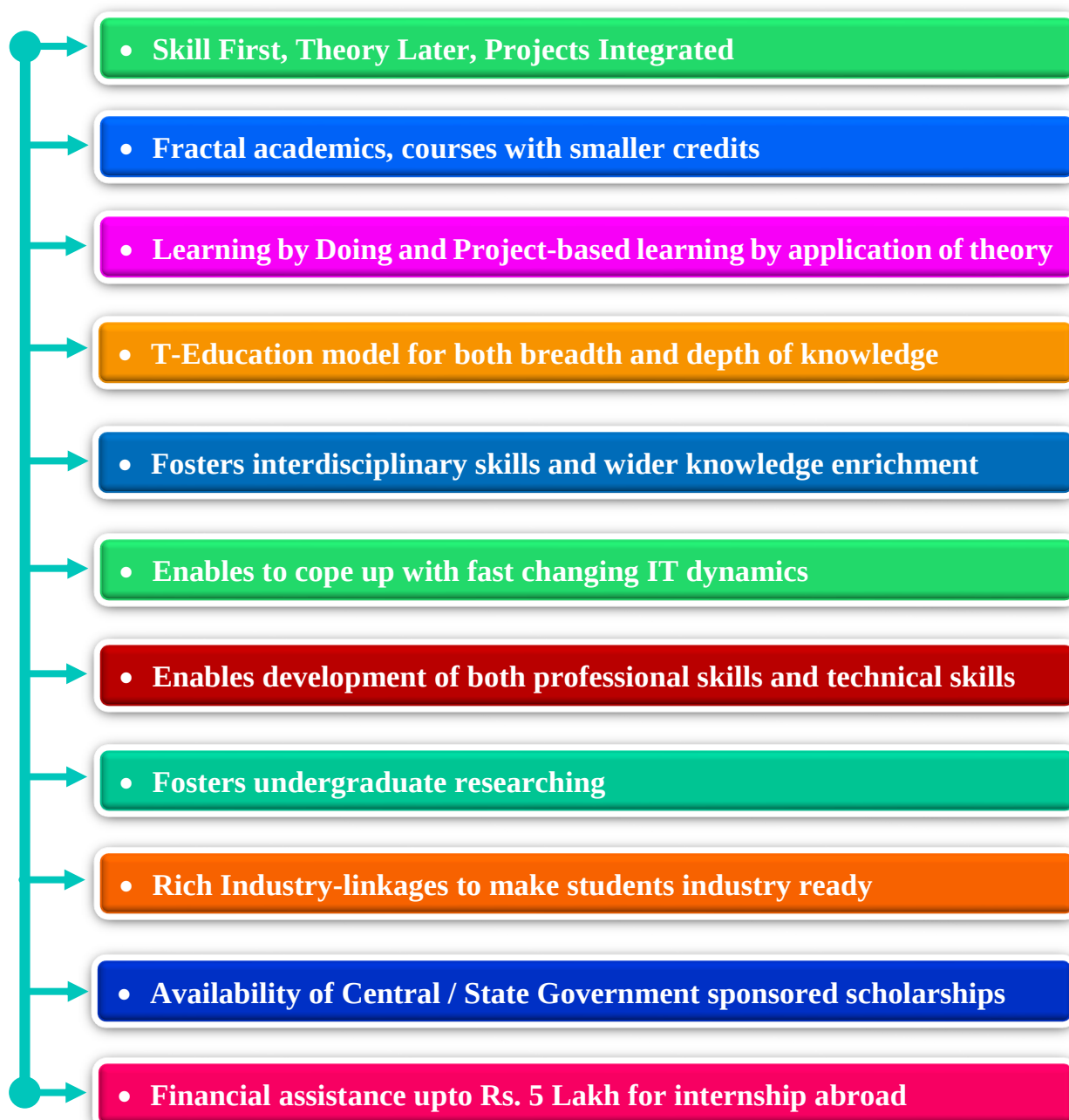
| Name                          | Designation                             |
|-------------------------------|-----------------------------------------|
| Lt. Col. Rajesh Kumar Mishra  | Registrar                               |
| Ms. Aakanksha Sharma          | Technical Assistant-Networking Hardware |
| Ms. Bharti Markam             | Junior Administrative Officer           |
| Ms. Kirti Naganpure           | Multi Tasking Staff                     |
| Ms. Kunita Meshram            | Multi Tasking Staff                     |
| Ms. Prema Patel               | Technical Assistant-CSE                 |
| Ms. Seema Sahu                | Multi Tasking Staff                     |
| Ms. Upasana Toppo             | Technical Assistant-ECE                 |
| Ms. Yogita Sahu               | Lab Assistant                           |
| Shri Abir Bhattacharya        | Senior Network Engineer                 |
| Shri Ajit Kumar Roy           | Assistant Librarian                     |
| Shri Amit Kumar Singh         | Junior Administrative Officer (SS)      |
| Shri Ayyub Ahmed              | Driver                                  |
| Shri Chandan Nishad           | Driver                                  |
| Shri Deepak Choudhary         | Estate Officer                          |
| Shri G. V. Srikant            | Lab Assistant                           |
| Shri Gopal K. Padhy           | Chief Finance & Accounts Officer        |
| Shri Gourav Sharma            | Technical Assistant-CSE                 |
| Shri Harish Kumar             | Assistant Registrar                     |
| Shri K. Sandeep               | Personal Staff Officer to VC & Director |
| Shri Kailash Chandra Agrawala | Assistant Finance Officer               |
| Shri Lalit Markanday          | Lab Assistant                           |
| Shri Lokendra Siriya          | Assistant Programmer                    |
| Shri Parvinder Singh          | Network Engineer                        |
| Shri Rajeev Sanmani           | Multi Tasking Staff                     |
| Shri Ranjeet Singh Thakur     | Junior Electrical Engineer              |
| Shri Ritesh Kujur             | Driver                                  |
| Shri Sourabh Jain             | Accountant                              |
| Shri Upendra Agrawal          | Technical Assistant-ECE                 |

(In alphabetical order – as on 31.03.2023)

## 08

## Academics

## Salient Features of Academics at IIIT-NR



## 8.1 Academic Programmes

IIIT-Naya Raipur is firmly focused on developing a different academic culture with many innovative ideas like fractal academics, project-based learning, industry engagements, entrepreneurship etc. The institute fosters excellence in education, research, and innovation and emphasizes “*Skill First, Theory Later and Projects Integrated*” learning pedagogy. IIIT-NR endeavours to transform students not only as good engineers or professionals but also as good human beings who can apply their knowledge for the betterment of the society and meet the aspirations of the citizens.

IIIT-Naya Raipur has an industry driven dynamic structure of curriculum designed to match the changing needs of industry and academia. The details of undergraduate (UG) and postgraduate (PG) programmes offered by IIIT-NR are as given below:

### UG and PG Programmes at IIIT-NR

#### 1. Undergraduate Programmes

- B.Tech. in Computer Science & Engineering (CSE)
- B.Tech. in Data Science & Artificial Intelligence (DSAI)
- B.Tech. in Electronics & Communication Engineering (ECE)

#### 2. Postgraduate Programmes

- M.Tech. in Computer Science & Engineering (CSE)  
**Specialisation**
  - (a) Data Science and Artificial Intelligence
  - (b) Information Security
- M.Tech. in Electronics & Communication Engineering (ECE)  
**Specialisation**
  - (a) Communication and Signal Processing
  - (b) VLSI and Embedded System
- Master of Science (by Research) [for Institute Project Staff]

#### 3. Ph.D. Programmes

IIIT-Naya Raipur offers Ph.D. programmes in all disciplines of Engineering, Science, English, Humanities and Management.

## Admission to Undergraduate Programmes (B.Tech.)

Admission to undergraduate (UG) programmes (B.Tech.) at IIIT Naya Raipur is based strictly on the merit of the candidates in JEE (Main). Based on the JEE (Main) Rank, available seats in the UG programmes (B.Tech.) at IIIT-Naya Raipur are divided in the following categories:-

|                        |                                                                                   |                                                                                     |
|------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Category</b>        |  |  |
|                        | All India Quota                                                                   | Chhattisgarh State Quota                                                            |
| <b>Share of Seats*</b> | 50%                                                                               | 50%                                                                                 |

\* Out of 50% All India Quota seats, 15% seats are for the wards of NTPC employees posted anywhere in India.

## Admission to Postgraduate Programmes (M.Tech.)

Admission to postgraduate (PG) programmes (M.Tech.) at IIIT-Naya Raipur is purely based on the GATE score. Based on the GATE score, available seats in the PG programmes (M.Tech.) at IIIT-NR are equally divided in 'All India Quota' and 'Chhattisgarh State Quota' similar to the UG programmes (B.Tech.).

### Salient Features of M.Tech. Programme at IIIT-NR

- Offers best of the faculty and resources for research to create solution to problems and products for needs of the country.
- Financial assistance to students to present their research work at reputed national and international conferences.
- Financial assistantship up to Rs. 5 Lakh to selected candidates for a semester long research programme abroad.
- Monthly scholarship to students as per the extant AICTE norms.
- State-of-the-art labs and Hi-Tech campus with research ecosystem

## Admission to Ph.D. Programmes

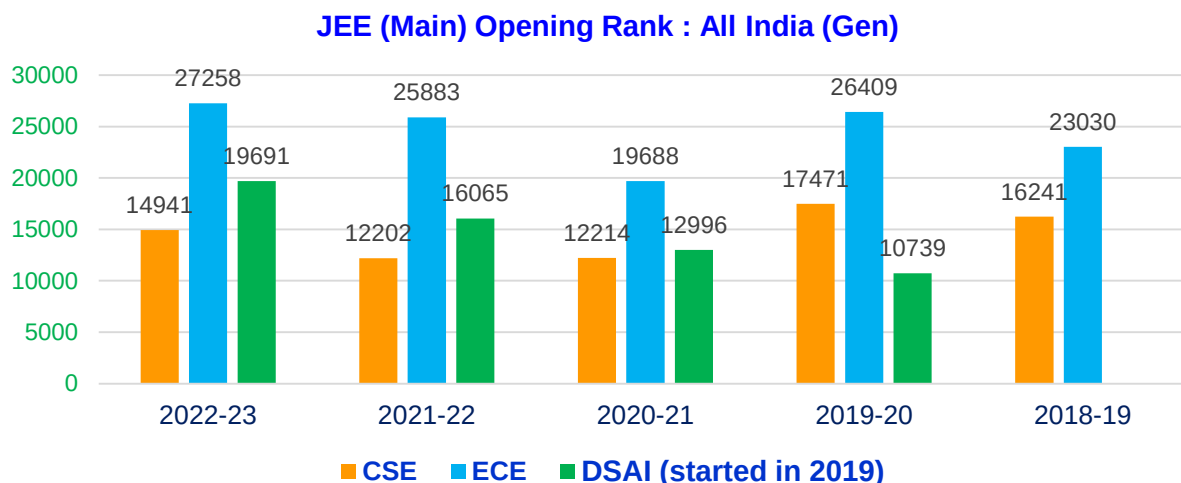
IIIT-Naya Raipur offers Ph.D. programmes in all disciplines of Engineering, Science, English, Humanities and Management. The goal of these programmes at IIIT-NR is to prepare scholars as leaders in the field of academia, industry and R&D. To facilitate quality research, scholars are provided with all support, including the best of the faculty, state-of-the-art labs and resources, financial assistance as monthly teaching/research assistantship as per the Ministry of Education, Govt. of India norms, and financial assistance for presenting research papers at reputed conferences.



## 8.2 Admission Statistics

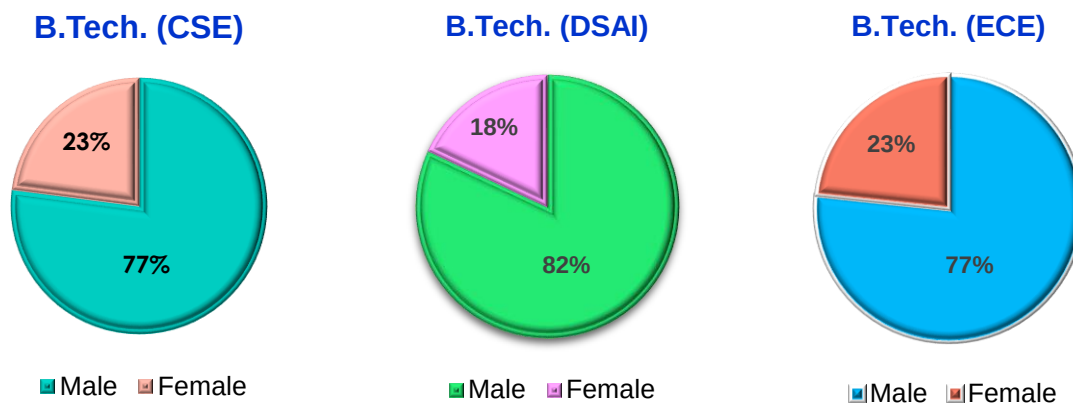
### JEE (Main) All India Opening Rank

Despite being a reasonably new academic institution, owing to our all-round exceedingly well performance in teaching-learning practices, research works, placement of students, and faculty accomplishments, IIIT-NR is successful in attracting top talent across the country. It is evident from the JEE (Main) opening rank for IIIT-NR over the years. All India opening rank of JEE (Main) for admission is better than many parallel institutes. The same is clearly visible below:-



### Gender Diversity – B.Tech. Admission 2022-23

The Institute also has a reasonably healthy students' gender diversity in admission in its undergraduate (UG) programmes compared to parallel institutions. Students' gender diversity in B.Tech. programmes admitted in the admission year 2022-23 is as follows:-



## Admission : 2022-23

The details of students admitted in various programmes in the academic year 2022-23 under different categories are as follows:-

### B.Tech. Admission : 2022-23

| B.Tech. – Computer Science & Engineering (CSE) |           |                         |              |      | Intake : 60     |
|------------------------------------------------|-----------|-------------------------|--------------|------|-----------------|
| S.No.                                          | Category  | Students Admitted Under |              |      | Total Admission |
|                                                |           | All India               | Chhattisgarh | NTPC |                 |
| 1.                                             | UR        | 13                      | 12           | 03   | 28              |
| 2.                                             | OBC (NCL) | 06                      | 07           | 02   | 15              |
| 3.                                             | SC        | 03                      | 04           | 01   | 08              |
| 4.                                             | ST        | 02                      | 06           | 01   | 09              |
| Grand Total                                    |           | 24                      | 29           | 7    | 60              |

| B.Tech. - Data Science & Artificial Intelligence (DSAI) |           |                         |              |      | Intake : 60     |
|---------------------------------------------------------|-----------|-------------------------|--------------|------|-----------------|
| S.No.                                                   | Category  | Students Admitted Under |              |      | Total Admission |
|                                                         |           | All India               | Chhattisgarh | NTPC |                 |
| 1.                                                      | UR        | 12                      | 14           | 03   | 29              |
| 2.                                                      | OBC (NCL) | 06                      | 04           | 04   | 14              |
| 3.                                                      | SC        | 03                      | 05           | 01   | 09              |
| 4.                                                      | ST        | 01                      | 06           | 01   | 08              |
| Grand Total                                             |           | 22                      | 29           | 9    | 60              |

| B.Tech. - Electronics & Communication Engineering (ECE) |           |                         |              |      | Intake : 60     |
|---------------------------------------------------------|-----------|-------------------------|--------------|------|-----------------|
| S.No.                                                   | Category  | Students Admitted Under |              |      | Total Admission |
|                                                         |           | All India               | Chhattisgarh | NTPC |                 |
| 1.                                                      | UR        | 18                      | 07           | 01   | 26              |
| 2.                                                      | OBC (NCL) | 11                      | 03           | 02   | 16              |
| 3.                                                      | SC        | 03                      | 05           | 02   | 10              |
| 4.                                                      | ST        | 01                      | 06           | 01   | 08              |
| Grand Total                                             |           | 33                      | 21           | 6    | 60              |

### M.Tech. and Ph.D. Admission : 2022-23

| S.No. | Category      | Intake Strength | Students Admitted Under |              |      | Total Admission |
|-------|---------------|-----------------|-------------------------|--------------|------|-----------------|
|       |               |                 | All India               | Chhattisgarh | NTPC |                 |
| 1.    | M.Tech. (CSE) | 20              | 04                      | 00           | 00   | 04              |
| 2.    | M.Tech. (ECE) | 20              | 01                      | 00           | 00   | 01              |
| 3.    | Ph.D.         | NA              | 09                      |              |      | 09              |

### Scholarship and Fellowship Granted

|             | Category of Student |     |    |    |     | Total Student |
|-------------|---------------------|-----|----|----|-----|---------------|
|             | General             | EWS | SC | ST | OBC |               |
| Scholarship | 20                  | -   | 21 | 71 | 19  | 131           |
| Fellowship  | 36                  | 1   | 5  | -  | 16  | 58            |

### Students Graduated : 2022-23

| S. No. | Programme      | Total Appeared in Final Examination | Total Passed the Final Examination |
|--------|----------------|-------------------------------------|------------------------------------|
| 1.     | B.Tech. (CSE)  | 59                                  | 59                                 |
| 2.     | B.Tech. (DSAI) | First batch graduates in 2023       |                                    |
| 3.     | B.Tech. (ECE)  | 55                                  | 54                                 |
| 4.     | M.Tech. (CSE)  | 15                                  | 15                                 |
| 5.     | M.Tech. (ECE)  | 03                                  | 03                                 |
| 6.     | Ph.D.          | NA                                  | 03                                 |



## Medal and Certificate Winners : 2022 Passout

### Master of Technology (M.Tech.)



**Ms. Isha Pali**  
Highest CGPA in M.Tech. (CSE)



**Ms. Maya Chandrakar**  
Highest CGPA in M.Tech. (ECE)

### Bachelor of Technology (B.Tech.)



**Mr. Gourav Chowdhary**  
Institute Gold Medal for All-Round Performance in B.Tech.



**Mr. Ritvik Muttreja**  
Chairman's Gold Medal for  
Best Innovation in B.Tech.



**Mr. Alluri L S V S Varma**  
Ajai Chowdhary's Gold Medal for  
Transformational Leadership



**Mr. Alluri L S V S Varma**  
Highest CGPA in B.Tech. (ECE)



**Mr. Ritvik Muttreja**  
Highest CGPA in B.Tech. (CSE)



**Mr. Anmol Sharma**  
Best thesis in B.Tech. (CSE)



**Mr. Harsh Verma**  
Best thesis in B.Tech. (CSE)

## All Round Development of Students at IIIT-NR



**PERSONAL : Preparing for Tomorrow**



**PROFESSIONAL : Sharpening Tech. Skills**



**PHYSICAL : Fitness and Team Work**



**INTELLECTUAL : Soft-skills and Leadership**



**CULTURAL : Indian Culture & Tradition**



**SOCIAL : Giving Back to Society**

## 8.3 Labs & Learning Infrastructure



**State-of-the-Art Labs for Hands-on Experience for Strong Practical and Technical Skills**



## Labs @ IIIT-NR

To provide students with hands-on experience for strong practical and technical skills with focus on real world problems and also to facilitate quality teaching, learning, and research, the IIIT-NR has setup various state-of-the-art labs. These are as follows:-

### ⦿ **The 3D Printing & Fabrication Lab**

The 3D Printing and Fabrication Lab is a state-of-the-art digital manufacturing space. It allows students to quickly fabricate parts from computer generated designs to rapidly iterate prototype designs.

### ⦿ **VLSI Embedded System Lab**

VLSI Embedded System Lab is well equipped with industry standard VLSI tools and hardware resources like Raspberry PI board, Universal FPGA/CPLD Card, XILINX FPGA -ADD ON Card, XILINX FPGA ZYNQ, etc. The primary objective of the lab is to provide hands-on experience to students in the field of analog, digital and mixed-signal circuit design so that they can conduct world class research, produce competitive results, create solutions tailored to society's needs, and to disseminate knowledge for public use.

### ⦿ **BigData Computing Lab**

BigData Computing Lab enables training and research in all aspects of large and distributed information systems (e.g. system development, testing, maintenance, data integration, networking, and application development). It prepares students for emerging cutting-edge technologies in the fast-growing IT industry such as BigData Computing, Healthcare Informatics, Data Integration and Data Analytics.

### ⦿ **Internet of Things & Sensors Lab**

IoT Lab is at the forefront of developing both hardware, including novel sensors, communication platform and analytics for Internet of Things (IoT) and smart environments. The IoT lab provides necessary software and equipment to develop and characterize novel smart sensors for IoT applications. Along with these, the lab is equipped with various kits like Smart Healthcare, Smart Agriculture, Smart Village/ City.

### ⦿ **Network Lab**

Fully equipped with Dell PCs and Netsim, Ns2 Simulators, the Network lab enables students to conduct research in the area of wired and wireless networks. The lab also emphasizes experimental research in the area of communication networking.

### ⦿ **Artificial Intelligence & Deep Learning (AIDL) Lab**

The AIDL Lab enables the students to carry out experiments and research on a variety of topics within Artificial Intelligence and Machine Learning with focus on using real world problems to discover computationally hard problems of modelling uncertainty, learning predictive models and enabling decision making.

### ⦿ **Communication Lab**

The Communication Lab provides necessary hardware and software resources to the undergraduate and research students of Electronics and Communication Engineering (ECE) for enhancing their skills in the communication field. Theory in the classroom is translated directly into practice with the help of the Ettus USRP software defined radio platform and Matlab/National Instruments LabVIEW. The emphasis is on physical layer concepts rather than implementation considerations.

### ⦿ **Electronic Circuit Lab**

The primary objective of this Lab is to provide engineering skills by way of breadboard circuit design with electronic devices and components and to analyse various electronic circuits and digital circuits such as multi-vibrators, Flip-Flop circuits, applications of operational amplifiers, RC coupled amplifiers, oscillators, etc. so that students are able to understand the practical aspects of basic electronics theory. It also enables the students to simulate and test the Analog, Digital and mixed electronics circuits using Verilog, Matlab and OrCAD software.

### ⦿ **Computer Programming Lab**

Computer Programming Lab is a hands-on project and assignment-based learning space where students can gain strong practical and technical skills in various programming languages, including C, Python, Java, UNIX/Linux: shell, tools, utilities and programming environments, user interfaces, and software engineering principles.

### ⦿ **Antenna & Microwave Circuit Design Lab**

This Lab supports intermediate and advanced courses in Electromagnetics and Microwave Engineering. The microwave laboratories provide the necessary support for training the students in the area of RF and Microwave Engineering. It offers analysis and simulation of various components and devices to understand the basics of RF and microwave engineering, to boost the quality of engineering education, deepen understanding, and provide the necessary practical skills to young minds.

### ⦿ **Data Science Lab**

Data Science Lab is configured to not only facilitate students to practice basic programming in C, C++ and JAVA; but also to learn advanced technologies of computer science such as Machine Learning, Artificial Intelligence, Deep Learning and Data Mining. The Lab facilitates database designing and management through dedicated tools such as Oracle and MySQL, and related computing paradigm and frameworks.

### ⦿ **Language Lab**

The blended methodology used at the Language Laboratory caters to the language needs of students by providing them suitable environment for self-paced effective language learning. The primary objective of the lab is to enhance language learning skills of the students based on their individual requirements. The lab is well-equipped with Digital Language Learning Software platform that aims to facilitate efficient teacher-student interaction to correct phonetic accents and dialects.

## IIIT-NR Library

IIIT-NR library is spanned over 2 floors acquiring plinth area of approximately 800 sq. meters with central air-conditioning system and open Wi-Fi. The primary aim of the IIIT-NR Library is to support the Institute's mission *"To create an Institute of excellence in IT education and research for meeting the emerging and challenging needs of the society."*



We go a step ahead when it comes to making available resources whether print or electronic by involving all the stake holders e.g, faculty members, students and researchers. It complements and supplements the scholarly and informational needs of faculty, researchers and students. Right from the day of inception, IIIT-NR Library is aggressively building on its collection in varieties of forms like printed text books, reference books, magazines, e- resources like IEEE xplore, e-Book collections etc. Digital Library

section of the Central Library has developed an institutional digital repository using open source software DSpace, which preserves the institute's intellectual output for the posterity. Library has a separate section for books on history and flora and fauna of Chhattisgarh.

In addition, the Institute library has membership of various professional associations and also online tools and courses for faculty, students, scholars and staff. These are :-

### Professional Membership

- National Digital Library of India (NDLI) Club
- ILA (Indian Library Association)
- INFLIBNET
- E-Shodhsindhu
- Shodhganga
- Shodhgangotri

### Online Tools in Library

| Name of the Online Tool | Description                                 |
|-------------------------|---------------------------------------------|
| Turnitin                | Plagiarism detection tool                   |
| Grammarly Premium       | Writing assistance (thesis / synopsis) tool |

## Online Courses at Library

| Name of the Online Course  | Description                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| IEEE E4TP                  | English for Technical Professionals                                                                               |
| SWAYAM<br>(Govt. of India) | Courses till PG level (Video lectures, Downloadable reading material, Self-assessment tests and discussion forum) |

## Print and e-Resources in IIIT-NR Library

| Description                                                        | Number |
|--------------------------------------------------------------------|--------|
| <b>Print Resources</b>                                             |        |
| • Books                                                            | 7814   |
| • Periodicals                                                      | 30     |
| • Newspaper Hindi                                                  | 7      |
| • Newspaper English                                                | 7      |
| <b>e-Resources</b>                                                 |        |
| • NDL (National Digital Library) - books, journals, video, dataset |        |
| • Journals - IEEE                                                  | 244    |
| • Journals - South Asia Archive                                    | 14180  |
| • Journals - Knimbus e-Library                                     | 5222   |
| • IEEE Conference papers                                           | 6043   |
| • IET Conference papers                                            | 2677   |
| • IEEE Standards                                                   | 2293   |
| • e-Books (perpetual) – [Pearson/McGraw Hills/Cambridge]           | 104    |
| • e-Books [South Asia Archive]                                     | 2751   |
| • e-Books [Knimbus e-Library]                                      | 7334   |
| • South Asia Archive - Reports                                     | 8943   |
| • South Asia Archive - Acts/Decisions etc.                         | 502    |
| • South Asia Archive - Statistical Publications                    | 550    |
| • South Asia Archive - Gazetteers                                  | 1768   |

09

# Training and Placement

## Premier Recruiting Partners

|                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |    |    |    |    |
|    |   |   |   |   |
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## Training and Placement

The Institute has a dedicated Training and Career Cell (TCC) which endeavours to provide good career opportunities to students. TCC is headed by Dr. Amit Agrawal, Assistant Professor (Management) and mentored by Dr. Pradeep K. Sinha, Director, IIIT-NR. TCC keeps close contact with potential employers for organising campus placements and internships drives for students. It also maintains lively relationships with industry partners through industry–academia interactions.



### Steps to Improve Placement

With rapid technological advancement in industry, the present-day campus placement of students has become more challenging and competitive. The Training and Career Cell (TCC) undertakes several initiatives to prepare students to successfully overcome these challenges and grab their desired career opportunities. Towards this, to make students employable and industry ready, the TCC facilitates an effective industry–institute interaction and organises various activities and programmes in coordination with Students Activity Centre (SAC) of the Institute. Some of these skill-set enhancing and career oriented programmes are:-



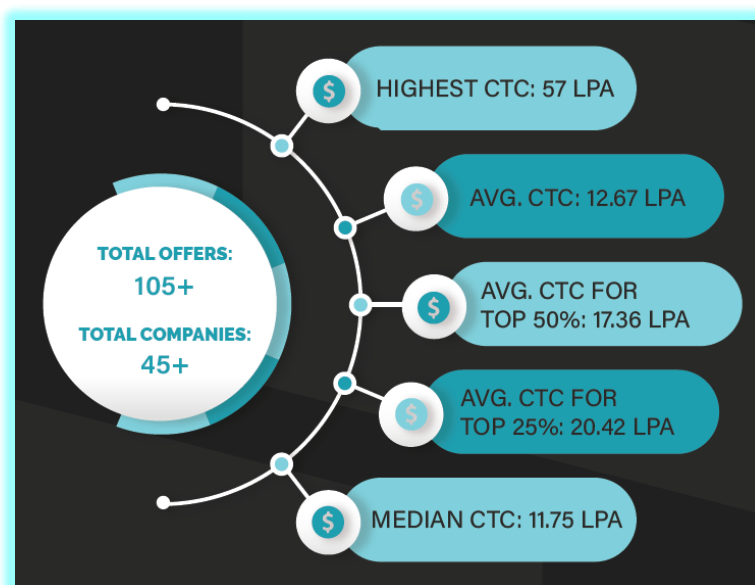
## Financial Assistance for Foreign Internship

As mentioned earlier, the Institute undertakes several initiatives to improve placement as well as career prospects of students. One of such unique initiatives is to provide financial assistance to desirous students for a semester long internship abroad. In the year 2022-23, six students applied for internship in various foreign universities. The brief details of these students are as follows:-

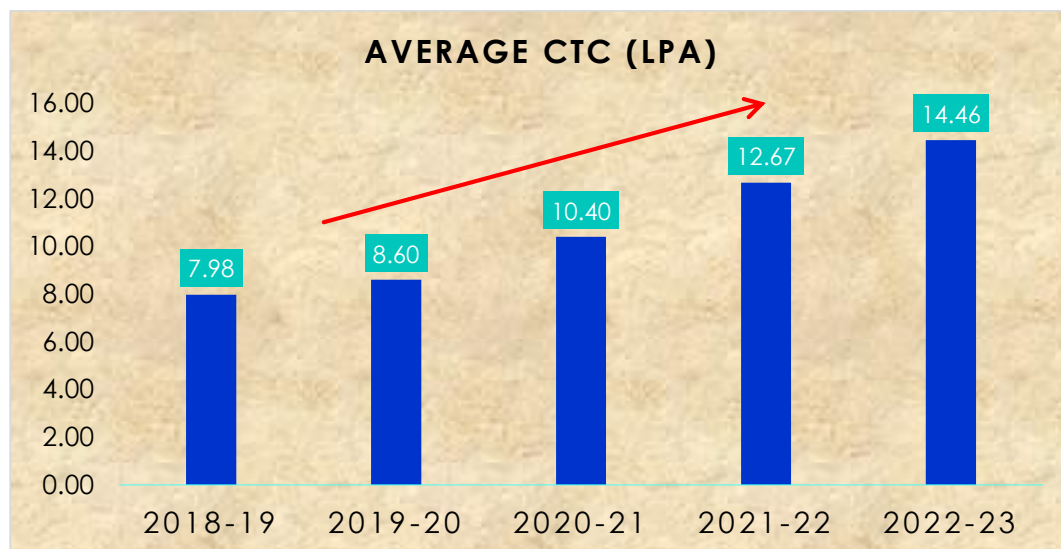
| S.No. | Name of Student     | Roll No.  | Name of the Foreign University   |
|-------|---------------------|-----------|----------------------------------|
| 1.    | Rishi Shounak       | 191020443 | Georgia State University (USA)   |
| 2.    | Aneesh Bhattacharya | 191000006 | PURDUE University (USA)          |
| 3.    | Varad Deouskar      | 191020457 | Washington University, St. Louis |
| 4.    | Aakash Agrawal      | 191010201 | Linkoping University (Sweden)    |
| 5.    | Prathistith Raj. M  | 191020442 | University of Calgary (Canada)   |
| 6.    | Marisetty Ashish    | 191020434 | INRIA (France)                   |

As a result of these placement oriented initiatives, the IIIT-NR has successfully achieved 100% placement of all its willing and eligible students for four years in a row.

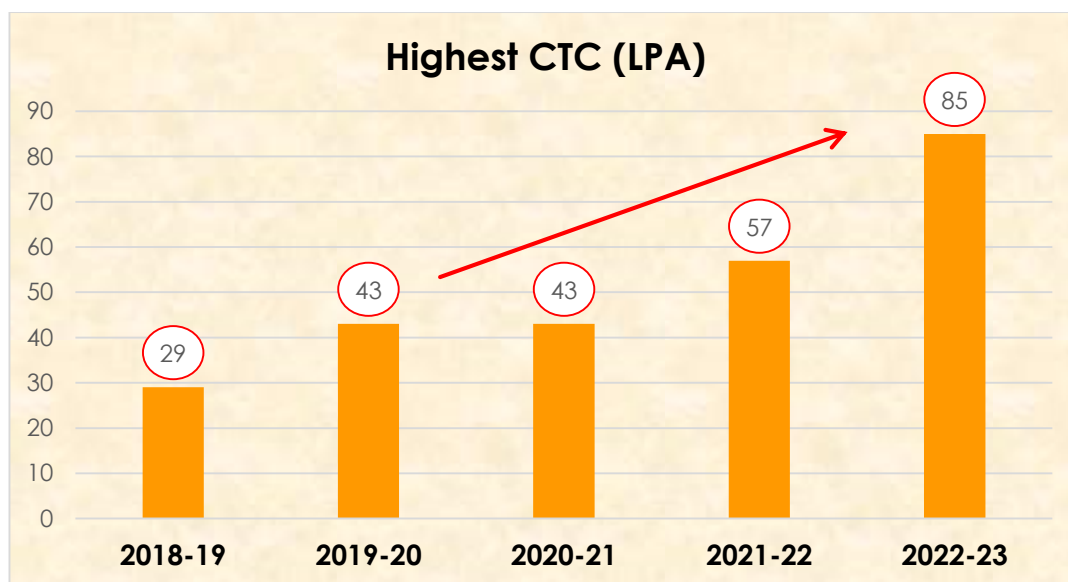
In addition, the Institute is also consistently improving the placement profile i.e. average CTC, median CTC, highest CTC and type of company profile i.e. students placed in various renowned and reputed company like Microsoft, Google, Goldman Sachs, Intel, Morgan Stanley, Amazon, American Express, Samsung, Deloitte, Directi, Carl Zeiss, Ericsson etc. A glimpse of the quality of placement of 2022 pass-out batch is as below:-



## Continuous Improvement in Placement Profile



### Continuous Improvement in Average CTC



### Continuous Improvement in Highest CTC

Those students who opt out of placement drives due to their strong inclination for higher studies in India (CAT / GATE) or abroad (GRE) are also successful in securing admission in world's top universities like University of Massachusetts Amherst (USA), University of Illinois Urbana-Champaign (USA), Cornell University (USA), Carnegie Mellon University (USA), IIT Bombay, IIT Delhi etc. Thus, IIIT-NR students are excelling in all fields whether it is industry or academia and bringing laurels to the Institute and to Chhattisgarh.

10

## Patents and Publications

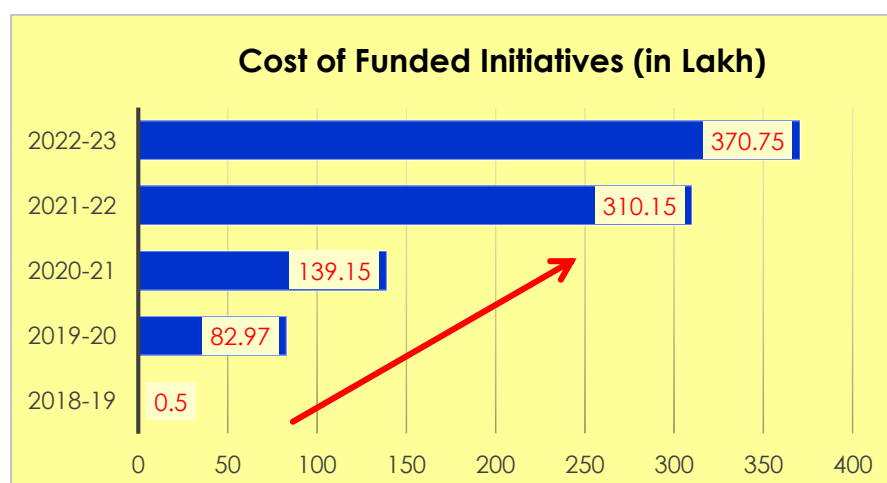
With the changing technological landscape and society needs, research has become increasingly important. Through research, educational institutions can make significant contributions to the society as well as the economy of the country. Further, quality of research are considered hallmark of quality of teaching, learning and research of any educational Institution. Considering the significance of research, IIIT-NR has been actively involved in quality research since its inception. As a result, the Institute has been making consistent progress in the field of research publications and patents.

Since its inception in 2015 till 2022-23, the Institute got awarded 23 patents and over 200 scholarly publications (research papers, books etc.) which is the testimony to the quality of research being done at IIIT-NR. In addition, there is consistent increase in funded initiatives. In the year 2018-19, the funded initiative was of Rs. 50 Lakh which has increased by manifold over the years. In the year 2022-23, the Institute got awarded 7 patents, published 84 scholarly publications and had funded initiatives of over Rs. 370 Lakh.



### Funded Initiatives

There has been a significant increase in funded initiatives at IIIT-NR. The total cost of funded initiatives has increased manifold over the years. The same is evident below:-



## Patents Published

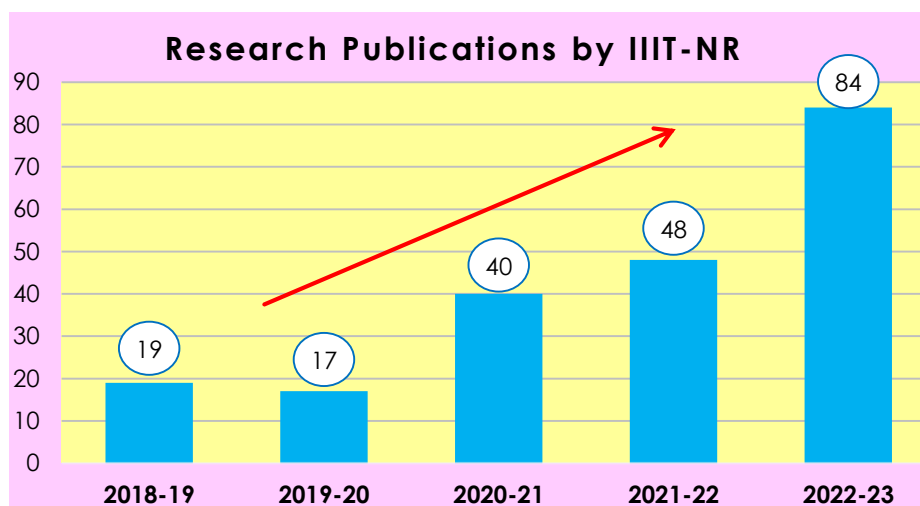
Since its inception in 2015, despite being a comparatively smaller Institute, the innovations done at IIIT-NR have been awarded 23 patents. In the year 2022-23, the Institute got awarded 7 patents from the Patent Office, Government of India. Brief details of recent patents are as follows:-

| S. No. | Name of Researchers                                                                                                                                                                                                                               | Patent Title                                                                                        |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1.     | <i>Faculty :</i><br>Dr. Maifuz Ali,<br>Dr. Deepika Gupta<br><br><i>Student :</i><br>Deepraj Chowdhury                                                                                                                                             | Smart Traffic Signal Management for Roadways                                                        |
| 2.     | <i>Faculty :</i><br>Dr. Manoj Kumar Majumder<br>Dr. Vivek Tiwari<br>Dr. Anurag Singh<br>Dr. Sresha Yadav<br>Dr. Muneendra Ojha<br>Dr. Shrivishal Tripathi<br><br><i>Student :</i><br>Divyansh Kushwaha<br>Deepanshu Pratik<br>Shivanshu Prajapati | System and Method for Determining Body Temperature Using Smartphone Coupled with USB Thermal Camera |
| 3.     | <i>Faculty :</i><br>Dr. Srinivas Naik<br><br><i>Student :</i><br>Yathin Prakash Kethepalli<br>Rony Joesph<br>Nandita Yadav                                                                                                                        | Smart Electricity Metering System and Method Thereof                                                |
| 4.     | <i>Faculty :</i><br>Dr. Mallikharjuna Rao<br><br><i>Student :</i><br>Yathin Prakash Kethepalli<br>Chahat Mittal                                                                                                                                   | System and Method to Segregate the Fruits by Machine Learning                                       |
| 5.     | <i>Faculty :</i><br>Dr. Lakhindar Murmu<br>Dr. Deepika Gupta<br><br><i>Student :</i><br>Deepraj Chowdhury                                                                                                                                         | Artificial Intelligence Controlled Door Access System for Dwellings                                 |
| 6.     | <i>Faculty :</i><br>Dr. Vivek Tiwari                                                                                                                                                                                                              | Smart Necklace and Wrist Band with Machine Learning-based Image Processing and GPS                  |

| S. No. | Name of Researchers                                                                                                        | Patent Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.     | <p><i>Faculty :</i><br/>Dr. Debanjan Das<br/>Dr. Venkanna U.</p> <p>in collaboration with IIT Kanpur and IIT Kharagpur</p> | <p>Following three different platforms for health, agriculture and education under the ‘Tribe Connect Project’ funded by the Ministry of Electronics and Information Technology (MeitY):-</p> <ul style="list-style-type: none"> <li>• ‘Tribe Aid’ platform for detecting Anemia and Lung Disease.</li> <li>• ‘Hamar Kisan’ platform for detecting diseases in crops.</li> <li>• ‘Tribe Guru’ platform for monitoring health, location and concentration of students by school and parents.</li> </ul> |

## Research Publications

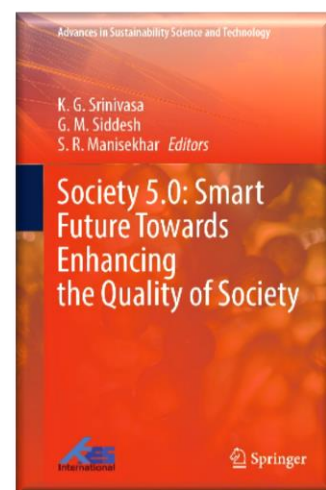
The progress made by the Institute in research publications over the years is evident from the data presented below.



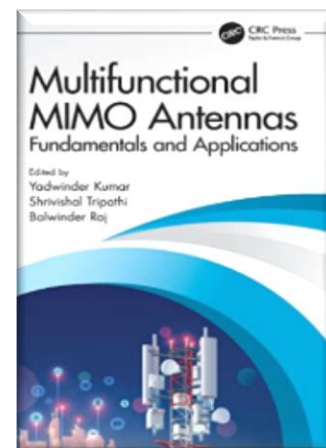
In the year 2022-23, 84 research publications were published by the Institute. Brief details of some of these research publications are as follows:-

- A research paper co-authored by students Pratyush Parashar, Vikalp Kumar Tripathi and Dr. Mallikharjuna Rao K, Assistant Professor (DSAI) on ‘*Missing Data: It’s Emergence in the Real-world - A Practical Review on Google Play Apps Dataset using Python*’, was presented at the ‘IEEE IAS Global Conference on Emerging Technologies (GlobConET), organized by IEEE IAS Society’, on May 20, 2022.

- ⊙ A research paper co-authored by students N. Meshram, G. Prasad, D. Sharma, and Dr. Bipin Chandra Mandi, Assistant Professor (ECE) on *'Low-Power and High-Speed SRAM for Ultra-Low Power Applications'*, was accepted for presentation at the '2022 IEEE International IoT, Electronics and Mechatronics Conference (IEMTRONICS)', organized from June 1-4, 2022.
- ⊙ A research paper by Dr. Srinivas Naik, Assistant Professor (CSE), on the topic *'Mobilizing Digital Volunteers to Support Underserved Communities in India During COVID-19 Lockdown'*, was published in the proceedings of 'ACM Conference on Computing Systems and Sustainable Societies (ACM COMPASS 2022)'. This work is in collaboration with CGNet Swara, IIIT Bangalore and Cornell University (USA).
- ⊙ A research paper co-authored by Kumar Shubham (student), Sakshita Jayaswal (student), Kunal Bose (student) and Dr. Vivek Tiwari, Assistant Professor (CSE) on *'US Traffic Sign Recognition by using Partial OCR and Inbuilt Dictionary'*, was published in the proceedings of '7<sup>th</sup> Springer International Conference on ICT for Sustainable Development (ICT4SD 2022)', Springer.
- ⊙ An edited book titled *'Society 5.0: Smart Future towards Enhancing the Quality of Society'*, edited by Dr. Srinivasa K. G., Professor (DSAI) was published by Springer, Advances in Sustainability Science and Technology Series (ISBN: 978-981-19-2160-5).
- ⊙ A research paper titled *'A Novel Multimodal Fusion Framework for Early Diagnosis and Accurate Classification of COVID-19 Patients Using X-ray Images and Speech Signal Processing Techniques'* (Manuscript Number: CMPB-D-22-01197R1), by Dr. Santosh Kumar, Assistant Professor, was published in the journal *'Computer Methods and Programs in Biomedicine'* (Impact factor: 7.027) Elsevier.
- ⊙ A research paper titled *'A Case Study on Marketing Strategy of boAt'* by Lakshya Singh, student of B. Tech. 2019 batch and Dr. Amit Agrawal, Assistant Professor (Management), was accepted for publication in *The Case Journal*.
- ⊙ A research paper co-authored by Kumar Shubham (student), Kunal Bose (student), Dr. Vivek Tiwari, Assistant Professor (CSE) and Dr. Kudip Patel, Assistant Professor (Mathematics) on *'Insect Image Semantic Segmentation and Identification using UNET and DeepLab V3+'*, was published in the proceedings of '7<sup>th</sup> Springer International Conference on ICT for Sustainable Development (ICT4SD 2022)', Springer.
- ⊙ A research paper by Dr. Amit Agrawal, Assistant Professor (Management) on *'Examining the Role of Gender on Family Business Entrepreneurial Intention: Influence of Government Support and Technology Usage'*, has been published in the *Journal of Family Business Management*.



⊙ A book titled '*Multifunctional MIMO Antennas: Fundamentals and Applications*', edited by Yadwinder Kumar, Dr. Shrivishal Tripathi, Assistant Professor (ECE), and Balwinder Raj was published by CRC Press, Taylor & Francis Group, ISBN: 978-1-032-19034-1, DOI: 10.1201/9781003290230.



⊙ A research paper titled '*Missing Data: It's Emergence in the Real-world - A Practical Review on Google Play Apps Dataset Using Python*' by Dr. Mallikharjuna Rao, Assistant Professor (DSAI) was published in the proceedings of '*IEEE IAS Global Conference on Emerging Technologies (GlobConET)*' organized by IEEE IAS Society, Arad, Romania.

⊙ A research paper co-authored by Nalini Pusarla, part-time Ph. D. scholar, Dr. Anurag Singh, Assistant Professor (ECE) and Dr. Shrivishal Tripathi, Assistant Professor (ECE), titled '*Normal Inverse Gaussian Features for EEG-based Automatic Emotion Recognition*', was published in '*IEEE Transactions on Measurements and Instrumentation*'.

⊙ A research paper titled '*A Decentralized KYC Based Approach for Microfinance using Blockchain Technology*' by students Bodicherla Digvijay Sri Sai, Ramisetty Nikhil, Shivangini Prasad and faculty Dr. Srinivas Naik, Assistant Professor (CSE), was published in the Journal Cyber Security and Applications (Elsevier).

⊙ A research paper titled '*Deep Learning based Holistic Speaker Independent Visual Speech Recognition*' by students Praneeth Nemani, G. Sai Krishna, Nikhil Ramisetty and B. Digvijay Sri Sai and faculty Dr. Santosh Kumar, Assistant Professor (CSE), was published in IEEE Transactions on Artificial Intelligence.

⊙ A research paper titled '*Fog Based Smart Healthcare: A Machine Learning Paradigm for IoT Sector*' by Prof. K. G. Srinivasa was published in Multim. Tools Appl. 81(26): 37299-37318 (2022).

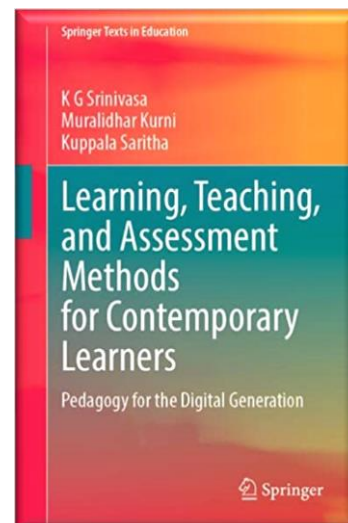
⊙ A research paper titled '*Performance Analysis using Air Gap Defected Through Silicon Via: Impact on Crosstalk and Power*' by M. Tech. student Maya Chandrakar, was published in IEEE Transactions on Components, Packaging and Manufacturing Technology.

⊙ A research paper titled '*Dual Broad Band-Stop Polarization-Insensitive and Scanning Angle Stability Frequency Selective Surface*' by Dr. Lakhindar Murmu, Assistant Professor (ECE), was published in AEU- International Journal of Electronics and Communications.

⊙ A research paper co-authored by a student Neeraj Gautam, and faculty Dr. Lakhindar Murmu, Assistant Professor (ECE) and Dr. Bipin Chandra Mandi, Assistant Professor (ECE) on '*Inverted V-Shape Slot Self Diplexing Antenna for X-band Applications*', was presented at URSI-Regional Conference on Radio Science (URSI-RCRS 2022), organized from December 01-04, 2022.

- A research paper titled '*An Energy-Efficient Event Reliability Protocol for Wireless Communication Networks*' by Dr. Sailesh Khapre, Assistant Professor (DSAI), has been published in *Mobile Information Systems*, Vol. 2022.
- A research paper co-authored by students Shristry Kashyap, Tavisha Thaware, Subham Raj Sahu and faculty Dr. Mallikharjuna Rao K, Assistant Professor (DSAI) on '*Multi-Crop Leaf Disease Detection using Deep Learning Methods*', was presented at IEEE INDICON 2022, organized by IEEE Kerala Section, at Cochin University of Science & Technology, on November 26, 2022.
- A research paper co-authored by student Kumar Shubham and Dr. Vivek Tiwari, Assistant Professor (CSE), on the topic '*Predictive Learning Methods to Price European Options using Ensemble Model and Multi-Asset Data*', was published in '*International Journal on Artificial Intelligence Tools*', World Scientific (SCI-E, Impact Factor: 1.059).
- A research paper co-authored by students Ghanta Saikrishna and Kundraput Supriya; and Dr. Mallikharjuna Rao, Assistant Professor (DSAI), on the topic '*Data Preprocessing Techniques: Emergence and Selection towards Machine Learning Models - A Practical Review using HPA Dataset*', was published in '*Multimedia Tools and Applications - An International Journal*', Springer Publisher, Impact Factor: 2.577, SCIE Indexed.
- A research paper authored by Dr. Lakhindar Murmu, Assistant Professor (ECE), on the topic '*A Dual-band Circularly Polarized Planner Monopole Antenna for Wireless Application*', was published in the proceedings of 'International Conference on Science, Technology and Engineering (ICSTE 23)' held on February 17-18, 2023, at NIT Manipur.
- A research paper authored by Dr. Lakhindar Murmu, Assistant Professor (ECE), on the topic '*A Compact Dual-Element MIMO Antenna with Low Mutual Coupling for Wideband Application*', was published in the proceedings of 'International Conference on Science, Technology and Engineering (ICSTE 23)' held on February 17-18, 2023, at NIT Manipur.
- A research paper authored by Dr. Shailesh Khapre, Assistant Professor (DSAI), on the topic '*Comparative Study in the Diagnosis of Parkinson's Disease Using Machine Learning*', was published in the proceedings of the 3rd International Conference on Emerging Trends and Technologies on Intelligent Systems (ETTIS-2023), organized by CDAC Noida in association with the Automatic Control, Computers & Electronics Department, Faculty of Mechanical and Electrical Engineering, Petroleum-Gas University of Ploiesti, Romania, and the University of Haute-Alsace, France.
- A research paper co-authored by students Aditi Verma, Mayank Lovanshi and Rahul Shrivastava; and Dr. Vivek Tiwari, Assistant Professor (CSE), on the topic '*A Human Body Part Semantic Segmentation Enabled Parsing for Human Pose Estimation*' was presented at the 'ACM 5th International Conference on Image, Video and Signal Processing (IVSP 2023)', on March 24-26, 2023, at NTU, Singapore. **Aditi Verma received the best paper award for this work.**
- A research paper co-authored by students Vedanshu Dewangan and Aditya Saxena; and Dr. Shrivishal Tripathi, Assistant Professor (ECE), on the topic '*Application of Image Processing Techniques for UAV Detection Using Deep Learning and Distance-Wise Analysis*', was published in '*Drones*'.

⊙ A book co-authored by Prof. Srinivasa K. G., titled *Learning, Teaching, and Assessment Methods for Contemporary Learners: Pedagogy for the Digital Generation*, was published by Springer Nature. The textbook tackles the matter of contemporary learners' needs, and introduces modern learning, teaching, and assessment methods. It provides a deeper understanding of these methods so that the students and teachers can create teaching and learning opportunities for themselves and others. This textbook showcases various modern learning, teaching, and assessment methods for contemporary learners in an increasingly digital environment. Each chapter presents insights and case studies that show how modern methods can be applied to classrooms, and how they can support the existing curriculum.



⊙ A research paper co-authored by students Anam Arshad, Mayank Lovanshi and Rahul Shrivastava and Dr. Vivek Tiwari, Assistant Professor (CSE), on the topic *'Human Activity Role Identification using Feature Vector and Encoding Techniques from Natural Language Sentences'* was presented at the 'ACM 5th International Conference on Image, Video and Signal Processing (IVSP 2023)', on March 24-26, 2023, at NTU, Singapore.

⊙ A research paper co-authored by students Harsh Verdhan Singh, D. Venkata Siva Prasad and Dr. Shrivishal Tripathi, Assistant Professor (ECE), on the topic *'Self-Isolated MIMO Antenna using Mixed Coupling by Close Coupling Technique'*, was published in 'Nature Scientific Reports'.



# 11

## Collaborations & Conferences

### Collaborations and MOUs

As the purpose of any collaboration or MoU with other higher educational institutions and industry is to develop co-operation and promote mutual understanding and excellence in practice-based education, research and knowledge exchange between the parties so that the research and knowledge developed is used for further research and its applications at regional, national and global level, IIIT-NR is proactively involved in increasing collaboration with other higher educational institutions and industry.

With its firm belief in advancing society through multidisciplinary collaborative research, IIIT-NR has signed MoUs with several premier academic and research institutes. These include All India Institute of Medical Sciences (AIIMS) Raipur, DRDO, C-DAC, Microsoft Research Lab, Samsung R&D, Louisiana State University (USA), Wright State University (USA), Indira Gandhi Krishi Vishwavidyalaya (IGKV), SECQAI Ltd. (UK) etc.

In the year 2022-23, IIIT-NR signed MoU with five reputed organisations. The brief details of these recent MoUs are as follows:-



#### © IIIT-NR and Samsung (SRI-B)

Samsung Research Institute Bangalore (SRI-B), Samsung's largest R&D facility outside Korea, is engaged in the development of software and applications for cellular phones, tablets, laptops, computers and other information technology and telecom products. It collaborates with students and faculty at partner institutes, giving them research as well as development projects to be executed over four to six months. With this MOU, students of IIIT-NR will get to work on projects in cutting edge technologies such as Artificial Intelligence (including Vision Tech), Machine Learning, Internet of Things and Connected Devices and 5G networks, to solve real-time problems of industry. Each project is taken up by a team of three students and one professor, with a mentor from SRI-B, who guides and trains them, and conducts regular reviews.

### ◎ IIIT-NR and SECQAI, London

SECQAI is a London based company inventing new hardware and software focused on solving the energy problem of Models, Simulations and Artificial Intelligence. As per the MOU, Research Internships of total Rs. 14,00,000/- were provided to the students of IIIT-NR to carry out few select research projects of SECQAI.

### ◎ IIIT-NR and GeeksforGeeks

GeeksforGeeks is India's biggest coding and tech-community. It conducted one of the largest and the most rewarding hackathon events in the country, sponsored by Google Cloud India, and this MoU was for mutual cooperation for the conduct of the Hackathon. The hackathon provided students with all necessary resources, such as mentors, access to Google's technical resources, and prize money. Top-performers were rewarded with exciting prizes and certificates. Hackathons are a great way for students to learn as well as demonstrate their skill-sets while working with industry-leading companies like Google.

### ◎ IIIT-NR and UiPath

An academic collaboration has been done between IIIT-Naya Raipur and UiPath, a global software company that makes robotic process automation software for collaboration on growing the Robotic Process Automation (RPA) knowledge ecosystem.

### ◎ IIIT-NR and IIIT-B COMET Foundation

IIIT-Naya Raipur and IIIT-Bangalore COMET Foundation entered into an MoU on June 8, 2022 in the field of Advanced Communication Systems ("Project") as per the Tripartite Agreement dated March 30, 2021 between IIIT-Bangalore and COMET with the Mission Director, Mission Office NM-ICPS, Ministry of Science & Technology, Department of Science & Technology, Government of India to create a strong foundation and a seamless ecosystem for Cyber Physical Systems technologies. Duration of the Project is 4 years with an outlay of Rs. 1,73,56,000/-.

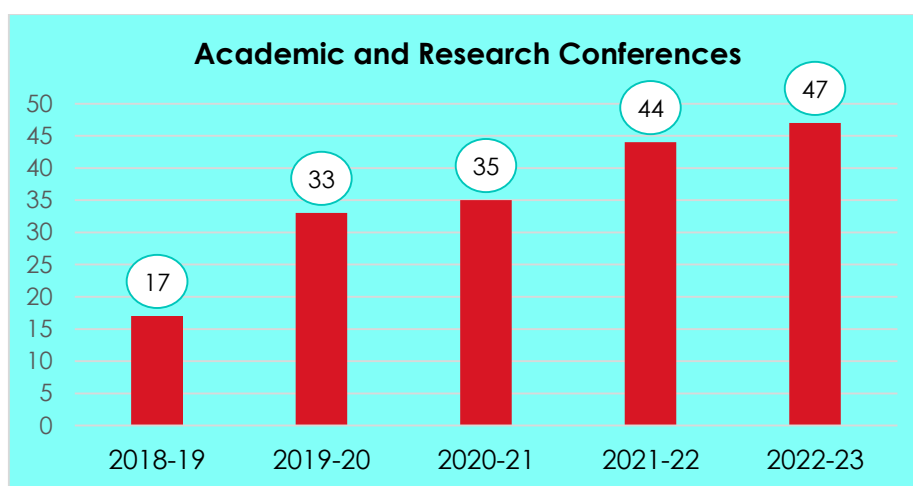
### ◎ IIIT-NR, IIT Kharagpur and IIT Kanpur collaboration for Tribe Connect Project

IIIT-NR team of researchers in collaboration with IIT Kanpur and IIT Kharagpur are working on 'Tribe Connect Project' funded by the Ministry of Electronics and Information Technology (MeitY), Govt. of India. In this project, the team is working on three areas, Health, Agriculture and Education. For Healthcare, 'Tribe Aid' platform has been developed, which helps in detecting Anemia and Lung Disease. For Agriculture 'Hamar Kisan' platform has been developed, which helps in detecting diseases in crops. For Education, 'Tribe Guru' platform has been developed, with the help of which school administration and parents can monitor the health, location and concentration of the students. The whole process has been patented.



## Academic and Research Conferences

Academic conferences help to accelerate research on a particular subject. They provide access to various research activities related to a particular subject with current findings and developments anticipated from them. Considering its significance, IIIT-NR has organised and participated in several such academic and research conferences in India and abroad since its inception.



In addition, for increased interactions with researchers across the globe, IIIT-NR started its own international conference named TRIBES (Technology, Research, and Innovation for BETterment of Society). To encourage its members to actively participate in professional societies globally, IIIT-NR has established IEEE Student Branch and IEEE Women in Engineering Affinity Group. In December 2022, it also hosted the IEEE International Women in Engineering Conference. With these initiatives, the institute and the State of Chhattisgarh are gaining increased popularity nationally and internationally as a research hub.

Brief details of some of the conferences organised and participated by the Faculty of IIIT-NR in the year 2022-23 are as follows:

- ❶ Dr. Shrivishal Tripathi, Assistant Professor (ECE), on invitation, took part as a ‘Technical Expert’ in the regional ‘*Security Strategies Conference*’, held on April 27, 2022. The conference was organized by the Intelligence Bureau (IB), Ministry of Home Affairs, Chhattisgarh.
- ❷ Dr. Amit Agrawal, Assistant Professor (Management), delivered an invited talk on ‘*Business Model*’, at an event ‘*Startup Weekend 2022*’, organized by ITM University, Raipur, on April 09, 2022.
- ❸ Dr. Amit Agrawal, Assistant Professor (Management), attended the conference and was Session Chair at the ‘*3rd International Conference on Building Resilience and Sustainability for India in an Ever-changing Turbulent World: A Vision for 2030*’, organized by IMS Ghaziabad, on April 21, 2022.

Dr. Bipin Chandra Mandi participated in a SERB-sponsored 5-day online short-term course on '*Advances in Signal Processing and VLSI Technologies*' from May 02-06, 2022.

Dr. Kuldip Singh Patel, Assistant Professor (Mathematics), was a resource person in a 2-weeks online Capacity Building Programme on '*Research Methodology in Social Sciences*', organized by the Department of Education, Guru Ghasidas Vishwavidyalaya, Bilaspur and sponsored by Indian Council of Social Science Research (ICSSR), from May 18-31, 2022. The programme was meant for faculty members and research scholars.

Dr. Shrivishal Tripathi, Assistant Professor (ECE), delivered an Invited talk on the topic '*Crop Protection using AI and IoT*', at a workshop organized by International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), Patancheru, Hyderabad (Sponsored by ICRISAT, India and University of Copenhagen). The objective of the workshop was '*Establishment of a Danish-Indian Partnership on Smart Plant Protection (SPP)*', based on the current progress of applying digital technologies and services for farmers in India.

Dr. Amit Agrawal and Dr. Lakhindar Murmu of IIIT-NR participated in a '*5-Days Orientation Program for Entrepreneurship Educators*' conducted at Entrepreneurship Development Institute of India (EDII), Ahmedabad from July 4-8, 2022. The objective of the programme was to help in building a climate for entrepreneurship education that enables students to consider entrepreneurship as a career option.

Dr. Rajarshi Mahapatra, Associate Professor, delivered a keynote lecture on the topic '*5G and Beyond*', at the International Conference on Human Machine Interaction (ICISHMC 2022), on July 09, 2022.

Prof. Rajesh Ingle, Professor (CSE), organized the 1<sup>st</sup> Edition of '*IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS) 2022*', from September 16-18, 2022, at Pune. The conference aimed at providing a platform for researchers, engineers, and practitioners to contribute to emerging research topics in Blockchain and present the latest advances and innovations in theories, applications, infrastructure and schemes for the Blockchain. Dr. Pradeep Sinha, Director IIIT-NR. was the Chief Guest of the inaugural session of the conference. Other eminent personalities present for the inaugural session included Dr. Vidyasagar Potdar, Director of the Blockchain R&D Lab, Curtin University, Australia, Mr. Sanjeev Fadnavis, CEO Nitor Infotech Pvt. Ltd., and Mr. Girish Khilari, Chair IEEE Pune Section. Dr. Nita Patel, IEEE Computer Society President Elect 2023, Dr. Mohammed Kassim, IEEE R10 CS Chair, Malaysia, and Dr. Rajkumar Buyya, University of Melbourne, Australia, were present online.



Prof. Rajesh Ingle, HoD CSE, was invited to deliver a talk on the topic '*Enhancing Smart City Initiatives Using Blockchain and Edge Computing*', as part of ATAL Faculty Development Program on '*Blockchain in Smart City: Emerging Technologies for the Next Decade and Beyond*', organized by the Department of Information Technology and Department of Computer Science & Engineering, National Institute of Technology, Raipur, on September 19, 2022.

**IEEE WIECON-ECE 2022 International Conference** : IIIT-Naya Raipur hosted a two-day '8<sup>th</sup> IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2022 (IEEE WIECON-ECE 2022)', on December 30-31, 2022 which was jointly organized by IEEE Bangladesh Section and IEEE Madhya Pradesh Section in collaboration with their WIE Affinity Groups and supported by Kalinga University. Prof. Rajiv Prakash, Director, IIT Bhilai was the Chief Guest of the inaugural session of the conference. In his address, Prof. Prakash emphasized the importance of female engineering students opting for various branches of engineering other than Computer Science, Information Technology and Electronics. He mentioned that in STEM (Science, Technology, Engineering and Mathematics) areas, the percentage of female graduates in India is one of the highest in the world. Immediately after the inaugural session, Dr. Tom Coughlin, IEEE President-Elect 2023, delivered a keynote address on *"The Benefits of Becoming an IEEE Member and the Various Opportunities that IEEE Provides to Professionals to Enable Them to Contribute for the Advancement of our Society"*. Among various other talks, Dr. Bozena Pasik-Duncan, Professor, University of Kansas, Lawrence, Kansas delivered a keynote on *"Diversity and Inclusion in Stochastic Systems and Control – Research and Education"*. In the valedictory session, Dr. P. K. Sinha, Director, IIIT-Naya Raipur, stressed upon empowering women for society upliftment by contributing different technology-driven solutions.



**General Chair at ICDMAI 2023** : Dr. Pradeep K. Sinha, Director, IIIT-NR, participated as General Chair in the '7<sup>th</sup> International Conference on Data Management, Analytics and Innovation (ICDMAI-2023)' on January 20-22, 2023, at Defence Institute of Advanced Technology (DIAT), Pune. The Annual Conference aims at bringing eminent academicians, corporate executives, researchers, technocrats and experts from the field of Computer Science, Information Technology, Electronics and Telecommunication Engineering, Electrical Engineering, Computer Applications and all relevant disciplines to a common forum for exploring and discussing the technology advances and innovations that play a key role in data management and analytics. Dr. Sinha also met and interacted with the Vice Chancellor and the faculty members of DIAT.





## 12

# Awards and Recognitions

## Director, IIIT-NR Awarded Asia-Pacific Artificial Intelligence Association Fellow

The Asia-Pacific Artificial Intelligence Association (AAIA), based in Hong Kong, has conferred its highest honour, the Fellow of the Association, on Dr. Pradeep K. Sinha, Director, IIIT-NR. Considering Dr. Sinha's contributions to the advancement of Science & Technology and Technical Education, the Executive Board of the AAIA decided to bestow this honour upon him after recognizing him among the top scientists in the world with outstanding achievements in the area of Distributed Computing.



## Start me-Up Award

IIIT-Naya Raipur has been awarded with 'Start me-Up Award (Silver Category)' by Wadhvani Foundation for the institute's active participation and contribution to global entrepreneurship initiatives. Wadhvani Foundation is a not-for-profit organization with the primary mission of accelerating economic development by driving job creation through large-scale entrepreneurship, small business growth, innovation, and skilling initiatives. The core objective of the Wadhvani Foundation is to inspire students to take entrepreneurship as a career.



## Gold Faculty Award

Dr. Amit Agrawal, Assistant Professor (Management), has been awarded "Gold Faculty Award" by Wadhvani Foundation for his efforts towards Educating, Inspiring and Guiding Global Jury Certified Student Ventures.



## Certificate of Excellence

Dr. Shrivishal Tripathi, Asst. Professor (ECE) was awarded with Samsung PRISM 'Certificate of Excellence' and Amazon Voucher for successful completion of the project titled 'IoT, ML Optimal UWB Sensor Placement in Smart Home Environment'.

### Elevation to 'IEEE Senior Member' Grade

The following faculty members of IIIT-NR have been elevated to 'IEEE Senior Member' grade during this year, an honour bestowed by IEEE for significant contributions to the profession:-

- Dr. Punya Prasanna Paltani, Assistant Professor (Physics)
- Dr. Ramakrishna Bandi, Assistant Professor (Mathematics)
- Dr. Shrivishal Tripathi, Assistant Professor (ECE)
- Dr. Srinivas Naik, Assistant Professor (CSE)

### IIIT-NR IEEE WIE Affinity Group

IEEE Women in Engineering (WIE) Affinity Group under IIIT-NR student branch was approved by IEEE. The IEEE Women in Engineering (WIE) is a global network of IEEE members and volunteers dedicated to promoting women engineers and scientists, and inspiring girls around the world to follow their academic interests in a career in engineering and science.

IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

### Prof. Rajesh Ingle Received Cloud Security Alliance Certificates

Prof. Rajesh Ingle (Dean Academics) received two certificates from Cloud Security Alliance (CSA) for his contributions to two modules namely "*Introduction to Zero Trust Architecture*" and "*Key Features and Technologies of Software Defined Perimeter*".

Cloud Security Alliance (CSA) is the world's leading organization dedicated to defining and raising awareness of best practices to help ensure a secure cloud computing environment. CSA team as a part of Cloud Security Alliance's training program, thanked Prof. Rajesh Ingle for his incredible contributions to the Zero Trust Training. This courseware was made possible by his commitment to vendor-neutrality, student understanding, and precision.



### Google Developer Student Club

IIIT-NR initiated Google Developer Student Club, supported by Google, at its campus. It will play an instrumental role in building a strong community in the campus through which IIIT-NR students can become industry ready by learning the latest trends in technology and build projects that solve real world challenges.



Google Developer Student Clubs

## Dr. Kuldeep Received SERB International Research Experience Fellowship

Dr. Kuldeep Singh Patel, Assistant Professor (Mathematics), was awarded the 'SERB International Research Experience (SIRE) Fellowship' by the Department of Science & Technology (DST), Government of India. Under this fellowship, Dr. Kuldeep visited University of Mauritius to work with scientists there on financial market problems, specifically 'High-Order Compact Schemes for Pricing Interest Rate Derivatives'.

The SIRE fellowship helped IIIT-NR to increase its visibility in India and abroad, and to open various research/intern/post-doctoral opportunities to the B. Tech./ M. Tech. students and research scholars of IIIT-NR.

### Dr Kuldeep receives International fellowship

■ Staff Reporter  
RAIPUR, May 28

THE SERB International Research Experience (SIRE) Fellowship proposal by Dr Kuldeep Singh Patel, Assistant Professor (Mathematics) in IIIT-Naya Raipur, has been accepted by the Department of Science & Technology (DST), Government of India.



Dr Kuldeep Singh Patel

Under this fellowship, Dr Kuldeep will visit the University of Mauritius to work on financial market problems, specifically 'High-Order Compact Schemes for Pricing Interest Rate Derivatives'. The researcher will collaborate with Professor Muddun Bhuruth, Department of Mathematics in University of Mauritius. The fellowship duration is four months, from June 2022-September 2022. These fellowships broaden the vision for future research collaboration and establish new research possibilities. It will also open various research/intern/post-doctoral opportunities to the B.Tech./M.Tech students and research scholars of IIIT-NR.

It is a prestigious fellowship by the government of India, which will significantly increase the visibility of IIIT-NR in India and abroad, said Dr Pradeep Kumar Sinha, Director of IIIT-NR. Dr Sinha also added that the main focus of IIIT-NR in the coming years would be on research and such international collaborations will strengthen the foundation for multi-national collaborative research in Chhattisgarh.

## IIIT-NR Received Appreciation Certificate from the Hon'ble Governor of Chhattisgarh

IIIT-NR received 'Certificate of Appreciation' issued by Indian Red Cross Society, CG State Branch, at the hands of then Hon'ble Governor of Chhattisgarh, Ms. Anusuiya Uikey, for the Voluntary Blood Donation camps organised by the institute during the last few years. The certificate was awarded on the occasion of 'World Blood Donor Day' on June 14, 2022. Dr. Pradeep K. Sinha, Director, IIIT-NR, received the certificate on behalf of the institute.



## Winner of Tally CodeBrewers

Kushal Jain, student of IIIT-NR, won 'Tally CodeBrewers 2<sup>nd</sup> Edition' in the 'Wizard of System Programming' track, amongst over 2000+ entries from across India. Kushal outshined in the high density online quiz and coding followed by product development challenge where he had to come up with a working solution for one of the challenges using a technology of his choice in 48 hours. Tally Code Brewers, by Tally Solutions, was started in the year 2021 to inculcate and promote the passion for innovation, logical thinking and problem solving in the next generation of engineers. The winners will get an opportunity to intern with Tally Solutions and direct entry for campus placements.

### Kushal Jain of IIIT outshined at Tally CodeBrewers 2nd edition

Raipur, Aug 18: Kushal Jain of IIIT Naya Raipur won Tally CodeBrewers 2nd edition in the 'Wizard of System Programming' track amongst over 2000+ entries from across India. Kushal outshined in the high-density online quiz and coding followed by product development challenge where he had to come up with a working solution for one of the challenges using a technology of his choice in 48 hours.

Tally CodeBrewers by Tally Solutions was started in the year 2021 to inculcate and promote the passion for innovation, logical thinking and problem solving in the next generation of engineers.

An annual initiative by Tally Solutions, Tally CodeBrewers is open to engineering students across all years pursuing a full-time B.Tech./B.E./Dual-Degree in any branch across India. In round one of Tally CodeBrewers, participants had to go through a time-bound online multiple-choice questions followed by online coding challenge where the 'Team Lead' had to solve two coding problem statements in 90 minutes. Top performing teams got a chance to par-

ticipate in the 'Weekend Hackathon' where they had to come up with a working solution for any of the two product development challenges - 'Wizard of System Programming' and 'Commander of Full Stack'. The differentiating factor of this hackathon was that shortlisted teams were assigned dedicated mentors to guide them in their journey by helping understand the challenge in detail, solve their queries, and provide feedback. The final demos were evaluated by a panel of experts from Tally Solutions.

The second edition of Tally CodeBrewers witnessed three winning categories - Winner, 1st runner up and 2nd runner up across both the product development challenges. Two participants got special mention for their near winning solutions and spirit. Winning teams for both the product development challenges were rewarded with ₹75,000 followed by ₹40,000 for the 1st runner up and ₹25,000 for the 2nd runner. The winners will also get an opportunity to intern with Tally Solutions and direct entry for campus placement.

## IIIT-NR Recognized for Swachhata Activities

Mahatma Gandhi National Council of Rural Education (MGNCRE), Ministry of Education, Government of India appreciated the initiatives of IIIT-NR for maintaining one of the cleanest and greenest campuses in the region. The Ministry has acknowledged that the Swachhata related activities of the institute have been highly progressive and motivating.

## Development of Software to Filter Toxic Comments on Social Media

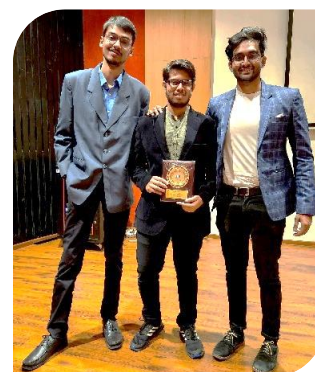
In order to keep the user safe on social media and to maintain positivity on social media, a group of three students namely Sayantan Roy, Rishi Shounak, and Vivek Kumar, under the guidance of Dr. Vivek Tiwari, worked on the development of a '*Software that Filters Toxic Comments on Social Media*'. This software works on all social media platforms including Facebook, Instagram, Twitter, and YouTube.

The software works on a machine learning model. It has its own data based on which it decides whether the comment is negative or positive and if anyone posts a negative comment, the filter automatically deletes it. The software also allows the user to decide first whether he/she wants to see the comment or wants to delete it immediately.



## Winners at NIT Raipur Entrepreneurship Summit

*UniPlug*, an entrepreneurial venture by IIIT-NR final year students Yashasva Raj Sherke, Ojas Ankit and Priyansh Pandey, was adjudged winner in the '*Ignition - the B Plan Competition*' conducted under the aegis of the annual '*Entrepreneurship Summit*' of NIT Raipur held from December 02-04, 2022. In this competition, the various startups shortlisted teams had to pitch their ideas to the esteemed judges. The team won amongst 100+ startups after a stirring competition of three rounds.



## A Few More Achievements

- IIIT-NR students Vinayak Srivastava, Aman Singh, Dhaval Kumar and Anisha Agrawal won the National Level Hackathon hosted by the Centre for Networked Intelligence (CNI) at the Indian Institute of Science (IISc), Bangalore. The hackathon had two tracks: Data Science Challenge and an AI Challenge. The Data Science task involved a bus fleet dataset obtained from BMTC (Bengaluru Metropolitan Transport Corporation). IIIT-NR Team came 3<sup>rd</sup> in this track. The AI challenge consisted of building a bot that competes with another bot to conquer more cells (area) in a grid. The IIIT-NR team of students came 2<sup>nd</sup> in this track.

- Team 'Grotesque' of IIIT-NR students presented the idea on 'Development of Online Projects Progress Monitoring System for Funded Higher Educational Institutions under Various Schemes and the Funding Agencies'. The problem statement was - To enhance the quality of financial assistance being provided by several funding agencies under respective notified schemes. The agencies were categorized into Government, Government-aided and Private.



- IIIT-NR team 'Code Titans', consisting of Vandit Tyagi and Devajit Patar won the 'Interrupt, Innovate, Iterate (III)' Hackathon, organized by 'Gravity', the technical society of IIIT Allahabad. This 36 hours long hackathon was scheduled from Jan 20-23, 2023. The aim of the event was to encourage the participants to dream big, enabling them to learn and build a project from scratch, be it blockchain, AI/ML, or future mobility.

- IIIT-NR team 'The Tag Ends' emerged winners at the 'Ethos Hackathon' organized by IIT Guwahat from December 17, 2022 to January 09, 2023. The hackathon focused on real-world problem statements proposed by Saptang Labs and the Assam Police. IIIT-NR team consisting of Devajit Patar, Vandit Tyagi, Ganesh Prajapati and Gautam Sarawagi were adjudged winners in the Assam Police problem statement named 'Smart Naka using Deep Learning and AI'. The project aimed at improving traffic management, safety, security and tracking criminals at Naka points (check-posts).

- Shri Dharmendra Pradhan, Cabinet Minister, Education and Skill Development, Government of India discussed India's Cyber Security matters with one of our alumni, Siddhant Dash. Siddhant Dash was a Batch 2015-2019 student of IIIT-NR, who did his Masters from Cornell University, USA after his B. Tech. from IIIT-NR.



- IIIT-NR team 'ChatGDP', consisting of Aman Sahu, Sanskar Singh, Rigved Desai, Shivaibhav Dewangan, Shivam Kushwaha and Ravil Patel, won the first prize under the problem statement category 'Driver Monitoring', at the National Level Hackathon 'CodeUtsava 6.0' organized by NIT Raipur. The 28-hour long hackathon focused on real-world problem statements. ChatGDP team of IIIT-NR had to detect driver drowsiness using an externally integrated web camera and implement the software in an Android application. The team created an app named 'DriveSharp'.

- IIIT-NR team of B. Tech. students 'Tokyo' secured 3<sup>rd</sup> position at the National Level Hackathon 'Academic Grand Challenge on Climate Change', organized by T-AIM in association with Capgemini. The team consisted of Harsh Pandey, Subhanshu Arya, Vishal Nitnaware and Aryaman and was mentored by Dr. Vivek Tiwari, Assistan Professor (CSE). The hackathon aimed at finding solutions to tackle some of the most pressing issues facing our planet today. The team worked for over two months, clearing four successive stages and competing against 7,500+ teams to develop a holistic solution that addressed climate change's challenges.

# 13

## Institute Events



**Hon'ble Governor of Chhattisgarh, Shri Biswabhusan Harichandan**

## Statutory Committee Meetings

| S.No. | Details of Meeting                                    | Date of Meeting    |
|-------|-------------------------------------------------------|--------------------|
| 1.    | Senate - 14 <sup>th</sup> meeting                     | June 09, 2022      |
| 2.    | Building & Works Committee - 29 <sup>th</sup> meeting | June 21, 2022      |
| 3.    | Finance Committee - 18 <sup>th</sup> meeting          | June 24, 2022      |
| 4.    | Board of Governors - 21 <sup>st</sup> meeting         | June 24, 2022      |
| 5.    | Finance Committee - 19 <sup>th</sup> meeting          | September 16, 2022 |
| 6.    | Board of Studies - meeting                            | September 24, 2022 |
| 7.    | Senate - 15 <sup>th</sup> meeting                     | November 08, 2022  |
| 8.    | Board of Recruitment – 7 <sup>th</sup> meeting        | December 09, 2022  |
| 9.    | Board of Governors – 22 <sup>nd</sup> meeting         | January 28, 2023   |

## Other Events at IIIT-NR

IIIT-NR Board Member, Shri Anil Kumar Pandey, Regional Executive Director, NTPC Ltd., superannuated on August 31, 2022. The institute felicitated him on August 25, 2022 to bid farewell to him. Shri Ashwini Kumar Tripathy assumed charge of RED NTPC (WR II) thereafter and become our Board Member.



## Industry Academia Meet 2022

IIIT-NR organized the fourth Industry Academia Meet (IAM), with the theme '*Technology Showcase Challenge*', at its campus on April 01, 2022. The event is organized every year to bridge the gap between industry and academic community. The event witnessed the participation of renowned judges from the industry, faculty, students, and staff members.



## Pariksha-Pe-Charcha 2022

Dr. Pradeep K. Sinha, Director, IIIT-NR, along with faculty, staff and students of IIIT-NR, attended the live broadcast of '*Pariksha-Pe-Charcha 2022*', on April 01, 2022, in which Hon'ble Prime Minister of India, Shri Narendra Modi interacted with approximately 1000 school students, at Talkatora Stadium, Delhi.



## DARPAN - A Unique Career Guidance Initiative of IIIT-NR

Annual Career Guidance and Self-introspection Workshop "**DARPAN**" was conducted on May 21, May 28, and June 4, 2022, for all the first year B.Tech. students who joined in the year 2021. The workshop provides a unique opportunity to the first year students to identify their strengths and interests and pursue opportunities as per their potential. During the workshop, the students develop a firm understanding of their long-term and short-term goals with the help of SWOT analysis, i.e. they identify and list down their strengths, weaknesses, threats and opportunities. Each student is guided to self-introspect and prepare and deliver a presentation before the participating members.

DARPAN is a unique initiative of IIIT-NR, which enables its students to plan their professional career right in their first year and then work towards their plans for the next three years. The workshop also helps the students in sharpening their presentation, public speaking and soft skills.

This noble initiative to prepare students to identify and pursue their passion for excellence in their profession is envisaged by the institute's Centre for Professional Development and is personally mentored by Dr. Pradeep K. Sinha, Director, IIIT-NR.



## World Environment Day

IIIT-NR organized a plantation drive at the Institute campus, on the occasion of World Environment Day. The volunteers enthusiastically participated in the initiative and planted saplings in the Institute premises. They also voluntarily took up the responsibility to take care of the planted saplings and ensuring that they are watered regularly. They pledged to work for the betterment of the environment and help in the conservation of flora in all possible ways.



## HackManthan

Chhattisgarh Police Department, in collaboration with IIIT-Naya Raipur, organized 'HackManthan', the first-ever Cybersecurity Hackathon of Chhattisgarh. The objective was to get the best young minds in the IT field working on real-life problems threatening Cybersecurity and to resolve to make the Internet a safer place.

Shri Ashok Juneja, Director General of Police (DGP), Chhattisgarh, was the Chief Guest and Dr. Pradeep Sinha, Director IIIT-NR, was the Guest of Honour for the inauguration ceremony, which was held on June 15, 2022, at Police Headquarters, Naya Raipur. Around 460 teams registered for the hackathon and prizes from a pool of Rs. 1,40,000/- were awarded to the winners. The winning project was team 'Third Eye' from Netaji Subhas University of Technology, Delhi, which presented a solution based on Live Crime Tracking from CCTV Surveillance. The second runner-up, team 'Vaak Bhavna' from IIIT-NR, won Rs. 20,000/- with their solution for Call Analysis to Emergency Services.



## International Yoga Day

IIIT-NR celebrated 'International Yoga Day' on June 21, 2022. Mr. Hitesh Kumar a yoga instructor from Raipur, took the session, which was attended by the Director, Dr. Pradeep Kumar Sinha, faculty, staff, their family members, and a large number of students of the institute. The session was filled with meditation and relaxation practices. The participants were told about the primary poses and their essence, do's and don'ts, the benefits of yoga, etc. The participants also pledged to make yoga a part of their life. Emphasizing on good health, Dr. Sinha, Director IIIT-NR, motivated the students and faculty members to practice yoga in their daily life and congratulated them for observing the International Yoga Day by practicing various exercises.



### Professional Development Programme

A Professional Development Programme cum Hands-on Workshop on '*Teaching, Learning, Assessment and Curriculum Development-An Accreditation Perspective*' was conducted in the campus on July 18, 2022. Director IIIT-NR and all the faculty members attended the workshop. Standard Operating Procedures for applying to NAAC, NIRF and ABET were discussed in detail. Revision of curriculum in compliance with accreditation agencies and finalization of Programme Educational Objectives (PEOs), Program Outcomes (POs) and Programme Specific Outcomes (PSOs) of all the B. Tech. and M. Tech. programmes offered by CSE, ECE and DSAI were also discussed during the workshop.

### Institution's Innovation Council (IIC) Regional Meet

To make the Institution's Innovation Council (IIC) network more vibrant, proactive, collaborative and productive, celebrating 75<sup>th</sup> Year of Independence of India or '*Azadi Ka Amrit Mahotsav*', the Ministry of Education's Innovation Cell and AICTE organized 15 IIC regional meets in different cities across the country. The programme provided a platform for participating IIC institutions to showcase institutions' innovation and entrepreneurship ecosystem preparedness, models/best practices, contributions, recent achievements and impacts made in the last two academic years 2020-21, 2021-22. Dr. P. P. Paltani, Dr. Amit Agrawal, Dr. Kuldip Singh Patel, and Dr. Vivek Tiwari of IIIT-NR attended the program at Nagpur on July 29, 2022.

### Hon'ble Governor's Review Meeting

To review the academic status and associated matters of the universities of the State, a review meeting of all Vice Chancellors of the State was conducted by the (then) Hon'ble Governor of Chhattisgarh, Ms. Anusuiya Uikey, on August 24, 2022. In the meeting, Dr. Pradeep K. Sinha, Director IIIT-NR presented the status of academic and other activities of the institute as well as the work done by the faculty, staff and students of the institute.



## IEEE Women in Engineering Affinity Group at IIIT-NR

IIIT-NR launched the IEEE WIE (Women in Engineering) Affinity Group on August 23, 2022, which is an international women empowerment initiative, focused on enlightening the society about the importance of women's role in social development, with special emphasis on their role in Science and Engineering. With the approval of IEEE, the members of IEEE IIIT-NR Student Branch formed this affinity group at IIIT-NR, to work towards empowering the girls and women of Chhattisgarh through their connections and collaboration with the IEEE network worldwide. Prof. Celia Shahnaz, the global IEEE WIE Committee Chair for the year 2023-24, was the Chief Guest and Ms. Sudeshna Chaudhary, Offering Owner, Partner Ecosystems and Alliances, TATA Consultancy Services, was the Guest of Honour of the launch event organized at IIIT-NR. The event was attended by the students, faculty and staff members of the institute.



## Chairman & Managing Director (CMD) of NTPC Visits IIIT-NR

Shri Gurdeep Singh, Chairman & Managing Director, NTPC Ltd., visited the institute on August 26, 2022. He also interacted with the faculty, staff members and students of IIIT-NR. In his address to them, he congratulated the IIIT-NR members for their efforts in bringing up the institute at par with other premier technical institutes, in a short span of time. He motivated the students and asked them to aim higher and to give back to the Institute and the State.



## 76<sup>th</sup> Independence Day Celebrated at IIIT-NR

IIIT-NR hoisted the tricolor in the campus on August 15, 2022 to mark the 76<sup>th</sup> Independence Day. Dr. Pradeep K. Sinha, Director IIIT-NR hoisted the National Flag, in the presence of faculty, staff and students.

While addressing the gathering, Dr. Sinha talked about Har Ghar Tiranga Campaign and threw light on Azadi ka Amrit

Mahotsav. He also congratulated the IIIT-NR team for their efforts in distributing Tiranga, door-to-door in nearby villages. Dr. Sinha also talked about IIIT-NR Vision 2.0 and threw light on the institute's achievements in the last seven years. He emphasized the students to become 21<sup>st</sup> century engineers and good human beings, which is very important for their better tomorrow. Dr. Sinha's address was followed by a Cultural program by the students, which was followed by prize distribution ceremony.



## Meeting with University of Mauritius Representatives

A meeting to discuss a possible collaboration and MoU between University of Mauritius and IIIT-NR for research and student exchange programme was organized on September 15, 2022. IIIT-NR focuses on such collaborations to improve the international visibility of IIIT-NR and to enable students to get an opportunity to explore their research abroad. Prof. D. Y. Tangman and Prof. M. Bhuruth of University of Mauritius participated in the meeting. From IIIT-NR, Dr. Pradeep Sinha, Director, Prof. Rajesh Ingle, Prof. K. G. Srinivasa, Dr. Rajarshi Mahapatra and Dr. P. P. Paltani participated in the meeting.

## Proposal for Atal Incubation Centre

NITI Aayog through its flagship scheme AIM supports the establishment of Atal Incubation Centers (AICs). IIIT-NR had applied for availing grant under this scheme. IIIT-NR was shortlisted for the presentation on September 30, 2022 at Atal Incubation Centre, Pondicherry Engineering College Foundation (AIC PECF). Dr. Pradeep K. Sinha, Director, IIIT-NR, Prof. K. G. Srinivasa, HoD, DSAI and Dr. Amit Agrawal, Assistant Professor (Management) presented IIIT-NR's proposal before the committee.



## Visit by Rakuten Representative

Mr. Ashish Nanjani, Vice President of Engineering, Rakuten-India, visited IIIT-NR on October 03, 2022 and interacted with the faculty and other stakeholders of IIIT-NR for potential collaboration, research, and campus relationship-building initiatives. The visit was part of IIIT-NR's initiatives to strengthen industry-academia ties and generate skilled human capital in Chhattisgarh.



## Run for Unity

National Service Scheme (NSS) unit of IIIT-NR in collaboration with SAC IIIT-NR organized 'Run for Unity', in the campus on October 31, 2022. Run for Unity is an initiative of the Government of India to mark the birth anniversary of Sardar Vallabhbhai Patel, known for his supreme patriotism, iron will, and courage of conviction. Faculty, staff members and students of the institute enthusiastically participated in the event and took the oath to protect the integrity of India. After the pledge, the unity march was organised. The enthusiastic students and faculty chanted the slogans of unity which filled the atmosphere with patriotism and brotherhood.



## Orientation Programme for B.Tech. 2022 Batch

The four-day Orientation Program for the newly admitted first year B. Tech. students of IIIT-Naya Raipur was conducted from November 14-17, 2022. The program was designed to cover detailed insight into IIIT-NR's culture, academics, administration, and personality development activities conducted in the institute. All newly admitted students, their parents, faculty and staff members of the institute attended the programme.

The Chief Guest of the program was Smt. R. Shangeetha, Secretary Technical Education Department, Government of Chhattisgarh, and Dr. Harold D'Costa, CEO of Intelligent Quotient Security, was the Guest of Honour for the event.

In her address to the students, Smt. R. Sangeetha, spoke about the importance of education in one's life. She also motivated the students by narrating her own life examples of being born in a remote village of Tamil Nadu, to becoming the first woman Collector of her district. Dr. Harold D'Costa, talked about the seven letters of ethics and motivated the students to work hard to achieve their dreams. In his address, Dr. Pradeep Sinha motivated the students to set up

### Orientation programme for first year B.Tech students inaugurated at IIIT

**■ Staff Reporter**  
RAIPUR, Nov 14

THE four-daylong orientation programme for the newly admitted first year B.Tech students of IIIT-Naya Raipur was inaugurated on Monday.

The programme has been designed to cover detailed insight into IIIT-NR's culture, academics, administration, and personality development activities conducted in the institute. As many as 180 students, their parents, faculty and staff members of the institute attended the event.

As scheduled, the first-day programme commenced with a curtain raiser ceremony with the lighting of the lamp and Saraswati Vandana. The chief guest of the programme was R Shangeetha, Secretary of Technical Education, Government of Chhattisgarh, while Dr Harold D'Costa, CEO of Intelligent Quotient Security, Dr Pradeep Kumar Sinha, Director IIIT-NR, and Dr Rajarshi Mahapatra, Dean (Academics) were present to grace the occasion as the guest of honours.

Addressing the programme, Dr Sinha motivated the students to set up their goals for the next four years and asked them to keep their curiosity alive so that they could work harder and fulfil all their dreams. He also emphasised becoming a good human being.

R Shangeetha spoke about the importance of education in an individual's life. She assured the students that she would provide maximum help from the Chhattisgarh Government to IIIT-NR for steering research in the institute.

She also motivated the students by narrating her own life examples of being born in a remote village of Tamil Nadu to becoming the first woman collector of her district.

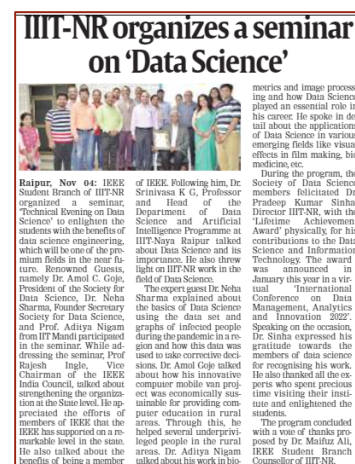
Dr Harold D'Costa talked about the seven letters of ethics and motivated the students to work hard to achieve their dreams.

The day-long programme included various sessions on culture and values of the institute, hostel rules, IT infrastructure and their usage guidelines and anti-ragging policies.

their goals for the next four years and asked them to keep their curiosity alive so that they could work harder and fulfil all their dreams. The inaugural session was followed by various sessions on culture and values of the institute, hostel rules, IT infrastructure and their usage guidelines and anti-ragging policies.

### Seminar on 'Data Science'

IEEE Student Branch of IIIT-NR organized a seminar, '*Technical Evening on Data Science*' to enlighten the students with the benefits of Data Science Engineering, which will be one of the premium fields in the near future. Dr. Amol C. Goje, President of the Society for Data Science, Dr. Neha Sharma, Founder Secretary Society for Data Science, and Prof. Aditya Nigam from IIT Mandi participated in the seminar. While addressing the seminar, Prof. Rajesh Ingle, Vice Chairman of the IEEE India Council, talked about strengthening the organization at the State level.



### Interaction with Members from ThinkTrans Foundations

An interactive session between faculty members of IIIT-NR and representatives of ThinkTrans Mr. Primal Khade, Mr. Akshat Joshi and Mr. Shobhit Gupta, was conducted on November 28, 2022. IIIT-NR had invited them to share their experience in building a robust innovation and entrepreneurship ecosystem at an academic institute. They also presented some successful case studies in this regard. ThinkTrans is working closely with FICCI on various initiatives promoting Internet Governance understanding among the student community with special focus on the Multilingual Internet. ThinkTrans Foundation is a not-for-profit organization, started with the passion to uncover the vast hidden talent within tier-II and tier-III towns and semi-urban and rural India.



### Jaan-Pehchan Sessions

IIIT-NR started its popular *Jaan-Pehchan* exercise for the Batch 2021 students from November 21, 2022. Due to Covid restrictions, this yearly exercise could not be carried out last year. The exercise consists of one-to-one interaction of all the first year students with Dr. Pradeep K. Sinha, Director, IIIT-NR. The sessions are used to create a database of the family background and other details new students, which is subsequently used for improving their weaknesses and building-up on their strengths, during their four years of stay in the campus. They are accordingly guided and helped to achieve their goals in life.



## Blood Donation Camp

National Service Scheme (NSS) unit of IIIT-NR and Student Activity Centre (SAC) of IIIT-NR, in coordination with AIIMS, Raipur, organized a Blood Donation Camp at the institute campus on November 08, 2022. About 53 individuals donated blood in the camp. A team of 18 doctors, under Dr. Sankalp Sharma, Additional Professor of the Department of Transfusion Medicine and Blood Bank, AIIMS Raipur, monitored the camp.



## Lecture on Innovations in Teaching

Dr. Pradeep K. Sinha, Director, IIIT-NR, delivered a talk on 'Innovations in Teaching', during the Special Training Programme organized by the Department of Technical Education, Government of Chhattisgarh on November 11, 2022, at Chhattisgarh Academy of Administration. The programme was organized for Training Officers of Employment Department of Government of Chhattisgarh. He was invited to speak about the various unique initiatives undertaken by IIIT-NR in training the students to become successful professionals.



## Expert Talks

-  An expert talk on 'Development of Compressed CNN Models for Plant Disease Identification' was delivered on October 19, 2022 by Prof. K. K. Biswas, retired Professor, IIT-Delhi. The talk was related to the work done in the area of *Automatic Plant Disease Identification* from leaf images, such as tomato, grapes, and cucumber. A lot of literature is available on applications of deep learning to solve this issue. However, the sizes of the CNN (Convolution Neural Networks) is so huge that it requires specialized professional equipment. His work relates to reducing the size of these networks, so that they can be easily deployed on small devices like Raspberry Pi or mobile phones. Using heuristic algorithms like differential evolution and Particle Swarm Optimization, the huge CNN models have been compressed to more than 80% of their size, while maintaining the accuracy of disease identification.



- Dr. Sasikant Tarai from NIT Raipur delivered expert lectures on '*Language, Mind, and Cognition*' on November 02, 2022 and '*Bilingual Mind: Conceptualizing Neuroplasticity*' on November 18, 2022.
- Dr. Aditya Nigam from IIT Mandi, delivered an expert talk on '*Computer Vision and Artificial Intelligence*' on November 03, 2022. Dr. Nigam has extensively worked in the field of Deep Learning, Biometrics and Computer Vision. He has published his research work in some of the top conferences in Vision domain like CVPR, ECCV, WACV, BMCV, AAAI, ISBI and many reputed journals. His current area of interest is medical imaging and biometrics.

## IIIT-NR's Annual Techno-Cultural Festival 'Technovate'

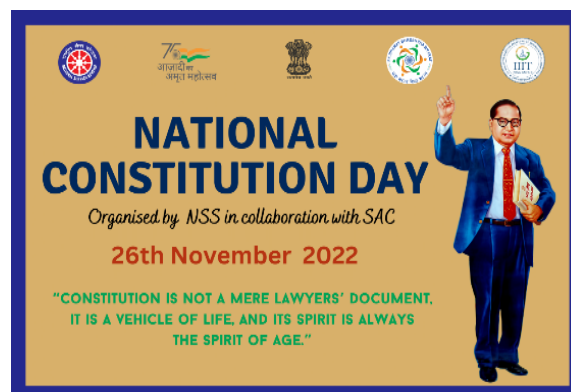
IIIT-NR organized its flagship Annual Techno-Cultural festival '*TECHNOVATE 2K22*' from November 04-06, 2022, which witnessed massive participation from students of various institutes. Technovate is Central India's most enthusiastic techno-cultural fest organized by IIIT-NR. Dr. Pradeep K. Sinha, Director IIIT-NR, inaugurated the event and in his inaugural address spoke about the genesis of this event and its benefits to the students. During the three days, various, events and celebrity nights were organized in the institute campus.



## Constitution Day Celebrations

‘Constitution Day’ Celebrations, an initiative to commemorate the adoption of the Constitution of India and to honor and acknowledge the contribution of the Founding Fathers of the Constitution, was celebrated in the institute campus on November 26, 2022.

The event began with NSS volunteers briefly introducing India’s constitution and its making. It was followed by reading of the preamble by Dr. Lakhindar Murmu, Program officer of NSS. Speaking on the occasion, Dr. Pradeep Sinha, Director IIIT-NR, emphasized on the need to know and learn the salient features of our constitution. He said that it is very important that the students who are the future of our country are aware of the history of India’s constitution and uphold its spirit in all situations in our lives. The students then participated in an online quiz organized by the Government of India on ‘*India’s Constitution and its History*’. The participants also received their participating certificates from the portal.

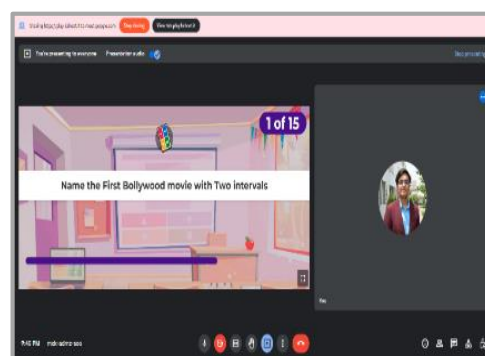


## Visit by FICCI Representative

Ms. Sarika Gulyani, Director and Head-ICT, New Age Technologies, Mobile Manufacturing and FICCI-ILIA Division at Federation of Indian Chambers of Commerce & Industry- (FICCI), visited the institute on December 27, 2022, to explore the possibilities of matching synergies. ThinkTrans, which is helping IIIT-NR in setting up its incubation centre, is working closely with FICCI on various initiatives promoting Internet Governance understanding among the student community with special focus on the Multilingual Internet. This also is a pre-cursor to the potential entrepreneurial opportunities for the students in the domain of NLP based products/services. She had a detailed interaction with Dr. Pradeep K. Sinha, Director IIIT-NR and Senior faculty members of the institute.

## Bharatiya Bhasha Diwas

The quiz club of IIIT-NR ‘*Inquizitive*’ organized a quiz competition to celebrate *Bharatiya Bhasha Diwas* on December 11, 2022. This event was organized online on Google Meet. The quiz was organized via Kahoot, which had 15 questions pertaining to the topic. The participant with the highest accuracy and lowest time was declared the winner.



## SCInTFIC 2k23

IIIT-Naya Raipur hosted its flagship Annual Science and Technology showcasing event ‘*SCInTFIC- Science & Technology over Fiction*’, designed for school students, on January 14, 2023. The event was inaugurated by the Chief Guest, Prof. Ram Kumar Kakani, Director, IIM-Raipur. The event showcased a total of 57 projects from schools across the country.

In his inaugural address, Prof. Ram Kumar Kakani, spoke to the students, encouraging them on the theme of ‘*Self-Experimentation: Learning through Conducting Experiments Themselves*’. He encouraged them to seek out explanations and new experiences on their own. In his welcome address, Dr. Pradeep K. Sinha talked about the genesis of the event and the Institute Social Responsibility (ISR) activities of the institute and further added that SCInTFIC is one such initiative.



## Republic Day Celebrations at IIIT-NR

The 74<sup>th</sup> Republic Day was celebrated in the institute with the unfurling of the tricolor by Dr. Pradeep K. Sinha, Director, IIIT-NR, in the presence of faculty, staff members, their families and students. The students presented a small cultural program consisting of patriotic songs and poems. The Student Activity Centre (SAC) office was also inaugurated by Dr. Pradeep Sinha, Director, IIIT-NR. The event ended with distribution of sweets.



### Orientation Session of the Institutions' Innovation Council (IIC 5.0)

Entrepreneurship Cell (E-Cell) of IIIT-NR organized an Orientation Session of the '*Institutions' Innovation Council (IIC 5.0)*' for the Academic Year 2022-23 on January 11, 2023. Mr. Dipan Kumar Sahu, Assistant Innovation Director at the Ministry of Education's Innovation Cell & AICTE, Government of India, delivered a welcome note and discussed the features of the 5<sup>th</sup> edition of IIC. He highlighted the achievements and new features of IIC, including a reduction in curriculum to generate space for critical thinking and entrepreneurship. He also discussed the reach of IIC coordinators across all states and union territories in India, the main mission and objective of establishing IIC cells under Ministry of Education (MoE) guidelines in higher education institutions, and ways to increase performance and award points. Additionally, he discussed the benefits of signing MoUs with external agencies, submitting good quality videos, training Innovation Ambassadors, and using the YUKTI platform for organizing innovation challenges and competitions. He also explained the levels of IIC activities and the scheme of awarding points according to institute's performance.



### Road Safety Awareness Programme

A '*Road Safety Awareness Programme*' for students, staff and faculty members of IIIT-NR was organized on March 24, 2023 by the Institute of Driving and Traffic Research, Chhattisgarh, (A joint venture between Transport Department, Govt. of Chhattisgarh and Maruti Suzuki India Ltd.). According to the World Health Organization, globally road traffic crashes kill about 1.3 million people and injure up to 50 million people every year. Road traffic injuries are projected to be the fifth leading cause of death globally by 2030. Part of the problem is the explosive growth in the number of vehicles, exacerbated by high speeds and lack of adequate infrastructure and regulations, especially in developing countries. The participants were told about professional and scientific driving practices as well as healthy road safety habits.

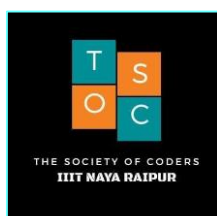


### Health-Checkup Camp

The Institute dispensary organized a health-checkup camp in the institute on March 24, 2023. The camp aimed to promote awareness about health and well-being among students, staff and faculty members of our institute.

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## Students Activity Centre



## Students Activity Centre and Clubs

IIIT-NR ensures that its students are not only transformed into good engineers or professionals but also good human beings so as to serve the society in better ways to meet the aspirations of the citizens. Towards this, for all round development of students, IIIT-NR has a dedicated 'Student Activity Centre (SAC)' which is run and managed by students of IIIT-NR.

The SAC aims to support, develop and promote students activities at IIIT-Naya Raipur to enhance their aesthetic, invasive, artistic, intellectual, and social development. It organises various seminars, expert talks, professional and skill oriented competitions, tech-fests, sports and games, extra-curricular activities and outreach programmes. All these programmes are organised by various students' clubs under the SAC. These clubs are divided into the following four verticals:-



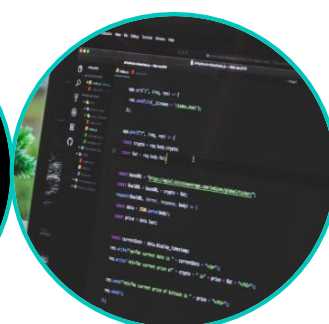
- ⦿ Technical and Entrepreneurship
- ⦿ Games and Sports
- ⦿ Art and Culture
- ⦿ Social Service and Upliftment

### ⦿ Technical and Entrepreneurship

The technical clubs under SAC promote hands-on and cooperative learning and also engage students in problem-solving and higher-order thinking. They regularly conduct workshops, seminars, hackathons, technical competitions on coding, problem solving, machine learning, circuits, and entrepreneurship for sharpening skill-sets and capacity building of the students of IIIT-NR. There are the following five students' clubs who organise technical events:-



**TSOC (The Society of Coders) Club :** The TSOC aims to foster a culture of coding at IIIT-NR by hosting various events such as hackathons, workshops, and coding competitions. The club emphasizes hands-on learning in a supportive and interesting environment. It's a club that welcomes everyone from first-time coders to experienced veterans of the trade.



**AIML (Artificial Intelligence and Machine Learning) Club :** The AIML Club is focused on helping individuals evolve from admirers of Artificial Intelligence and Machine Learning technology to capable developers. The club provides an opportunity to learn and build projects from the ground up using open-source tools such as TensorFlow and Keras, with a focus on application development using Python. Additionally, members have the opportunity to participate in competitions, hackathons, and even contribute to the open-source technology they work with. As we live in the internet age and now moving towards the age of AI, this club is an ideal place to learn, explore and develop AI-based projects.



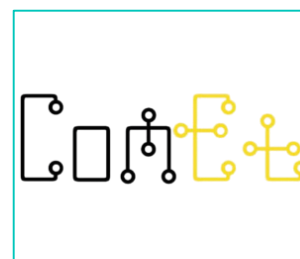
**CipherCell Club :** CipherCell is an official cybersecurity club at IIIT-NR. The goal of the club is to raise awareness about hacking and to dispel negative stereotypes associated with the field while educating those who are interested in cybersecurity.



**Inquizitive Club :** True to its name, Inquisitive is the abode of the most curious minds of IIIT-NR. Inquisitive brings current affairs, and general knowledge to you in a fun way. Thus, contributing to increase the IQ of the students in the campus. Inquisitive regularly organises quizzes and competitions on various topics, from general knowledge and trivia to specific subject areas such as history, science, literature, and more. The club provides a fun and engaging environment for individuals to learn while building camaraderie and friendly competition with others.



**ComEt Club :** The main objective of the club is to bring together like-minded individuals and find solutions to real-world problems using the available resources. The club encourages members to participate in idea-generating events and develop new and innovative ideas.



## ⦿ Games and Sports

Atharv is the sports club at IIIT-NR. It is responsible for organizing and conducting sports and eSports events and competitions such as basketball leagues, cricket, football, CS:GO Valorant and Fifa tournaments.



## Art and Culture

Cultural clubs under the SAC, provide ways for students to explore and engage in activities outside their academic studies. These clubs foster sense of community and belongingness, which are important for students in technical fields who may spend much time working independently. These clubs regularly organise programmes and competitions on Music, Art, Dance, Photography, Debate, Discussions, Poetry and Drama. There are the following eight students' cultural clubs which organise art and cultural events:-

**Indradhanush :** Indradhanush, as the name suggests, is a club made to add colors to the routine college life so that creativity follows right behind logic, symmetry, and expression. Indradhanush is a cultural club of the institute and is behind the beautiful artistic work behind every event. This club works on the power of imagination and visualization which is a primary requirement even while working with codes.



**Capriccio :** Capriccio is a music club under SAC that encourages students to become immersed in the art and science of music.

**ShutterBug :** Shutterbug is IIIT-NR's official photography club. This club intends to help interested students of photography to learn about the fundamentals of this field. It provides a supportive environment to students of IIIT-NR to share their knowledge and passion for photography. The club also organizes photography workshops and exhibitions.



**Club De Theatre :** Club De Théâtre is the drama club at IIIT-Naya Raipur that encourages students to participate in dramatic presentations such as skits, mime, plays, and street theatre. The club promotes the culture of drama, comedy, and other related forms of performance among the students. The club serves as a platform for students to showcase their talents, develop their skills, and experience the joy of performance.



**Igniters :** Igniters is a dance club that aims to create excitement and energy through the art of dance. The goal of the club is to provide a space for students to explore and express themselves through dance, and to celebrate this art form together as a community. By joining Igniters, students get the opportunity to develop their dancing skills, share their passion for dance with others, and perform on and off campus.



**Cover to Cover :** C2C is a literature club at IIIT-Naya Raipur that aims to promote the love and appreciation of literature among its members. The club creates a nurturing environment that encourages creativity and liberation by exploring different literary traditions and perspectives.

**Take da Bait :** Take da Bait is a debate club at IIIT-Naya Raipur. The club's purpose is to provide a platform for students to discuss and debate current trends and issues in the world, and to help members form their own opinions on these topics. The club is intended to help students connect to the changing socio-political environment of the world and to understand the potential advantages and disadvantages of technology and its impact on the world. The club believes that by discussing and debating important issues, members will be better equipped to be informed citizens and future leaders.



**Model United Nations :** The IIIT-NR's MUN is a Model United Nations conference that provides delegates with a platform to discuss current global issues and to develop a range of skills such as public speaking, debate, writing, critical thinking, collaboration, and leadership. The conference simulates the roles of UN ambassadors and delegates engaging in activities such as drafting resolutions, strategizing, and negotiating with other delegates. The conference provides an opportunity to think critically and creatively, and to develop solutions to global issues.



## 🕒 Social Service and Upliftment

**National Service Scheme (NSS) Club** and **Institute Social Responsibility (ISR) Club** of IIIT-NR work with a vision to create community awareness among the students and to develop a sense of social and civic responsibility among themselves. It encourages students to utilize their knowledge in generating practical solutions to individual and community problems with the motto of '**Not Me But You**'. These clubs have undertaken many initiatives towards conserving the environment, educating the underprivileged sections of society, and observing cleanliness and hygiene in the locality.



## Activities of Students' Clubs under SAC

As explained earlier, SAC organises various activities for overall growth and development of students of IIIT-NR. Accordingly, these activities are primarily in the following areas:-

- Sharpening technical skill-sets of students
- Fostering soft-skills and leadership qualities
- Promoting art and culture
- Sports and games
- Social service and upliftment

Details of some of the noteworthy and useful activities organised by the SAC in the above-mentioned areas in the year 2022-23 are as follows.

## Technical Skills Activities

### Blockchain Master Class

Cyber Security Club of IIIT-NR, *Cipher Cell*, conducted a series of sessions '*Blockchain Master Class*', to create interest, foster Blockchain Technology development skills and provide a learning opportunity for enthusiastic students in Blockchain. The sessions were conducted on April 01, April, 08 and April 29, 2022.

### Bit to Byte

The coding club of IIIT-NR, *The Society of Coders (TSOC)*, organized a series of coding related competitions, '*Bit to Byte*', on April 09-10, 2022. The competitions covered various aspects of an interview, from coding round to Computer Science Fundamentals round. The event mimicked the corporate environment, and teamwork was essential to compete in the event. The competition events included Competitive Programming contest (Based on ICPC format), Coding Speed Run, Machine Learning Contest, Aptitude and Math Contest, Computer Science Fundamentals, English, Coding for Girls, etc. Teams for the events were created through an IPL-like auction. The events were conducted in Codeforces, Vjudge and Kaggle platforms.

### Internal Smart India Hackathon - SIH 2022

AIML Club of IIIT-NR organized an internal Smart India Hackathon – SIH 2022, on April 20, 2022. The event aimed to foster development skills in the students, and provide a learning opportunity for enthusiastic students of AIML. The problem statements were selected based on real-life problems and aimed at improving the basic understanding and problem solving skills of students.



## Hackathon on Computer Vision

AIML Club of IIIT-NR organized a Hackathon on Computer Vision on April 24, 2022. The event aimed to foster development skills in the students and provide a learning opportunity for enthusiastic students of AIML. The problem statement given was - Recreating Snapchat filters using basic Computer Vision techniques. All the groups completed the task with better understanding of Python and OpenCV.

## Workshop on Robotics

Robonite Club of IIIT-NR in collaboration with Indian Institute of Technology, Bombay, conducted a workshop on 'Working of Robots' on September 17, 2022. The workshop provided introduction to robots and their working with demonstration of two robots Meshmerize and Cozmo Clench. Robonite as a CS-STEAM Education platform brings forth an activity based hands-on learning system that can be applied to involve students in designing and building exciting robots, using LEGO Mind storm robotics kits; thereby enhancing their mathematical reasoning and logical skills.

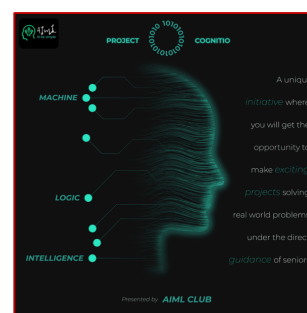


## Machine Learning Mania-2k22

AIML Club of IIIT-NR organized a Hackathon 'Machine Learning Mania-2k22' from May 14-16, 2022. The event aimed to foster development skills in the students and provide a learning opportunity for enthusiastic students of AIML. During the contest, participants developed a Machine Learning model to predict house prices. The event improved the students' understanding and problem-solving skills in Keras, Python, Numpy, Pandas, Seaborn and Matplotlib.

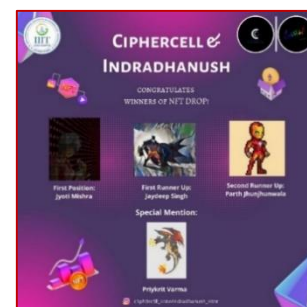
## Machine Learning Project 'Cognito'

AIML Club of IIIT-NR organized a machine learning project building event 'Cognito', from June 15, 2022 to July 30, 2022. The unique initiative gave students an opportunity to make exciting projects for solving real-world problems, under the direct guidance of AIML team members. Candidates had to complete their allotted projects within the given time.



## The NFT Drop

The cybersecurity cell of IIIT-NR, *CipherCell* and the cultural club of IIIT-NR *Indradhanush*, jointly organized a NFT (Non-Fungible Tokens) design competition 'The NFT Drop' from August 16-23, 2022. The motive of the event was to enhance the designing skills of students, giving them a platform to display their NFTs and to give them an opportunity to delve deeper into the realm of NFTs. NFTs are cryptographic assets on a Blockchain with unique identification codes and metadata that distinguish them from each other and cannot be replicated. NFTs can represent real-world items like artwork and real estate.



## IDEACRYPT

The Electronics Club of IIIT-NR *ComEt* organized an event 'Ideacrypt' on September 18, 2022 to provide a platform for students to present their innovative ideas for solving challenging problems through technological means. Students were given problem statements related to health, sports and fitness, agriculture, smart automation, smart education, and clean and green energy. Their job was to pitch an idea/ concept to solve that problem by a hardware solution and to present their solution (with concept level diagram/architecture), advantages/ disadvantages, implementation, etc.



## Code Nubs 3.0

The coding club of IIIT-NR, *The Society of Coders (TSOC)*, organized 'Code Nubs 3.0', on January 08, 2023, for the freshers to help them begin their journey in competitive programming. The competition was held in Codeforces platform.



## Disrupt 3.0

Entrepreneurship Cell (E-Cell) of IIIT-NR hosted 'Disrupt 3.0' on January 18, 2023. Participating team had to pitch their ideas on the spot on how to market and sell a product made from an object that was given to their team. To keep the competition challenging for participants, business models were assigned to the teams during the event and they were judged based on the innovation and feasibility of their idea, as well as the pitching abilities of the speakers.



## DesCode

The coding club of IIIT-NR *The Society of Coders (TSOC)* organized a coding competition 'DesCode' on February 05, 2023, to help the students improve their competitive programming skills. The contest had three Data Structure and Algorithm (DSA) problems and participants were given two hours to solve the problem.



## Kaggle Competition

The AIML club of IIIT-NR organized Kaggle Competition 'ML Mania 2023' on February 07, 2023, to introduce the Batch 2022 students to Kaggle competitions and improve their skills in the same to foster a competitive spirit. A session was arranged for the students to make them familiar with Kaggle and basic ML algorithms.



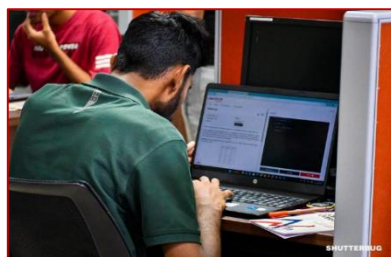
## The Big Bull: A Virtual Stock Market Challenge

E-Cell of IIIT-Naya Raipur organized a virtual stock market simulation challenge '*The Big Bull*', from February 06-10, 2023, to enable the participating students to get a head-start to their stock market journey.



## Machine Learning Hackathon

AIML club of IIIT-NR organized '*ML Hackathon*' on March 18, 2023. The event's objective was to familiarize the first-year students with Kaggle and basic ML algorithms. This competition was a continuation of the last session arranged, which primarily focused on the basics of Data Analysis using Pandas. It ended with a Code-along session to develop a basic Linear Regression model and submit the output to the competition.



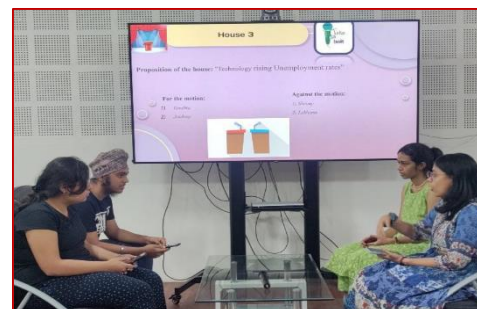
## CodeRush Challenge 2023

The coding club of IIIT-NR, *The Society of Coders (TSOC)*, organized a coding competition '*CodeRush Challenge 2023*', in collaboration with Algo University, on March 22-23, 2023. On the first day of the event, a webinar was organized, in which Mr. Swapnil Daga (Co-Founder of Algo University), shared the placement preparation roadmap and tips. He shared his insights on what makes a project presentable. The programming competition took place on the second day.

## Soft-skills and Leadership Activities

### Word War Edition 1.0

The Debating Society of IIIT-NR, *Take-da-bait*, organized an event "*Word War Edition 1.0*", on April 02, 2022. The debate was held in standard oxford template. The aim of the competition was to give the students an experience of speaking in a formal debate setup and to make them familiar with oxford format. The competition also trained the students to think creatively and generate relevant arguments and helped them to develop persuasive responses to simple questions.



### Group Discussion on India's Journey from Partition to Present

The Model United Nations club of IIIT-NR conducted a group discussion on 'What If - India's Journey from Partition to Present' on September 09, 2022. The objective of the event was to provide a platform to students to share their opinions on some of the most important events that have defined India's journey since independence. This event proved to be helpful in encouraging students to learn about the major political, social and economic events of our post-independence history. A platform for group discussion also helped students to develop leadership and public speaking skills.



### Model United Nations (MUN) Conference

The Model United Nations (MUN) Club of IIIT-NR conducted a 2-day conference under the two committees namely- All India Political Parties Meet (AIPPM) and Disarmament and International Security Committee (DISEC).

The AIPPM discussed on the topic 'Safeguarding the Integrity of the Democracy in Recent Times', by discussing developments, including the persecution of journalists and independent media and analyzing the significance and aptness of the laws like UAPA, NRC and CAA. On the other hand, DISEC focused on the 'Illicit Trade of Small Arms and Light Weapons' and the 'Weaponization of Big Data'. The motive of the event was to instigate interest among the students in world politics as well as in various political and civic affairs of our democracy. The event was very useful for students to develop leadership and public speaking skills.



### Chai Pe Charcha

Model United Nations (MUN) club of IIIT-NR organized a group discussion 'Chai Pe Charcha', on January 19, 2023. The topic of discussion was 'The State of Media in Contemporary Indian Democracy and Social Media's Relevance and Role in Shaping Our Country's Future Politics'. The participants discussed various aspects like the independence of media, the neutrality of facts, the role of media and social media in contemporary policy, the relevance of social media trends, and media reforms, among others. Overall, it was a healthy, engaging, and insightful discussion that enhanced the understanding of the current political and geopolitical scenario of the participants.



## Art and Culture Activities

### Cultural Night 'Mellow Repose'

SAC clubs together organized an electrifying fun-filled cultural night '*Mellow Repose*', on May 09, 2022, for the students to relish the amazing performances by some of the B. Tech and M. Tech students. The event consisted of performances from various genres ranging from dance, singing to literature.



### RangDe Aazadi

The cultural club of IIIT-NR, *Indradhanush*, organized a contest, '*RangDe Aazadi*', from August 8-12, 2022, as a part of '*Azadi ka Amrit Mahotsava*' celebration activities. The theme of the contest was '*Patriotism*'. The objective of the event was to gauge the participants' thought processes and awareness of patriotism. It aimed to keep the students engaged, explore and encourage creativity in students and offer them a platform to showcase their talent. The participants were given 5 days to showcase their arts. The contest was open to all art forms and mediums. This event inspired students to think and work creatively in order to promote artistic excellence.



### Azaadi Ke Paar

The theatre club of IIIT-NR, *Club De Theatre*, organized '*Azaadi Ke Paar*', on August 09, 2022, as part of '*Azaadi Ka Amrit Mahotsav*' celebration activities. In the event students were assigned a freedom fighter and they were required to enact him/her. The audience was filled with patriotism which was the main motive of the event.

### Bhavarth - Independence Day Quiz

The literature club of IIIT-NR *Cover to Cover* organized a quiz event called '*Bhavarth*' on August 12, 2022, as a part of '*Azadi ka Amrit Mahotsava*' celebration activities. The topic was '*Our Country: Azadi Ka Amrit Mahotsav*'. The quiz was held to make students appreciate the vibrant and long history of our country, with major leaps in cinema, culture, politics, and administration. The questions spanned the length and breadth of India, with genres like famous personalities, geography, businesses, and government policies. The questions were thought-provoking and drawn so as to be engaging and a learning experience for the students.

## Birsa Munda Quiz

Cover to Cover club of IIIT-NR organized a quiz event called '*Birsa Munda*' on the occasion of Birsa Munda's Birthday, as a part of "*Azadi ka Amrit Mahotsava*" celebrations being conducted by Student Activity Centre of IIIT-NR. The quiz was held to make students aware of the history of tribal leaders in India's freedom fight.



## Ganesh Chaturthi Celebrations

Students, faculty and staff members of IIIT-NR celebrated '*Ganesh Chaturthi*' in the institute from August 31 to September 04, 2022. Various cultural programs were organized during these five days. A mythological quiz named '*Pratispardha*' was also organized by the '*Inquisitive*' club. Cultural performances included melodies from members of '*Capriccio*' club and ground breaking dance performances by members of '*Igniters*' club. The 5-day celebrations concluded with the Visarjan on Sept. 04, 2022.



## Rampage

The literary club of IIIT-NR '*Cover to Cover*' organized an event '*Rampage*' on September 13, 2022. In the event, each participant had to pick a genre, and a topic of their choice within that genre. They then had to get 2 out of 3 judges to side with them to win a round. The one who wins the most rounds was the winner. The participants got to argue for categories like anime, superhero media, web series, sports, gaming, etc.



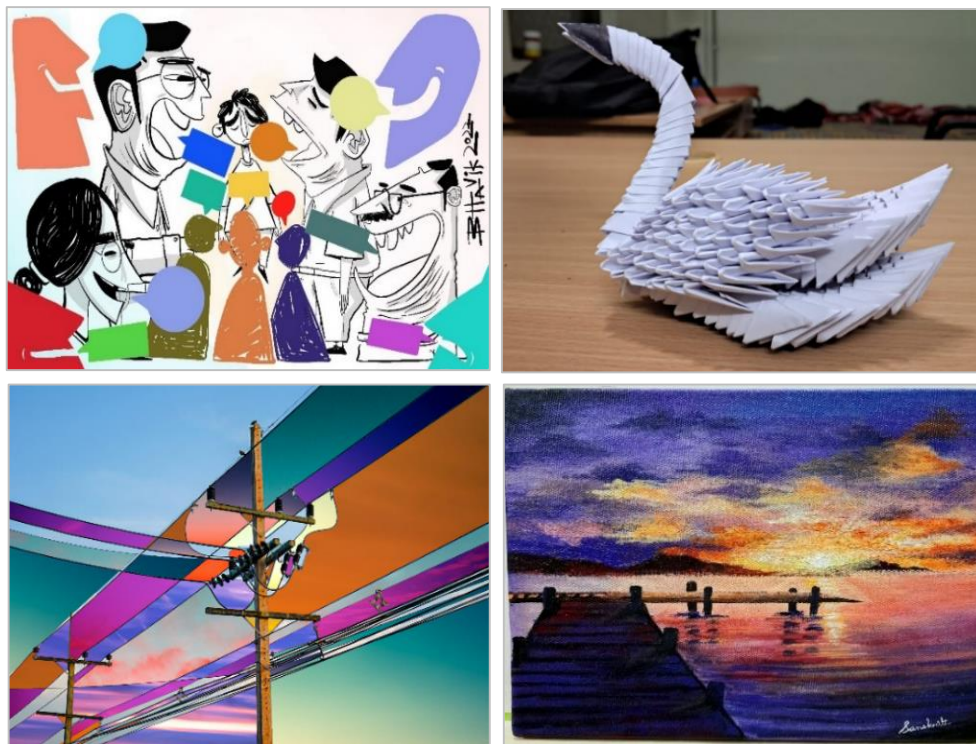
## Garba Night Celebration

'*Garba Night*' was organized in the campus by Student Activity Centre (SAC) on September 30, 2022, as part of the Navratri Celebrations. Students dressed in Gujarati attires played garba and displayed their dancing skills.



## Rang-Express

The fine art club of IIIT-NR, *Indradhanush*, organized '*Rang Express 5.0*', from Dec 13 to 25, 2022 to uncover the talents and to bring out innovative ideas of Batch 2022. The event consisted of competitions in Sketching, Painting, Digital Art and Craft.



## Dance Workshop – Freestyle

The dance club of IIIT-NR, *Igniters*, conducted a '*Dance Workshop*' on January 07, 2023, for introducing the students to the world of dance. The workshop began with demonstration of different dance moves by the instructor. Participants were encouraged to explore their own creativity and style while learning the basic steps and techniques. The participants were able to learn and understand the basics of Freestyle dancing and were able to apply their knowledge in a creative way.



## Sports and Games Activities

### Inter-College Cricket Tournament ‘Aagaaz’

The sports club of IIIT-NR, *Atharv*, organized an inter-college cricket tournament, ‘*Aagaaz*’, from April 14-17, 2022. Teams from various institutes of the region, like NIT Raipur, BIT Raipur, BIT Durg, AIIMS and GEC Bilaspur, participated in the tournament. IIIT-NR (Team B) emerged as the winner and NIT-Raipur as the runner up in this tournament.



### Badminton Tournament (Girls)

Sports Club of IIIT-NR *Atharv* organized a cordial intra badminton tournament from May 02-05, 2022, to identify the best badminton players in IIIT-NR.



### Treasure Hunt

*Inquisitive Club* of IIIT-NR organized a treasure hunt competition on May 16, 2022, in the Institute campus. It was a clue-based treasure hunt competition consisting of two rounds. Each team had to solve 5 clues to reach the destination. By solving each clue, the team got the location of the next clue.

### LAKSHYA - Shaheed Veer Narayan Singh Trophy

SAC (Students Activity Centre) conducted a Sports tournament ‘*Lakshya*’ from December 14-15, 2022. The event witnessed 200+ participants in various sports disciplines, including Football, Cricket, Basketball, Kabaddi, Badminton, Volleyball, Athletics and Table Tennis. The event was played between 4 teams and the team with the highest points was awarded Shaheed Veer Narayan Singh trophy, in honour of the Chhattisgarh’s first freedom fighter.



## Aarambh 5.0

Sports Club of IIIT-NR, *Atharv*, organized the annual intra sports event '*Aarambh 5.0*', to discover the best sportspersons in the institute and to promote sports activities in the campus. The event was inaugurated on January 26, 2023 and was conducted from January 27-29, 2023. With the help of auction process, all the participants were divided into 4 teams namely (Titans, Mavericks, Renegades, Spartans), and many sports events were conducted between them, which included cricket, football, basketball, kabaddi, volleyball, chess, table tennis, badminton athletics and valorant.



## Sports Fest (ASTRA) at IIIT Delhi

IIIT-NR teams participated in '*Astra23*', a three-day sports event organized by IIIT Delhi, from March 24-26, 2023, in which 9 college teams participated. Our team participated in football and badminton and gave a good fight to the opponents.



## Social Service and Upliftment Activities

### Seminar on 'Women in Engineering'

IEEE students branch of IIIT-NR, organized a seminar on '*IEEE Women in Engineering (WIE)-Towards Technology and Leadership*,' at the institute campus, on April 27, 2022, to promote women engineers and scientists and inspire girls around the world to motivate them to pursue their academic career in engineering.

IEEE Women in Engineering (WIE) is one of the largest international professional organizations, dedicated to promoting women engineers and scientists, which facilitates the recruitment and retention of women in technical disciplines globally.

## NSS Orientation Programme

The National Service Scheme (NSS) unit of IIIT-NR organized a first-of-its-kind NSS Orientation program for the B. Tech. Batch 2021, on June 30, 2022. The program aimed at educating the new NSS volunteers about how NSS offers excellent opportunities to serve the society and for personal development.

The event had the presence of Dr. Pradeep Sinha, Director IIIT-NR; Dr. Neeta Bajpai, NSS Coordinator Chhattisgarh; Mr. Satyendra Sahu, President Awardee, NSS Senior Volunteer and Mr. Nikhil Kumar, State Awardee, Senior NSS Volunteer.



Dr. Sinha in his address to the new NSS Volunteers emphasized the importance of becoming good human beings in life and not just professionally successful human beings. He encouraged students to utilize their knowledge and skills for the betterment of society.

## Sabka Bharat, Nikharta Bharat

As part of the celebrations to mark the '7<sup>th</sup> Anniversary of Smart City Mission', Nava Raipur Atal Nagar Vikas Pradhikaran (NRANVP), organized an event 'Sabka Bharat, Nikharta Bharat' from June 24-25, 2022. The event consisted of various competitions like Painting Competition, Know Your Smart City Quiz, Essay Writing, Give Your Suggestions (Speech format) and Debate. The event was organized in Smart Cities around the country to educate the young masses of the country about the plans of the mission and also the upcoming tasks to be done. Teams from institutions like IIM Raipur, KPS International Naya Raipur, Kalinga University and many more, participated in the competitions. IIIT-NR teams won two prizes.



## Classes by IIIT-NR for Underprivileged Children

As a part of the 'Rural Transformation Through Education Initiative' of National Service Scheme (NSS), the NSS unit of IIIT-NR, in collaboration with the Institute Social Responsibility (ISR) cell conducts weekly classes in the local schools of Uparwara and Nawagaon villages. The initiative focuses on the overall personality development of underprivileged students through education, awareness and physical fitness. The volunteers of various subjects like Mathematics, Basic Sciences, Humanities etc. take turns visiting the schools every Saturday and Sunday. They also organize multiple games for the students and spread awareness about the importance of games and sports for their physical well-being and overall personality development. The drive is a part of various initiatives taken by NSS, IIIT-Naya Raipur, under the 'Azadi ka Amrit Mahotsav' celebrations.



## Education through Social Work

IIIT-NR conducts various programs as part of its social responsibility for the development of children of nearby villages. As part of this, the ISR team of IIIT-NR distributed stationery items and chocolates to the children of the Government Primary School, Nawagaon, Naya Raipur. The team members also motivated the children for planting trees and educated them about the importance of plants and how they should save trees for the benefit of mankind. Mr. R K Sinha, Head Master of the school, thanked IIIT-NR team. Panchayat President, Mr. Sujeet Ghidoude, also expressed his gratitude to the IIIT-NR for such initiatives.

### ISR cell of IIIT-NR conducts various events for development of children



The members of Institute's Social Responsibility cell distributed stationery among school children.

#### ■ Staff Reporter

RAIPUR, July 28

THE Institute's Social Responsibility (ISR) cell of IIIT-NR conducts various programmes as a part of its social responsibility for the development of children of nearby villages.

Recently the institute's ISR team distributed stationery and chocolates to the children of the Government Primary School, Nawagaon, Sector 28, Naya Raipur. They also motivated children to study and motivated them to aim big in life.

The team members also

encouraged the children to plant trees and educated them about the importance of plants and how they should save trees for the benefit of mankind.

Mentioning the famous quote by Nelson Mandela "Education is the most powerful weapon to change the world," officer-in-charge ISR cell Dr Mithlesh Chaube encouraged the children to study hard so that they can pursue their dreams. He further added that IIIT-NR has been making such contributions to society since the day it opened in Naya Raipur.

The institute is contributing to the community in several ways, one of which is through ISRA activities said that the institute will continue doing this for the benefit of society, he added. R.K.Sinha, Headmaster of the school, extended greetings to the ISR team for distributing stationery among students.

## Awareness Camp on Poshan Abhiyan

Under the aegis of Poshan Abhiyan of the Government of India, the National Service Scheme (NSS) unit of IIIT-NR, in collaboration with Institute Social Responsibility (ISR) cell, organized an awareness drive in the nearby Government School of Nawagaon, Naya Raipur. The campaign aimed to disseminate knowledge about the importance of nutrition in our lives and to promote practices that nurture health, wealth, and immunity to disease and malnutrition. The Poshan Abhiyan is a Government of India's flagship program to improve the nutritional outcome for children, adolescent girls, pregnant women, and lactating mothers. The school administration appreciated the efforts and contributions of the NSS unit of IIIT-NR towards various schemes such as the *Swachchha Bharat Abhiyaan*, *Azadi ka Amrit Mahotsav*, *Poshan Abhiyan*, and so on, thereby invoking a sense of responsibility and social service among the volunteers.



## Annual Cultural Programme at Learning Centre

The Institute Social Responsibility (ISR) cell of IIIT-NR organized the Annual Cultural Programme by nearby village school children studying in the Learning Centre run by IIIT-NR, on the occasion of Netaji Subhash Chandra Bose's Birthday, on January 23, 2023. The Learning Center at Uparwara, Naya Raipur, named as 'Vidyasagar - Phule Learning Center' has been continuing since 2021. Students, research scholars, and faculties of IIIT-NR teach at this Learning Center every Saturday and Sunday with the aim of imparting better scientific education to village school students. The programme was attended by faculty, staff, students of IIIT-NR and the villagers.



### BMC Walkathon

IIIT-NR students participated in a walkathon organized by Balco Medical Centre, Nava Raipur on the occasion of its foundation day on March 23, 2023. The event aimed to raise awareness about cancer. It began with a presentation about the disease and its prevalence in society. The speakers highlighted the importance of early detection and prevention measures to fight against cancer. They also shared some of the current research and progress being made in the field of cancer treatment. The presentation was followed by the walkathon.



### World Consumer Rights Day 2023

The Student Activity Center (SAC) and National Service Scheme (NSS) of IIIT-Naya Raipur in association with the Bureau of Indian Standards (BIS), Raipur, as part of BIS's '*Quality Connect Campaign 3.0*' conducted a door-to-door survey at its campus at Nava Raipur from March 15 – 19, 2023, to spread awareness about the importance of standardization. BIS is the National Standards Body of India, established under the BIS Act 2016 for the harmonious development of the activities of standardization, marking and quality certification of goods and for matters connected therewith or incidental thereto.



## 15

## Infrastructure Development

## Status of Various Infrastructure Development Projects

- ① **Construction of Residential Blocks :** Construction of three residential blocks i.e. 11C, 13C and 14C for faculty and staff of the Institute is nearing completion. Structure, slab work, brick work, putty work etc. are already completed and following are in progress:

| Description        | % Completion Status |
|--------------------|---------------------|
| Plumbing Work      | 90%                 |
| Electric Work      | 90%                 |
| Painting Work      | 50%                 |
| Tiles Fitting Work | 90%                 |
| Lift Work          | 80%                 |



Residential Block - 11C



Residential Block - 13C



Residential Block - 14C

- ① **Construction of Sports Complex :** Construction of sports complex is in progress. Excavation and PCC work have been completed. Other structure work i.e. footing, plinth beam, column, brick work etc. are nearing completion.



- Construction of Admin Block :** Presently all administrative offices are in the academic building only. Construction of a separate Administrative Building (Block) of the Institute is in progress and nearing completion. Structure, slab work, brick work, putty work etc. are already completed and following are in progress:

| Description             | % Completion Status |
|-------------------------|---------------------|
| Putty Work              | 95%                 |
| Plumbing Work           | 95%                 |
| Electric Work           | 85%                 |
| Painting Work           | 30%                 |
| HVAC Work               | 90%                 |
| Partitioning for cabins | 80 %                |
| Tiles Fitting Work      | 90 %                |
| Lift Work               | Yet to start        |



Administrative Building (under construction)

- Construction of Vehicle Parking :** Construction of an open parking for vehicles of faculty, staff and visitors in front of the academic building is nearing completion. Only stone pitching and tiles fitting are remaining which are also in progress.



Vehicle Parking (under construction)

# 16

## Financial Statement

**Placed as Annexure - 1**

**INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY , NAYA RAIPUR**  
**PLOT NO -7, SECTOR -24, NAYA RAIPUR (C.G)**  
**BALANCE SHEET AS ON 31 MARCH 2023**

| SOURCES OF FUND                            | SCHEDULES | 2022-23               | 2021-22               |
|--------------------------------------------|-----------|-----------------------|-----------------------|
| <b>UNRESTRICTED FUND</b>                   |           |                       |                       |
| Corpus                                     | 1         | 46.00                 | 46.00                 |
| General Fund                               | 2         | 537,107,097.67        | 468,190,017.60        |
| Designated / Earmarked Funds               | 3         | 111,204,282.00        | 98,563,656.00         |
| <b>RESTRICTED FUND</b>                     | 4         | 180,000,000.00        | 180,000,000.00        |
| <b>LOAN &amp; BORROWINGS</b>               | 5         | -                     | -                     |
| Secured                                    |           | -                     | -                     |
| Unsecured                                  |           | -                     | -                     |
| <b>CURRENT LIABILITIES &amp; PROVISION</b> | 6         | 31,772,743.05         | 27,690,125.05         |
|                                            |           |                       |                       |
| <b>TOTAL(A)</b>                            |           | <b>860,084,168.72</b> | <b>774,443,844.65</b> |
|                                            |           |                       |                       |
| <b>APPLICATION OF FUND</b>                 |           |                       |                       |
| <b>FIXED ASSETS</b>                        |           |                       |                       |
| Tangible Assets                            | 7         | 358,687,627.07        | 331,314,484.50        |
| Intangible Assets                          |           | -                     | -                     |
| Capital Work in Process                    |           | 25,348,000.00         | 26,174,000.00         |
| <b>INVESTMENTS</b>                         |           | -                     | -                     |
| Long Term                                  | 8         | 279,598,782.00        | 268,350,305.00        |
| Short Term                                 |           | -                     | -                     |
| <b>CURRENTS ASSETS</b>                     | 9         | 140,915,866.57        | 107,342,412.70        |
| <b>LOAN,ADVANCE &amp; DEPOSITS</b>         | 10        | 55,533,893.08         | 41,262,642.45         |
|                                            |           |                       |                       |
| <b>TOTAL(B)</b>                            |           | <b>860,084,168.72</b> | <b>774,443,844.65</b> |
| Notes on Accounts                          | 22        |                       |                       |

CFAO  
IIIT N.R.

*B. Sinha*  
DIRECTOR  
IIIT N.R.

**INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY , NAYA RAIPUR**  
**PLOT NO -7, SECTOR -24, NAYA RAIPUR (C.G)**

**INCOME & EXPENDITURE ACCOUNTS ON 31 MARCH 2023**

|                                                                | SCHEDULES | 2022-23           |                 |                       |                       | 2021-22               |                       |
|----------------------------------------------------------------|-----------|-------------------|-----------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                |           | UNRESTRICTED FUND |                 |                       | RESTRICTED FUND       | TOTAL                 |                       |
|                                                                |           | CORPUS            | DESIGNATED FUND | GENERAL FUND          |                       |                       |                       |
| <b>INCOME</b>                                                  |           |                   |                 |                       |                       |                       |                       |
| Academic Receipts                                              | 11        |                   |                 | 126,833,202.91        |                       | 126,833,202.91        | 121,446,142.18        |
| Grants & Donations                                             | 12        |                   |                 | 148,000,000.00        |                       | 148,000,000.00        | 100,000,000.00        |
| Income From Investments                                        | 13        |                   |                 | -                     | 111,204,282.00        | 111,204,282.00        | -                     |
| Other Incomes                                                  | 14        |                   |                 | 9,495,492.88          |                       | 9,495,492.88          | 5,758,975.57          |
| <b>TOTAL(A)</b>                                                |           |                   |                 | <b>284,328,695.79</b> | <b>111,204,282.00</b> | <b>395,532,977.79</b> | <b>227,205,117.75</b> |
| <b>EXPENDITURE</b>                                             |           |                   |                 |                       |                       |                       |                       |
| Staff Payments & Benefits                                      | 15        |                   |                 | 113,035,181.17        |                       | 113,035,181.17        | 90,189,048.00         |
| Academic Expenses                                              | 16        |                   |                 | 12,776,761.09         |                       | 12,776,761.09         | 13,930,593.00         |
| Administrative & General Expenses                              | 17        |                   |                 | 55,292,776.37         |                       | 55,292,776.37         | 30,582,127.55         |
| Transportation Expenses                                        | 18        |                   |                 | 1,236,842.00          |                       | 1,236,842.00          | 1,291,885.00          |
| Repairs & Maintenance                                          | 19        |                   |                 | 6,370,226.43          |                       | 6,370,226.43          | 13,379,110.00         |
| Finance Costs                                                  | 20        |                   |                 | 1,376.71              |                       | 1,376.71              | 10,737.28             |
| Depreciation                                                   |           |                   |                 | 23,164,822.95         |                       | 23,164,822.95         | 31,416,522.53         |
| Other Expenses                                                 | 21        |                   |                 | 3,533,629.00          |                       | 3,533,629.00          | 1,192,929.00          |
|                                                                |           |                   |                 | <b>215,411,615.72</b> | <b>-</b>              | <b>215,411,615.72</b> | <b>181,992,952.36</b> |
| <b>Balance Being Surplus (Deficit) Carried To General Fund</b> |           |                   |                 | <b>68,917,080.07</b>  |                       | <b>180,121,362.07</b> | <b>45,212,165.39</b>  |
| Notes To Account                                               | 22        |                   |                 |                       |                       |                       |                       |

  
**CFAO**  
**IIIT N.R.**

  
**DIRECTOR**  
**IIIT N.R.**

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY , NAYA RAIPUR  
PLOT NO -7, SECTOR -24, NAYA RAIPUR (C.G)

Schedules

**Schedule -1 Corpus**

|       |                                                 | 2022-23 |              | 2021-22 |              |
|-------|-------------------------------------------------|---------|--------------|---------|--------------|
|       |                                                 | (Rs).   | (Rs).        | (Rs).   | (Rs).        |
|       | Balance as at the beginning of the year         | -       | 46.00        | -       | 46.00        |
| Add.: | Contribution Towards Corpus                     | -       | -            | -       | -            |
| Less: | Asset written off during the year out of corpus | -       | -            | -       | -            |
|       | <b>BALANCE AT THE YEAR END</b>                  | -       | <b>46.00</b> | -       | <b>46.00</b> |

**Schedule -2 General fund**

|       |                                                                                  | 2022-23 |                       | 2021-22 |                       |
|-------|----------------------------------------------------------------------------------|---------|-----------------------|---------|-----------------------|
|       |                                                                                  | (Rs).   | (Rs).                 | (Rs).   | (Rs).                 |
|       | Balance as at the beginning of the year                                          | -       | 468,190,017.60        | -       | 422,977,852.21        |
| Add.: | Contribution Towards General fund                                                | -       | -                     | -       | -                     |
| Less: | Balance of net Income(Expenditure) Transferred From Income & Expenditure Account | -       | 68,917,080.07         | -       | 45,212,165.39         |
|       | <b>BALANCE AT THE YEAR END</b>                                                   | -       | <b>537,107,097.67</b> | -       | <b>468,190,017.60</b> |



**Schedule -3 Designated/Earmarked Funds****Fund wise break up**

|                                                                  | Fund AA | Fund BB | Fund CC | Fund DD | 2022-23        | 2021-22       |
|------------------------------------------------------------------|---------|---------|---------|---------|----------------|---------------|
| <b>a) Opening balance of the funds</b>                           | -       | -       | -       | -       | 98,563,656.00  | 84,731,388.00 |
| <b>b) Additions to the funds</b>                                 | -       | -       | -       | -       | -              | -             |
| 1. Donation/ Grants                                              | -       | -       | -       | -       | -              | -             |
| 2. Income From Investment Made of the funds                      | -       | -       | -       | -       | 13,778,057.00  | 14,132,268.00 |
| 3. Accrued Investment of the funds                               | -       | -       | -       | -       | -              | -             |
| 4. Other Additions                                               | -       | -       | -       | -       | -              | -             |
| <b>Total (a+b)</b>                                               | -       | -       | -       | -       | -              | -             |
| <b>c) Utilisation/Expenditure towards objective of the funds</b> | -       | -       | -       | -       | -              | -             |
| 1. Capital Expenditure                                           | -       | -       | -       | -       | -              | -             |
| - Fixed Asset                                                    | -       | -       | -       | -       | -              | -             |
| - Others                                                         | -       | -       | -       | -       | -              | -             |
| <b>Total</b>                                                     | -       | -       | -       | -       | -              | -             |
| 1. Revenue Expenditure                                           | -       | -       | -       | -       | -              | -             |
| - Salaries, Wages And allowances etc.                            | -       | -       | -       | -       | 1,137,431.00   | 300,000.00    |
| - Rent                                                           | -       | -       | -       | -       | -              | -             |
| - other administrative expenses                                  | -       | -       | -       | -       | -              | -             |
| <b>Total (c)</b>                                                 | -       | -       | -       | -       | -              | -             |
| <b>Net balance as at the year end (a+b+c)</b>                    | -       | -       | -       | -       | 111,204,282.00 | 98,563,656.00 |

**Schedule -4 Restricted fund****Fund wise break up**

|                                                                  | Fund AA | Fund BB | Fund CC | Fund DD | 2022-23        | 2021-22        |
|------------------------------------------------------------------|---------|---------|---------|---------|----------------|----------------|
| <b>a) Opening balance of the funds</b>                           | -       | -       | -       | -       | 180,000,000.00 | 180,000,000.00 |
| <b>b) Additions to the funds</b>                                 | -       | -       | -       | -       | -              | -              |
| 1. Donation/ grants                                              | -       | -       | -       | -       | -              | -              |
| 2. Income From Investment made the fund                          | -       | -       | -       | -       | -              | -              |
| 3. Accrued Investment of the funds                               | -       | -       | -       | -       | -              | -              |
| 4. Other Additions                                               | -       | -       | -       | -       | -              | -              |
| <b>Total (a+b)</b>                                               | -       | -       | -       | -       | -              | -              |
| <b>c) Utilisation/Expenditure Towards objective of the funds</b> | -       | -       | -       | -       | -              | -              |
| 1. Capital Expenditure                                           | -       | -       | -       | -       | -              | -              |
| - Fixed Asset                                                    | -       | -       | -       | -       | -              | -              |
| - Others                                                         | -       | -       | -       | -       | -              | -              |
| <b>Total</b>                                                     | -       | -       | -       | -       | -              | -              |
| 1. Revenue Expenditure                                           | -       | -       | -       | -       | -              | -              |
| - Salaries, Wages And allowances                                 | -       | -       | -       | -       | -              | -              |
| - Rent                                                           | -       | -       | -       | -       | -              | -              |
| - other administrative expenses                                  | -       | -       | -       | -       | -              | -              |
| <b>Total</b>                                                     | -       | -       | -       | -       | -              | -              |
| <b>Net balance as at the year end (a+b+c)</b>                    | -       | -       | -       | -       | 180,000,000.00 | 180,000,000.00 |

**Schedule -5 Loans/Borrowing**  
**SECURED LOANS**

|                                          | 2022-23 |       | 2021-22 |       |
|------------------------------------------|---------|-------|---------|-------|
|                                          | (Rs).   | (Rs). | (Rs).   | (Rs). |
| 1. Central Government                    | -       | -     | -       | -     |
| 2. State Government (Specify)            | -       | -     | -       | -     |
| 3. Financial Institution                 | -       | -     | -       | -     |
| a) Term Loan                             | -       | -     | -       | -     |
| b) Interest Accured and due              | -       | -     | -       | -     |
| 4. Banks                                 | -       | -     | -       | -     |
| a) Term Loan                             | -       | -     | -       | -     |
| b) Interest Accured and due              | -       | -     | -       | -     |
| 5. Other Institution and agency          | -       | -     | -       | -     |
| 6. Debenture and Bonds                   | -       | -     | -       | -     |
| 7. Others (Specify)                      | -       | -     | -       | -     |
| <b>Total</b>                             | -       | -     | -       | -     |
| <b>Note: Amount Due with in one year</b> |         |       |         |       |

| UNSECURED LOANS                          | 2022-23 | 2021-22 |
|------------------------------------------|---------|---------|
|                                          | (Rs).   | (Rs).   |
| 1. Central Government                    | -       | -       |
| 2. State Government (Specify)            | -       | -       |
| 3. Financial Institution                 | -       | -       |
| 4. Banks                                 | -       | -       |
| a) Term Loan                             | -       | -       |
| b) Other Loans(Specify)                  | -       | -       |
| 5. Other Institutions and agencies       | -       | -       |
| 6. Debentures and Bonds                  | -       | -       |
| 7.Others( Specify)                       | -       | -       |
| <b>Total</b>                             | -       | -       |
| <b>Note: Amount Due with in one year</b> |         |         |

**Schedule -6 Current Liabilities and provision**

|                                                        | 2022-23 |               | 2021-22 |               |
|--------------------------------------------------------|---------|---------------|---------|---------------|
|                                                        | (Rs).   | (Rs).         | (Rs).   | (Rs).         |
| <b>A. Current Liabilities</b>                          | -       | -             | -       | -             |
| 1. Deposit From staff                                  | -       | -             | -       | -             |
| 2. Deposit From students                               | -       | 15,867,529.86 | -       | 14,612,326.86 |
| 3. Sundry creditors                                    | -       | -             | -       | -             |
| a) For Goods & services                                | -       | 4,449,166.19  | -       | 1,517,203.19  |
| b)Others                                               | -       | 7,248,098.00  | -       | 7,248,098.00  |
| 4.Advances Received                                    | -       | -             | -       | -             |
| 5.Interest accrued but not due on:                     | -       | -             | -       | -             |
| a)Secured Loans/borrowings                             | -       | -             | -       | -             |
| b)Unsecured Loans/borrowings                           | -       | -             | -       | -             |
| 6.Statutory laibilities(GPF,TDS,WC,TAX,CPE,GIS,NPS)    | -       | -             | -       | -             |
| a)Ovrdue                                               | -       | -             | -       | -             |
| b)Others                                               | -       | -             | -       | -             |
| 7.Other current laibilities                            | -       | -             | -       | -             |
| a) Salaries                                            | -       | -             | -       | -             |
| b)Receipts against sponsord projects                   | -       | -             | -       | -             |
| c)Receipts against sponsord fellowships & scholarships | -       | -             | -       | -             |
| d)Unutilised grants                                    | -       | -             | -       | -             |
| e)Grants in advance                                    | -       | -             | -       | -             |
| f)Other funds                                          | -       | -             | -       | -             |
| g)Other laibilities                                    | -       | 251,323.00    | -       | 629,872.00    |
| h)Earnest Security Deposit                             | -       | 3,888,611.00  | -       | 3,667,231.00  |
| <b>TOTAL (A)</b>                                       | -       | -             | -       | -             |
| <b>B. PROVISIONS</b>                                   | -       | -             | -       | -             |
| a)For Taxation                                         | -       | 68,015.00     | -       | 15,394.00     |
| b)Gratuity                                             | -       | -             | -       | -             |
| c)Superannuation/Pension                               | -       | -             | -       | -             |
| d)Accumulated Leave Encashment                         | -       | -             | -       | -             |
| e)Expenses Payable                                     | -       | -             | -       | -             |
| f)Trade warranties/claims                              | -       | -             | -       | -             |
| g)Other (Specify)                                      | -       | -             | -       | -             |
| <b>TOTAL (B)</b>                                       | -       | -             | -       | -             |
| <b>TOTAL (A+B)</b>                                     | -       | 31,772,743.05 | -       | 27,690,125.05 |

| Head Of Account                | Opening Balance | Transaction during the year | Closing Balance |
|--------------------------------|-----------------|-----------------------------|-----------------|
| 1. University Grant Commission | -               | -                           | -               |
| 2. Ministry.....               | -               | -                           | -               |
| 3.....                         | -               | -                           | -               |
| <b>Total</b>                   | -               | -                           | -               |

## Schedule -7 FIXED ASSETS

| DESCRIPTION                                                        | GROSS BLOCK                      |                          |                            |                                  | DEPRECIATION                     |                           |                              | NET BLOCK        |                  |
|--------------------------------------------------------------------|----------------------------------|--------------------------|----------------------------|----------------------------------|----------------------------------|---------------------------|------------------------------|------------------|------------------|
|                                                                    | Opening balance as on 01.04.2022 | Addition during The Year | Deductions during the year | Closing balance as on 31.03.2023 | Opening balance as on 01.04.2022 | Depreciation for the year | Closing dep as on 31.03.2023 | As on 31.03.2023 | As on 01.04.2022 |
| 1. Land:                                                           |                                  |                          |                            |                                  |                                  |                           |                              |                  |                  |
| a) Freehold                                                        | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| b) Leasehold                                                       | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| 2. Buildings:                                                      |                                  |                          |                            |                                  |                                  |                           |                              |                  |                  |
| a) On Freehold Land                                                | 154,265,826.00                   | 9,956,892.00             | -                          | 164,222,718.00                   | 10,757,942.76                    | 3,069,655.50              | 13,827,598.26                | 150,413,119.74   | 143,525,883.24   |
| b) On leasehold Land                                               | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| c) Ownership Flats/Premises                                        | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| d) Superstructures on land not belonging to education institutions | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| 3. Plants, machinery & equipment                                   | 13,621,731.00                    | 3,217,449.00             | -                          | 16,839,180.00                    | 2,836,121.65                     | 687,731.85                | 3,523,853.50                 | 13,066,905.09    | 10,537,187.94    |
| 4. Vehicle                                                         | 1,795,342.00                     | -                        | -                          | 1,795,342.00                     | 897,671.00                       | 71,813.68                 | 969,484.68                   | 646,323.12       | 718,136.80       |
| 5. Furniture & fixtures                                            | 44,492,393.20                    | 3,571,154.00             | -                          | 48,063,547.20                    | 13,093,449.47                    | 2,806,714.29              | 15,900,163.76                | 34,616,142.91    | 33,851,703.20    |
| 6. Office equipment                                                | 2,025,204.00                     | 106,503.00               | -                          | 2,131,707.00                     | 584,876.10                       | 242,619.27                | 827,495.37                   | 2,992,304.33     | 3,128,420.60     |
| 7. Computers & peripherals                                         | 40,869,104.00                    | 7,973,800.00             | -                          | 48,842,904.00                    | 26,561,040.40                    | 4,781,423.96              | 31,342,464.36                | 19,029,695.64    | 15,837,319.60    |
| 8. Electric installations                                          | 4,513,949.00                     | 172,740.00               | -                          | 4,686,689.00                     | 1,276,693.15                     | 200,513.37                | 1,477,206.52                 | 3,809,753.98     | 3,837,527.35     |
| 9. Library books                                                   | 6,785,192.00                     | 751,077.52               | -                          | 7,536,269.52                     | 2,485,694.40                     | 489,541.94                | 2,975,236.34                 | 4,405,877.48     | 4,144,341.90     |
| 10. Tube well & Water Supply                                       | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| 11. Lab Equipment                                                  | 115,559,186.92                   | 24,788,350.00            | -                          | 140,347,536.92                   | 22,829,511.68                    | 10,103,188.51             | 32,932,700.18                | 116,186,667.86   | 101,501,506.37   |
| 12. Solar power plant                                              | 18,463,766.00                    | -                        | -                          | 18,463,766.00                    | 3,308,166.20                     | 711,620.58                | 4,019,786.78                 | 13,520,790.93    | 14,232,411.50    |
| 13. Software                                                       | 14,902,126.00                    | -                        | -                          | 14,902,126.00                    | 14,902,126.00                    | -                         | 14,902,126.00                | -                | -                |
| 14. Other fixed assets(NTPC)                                       | 46.00                            | -                        | -                          | 46.00                            | -                                | -                         | -                            | 46.00            | 46.00            |
| A) Total of CURRENT YEAR PREVIOUS YEAR                             | 417,293,866.12                   | 50,537,965.52            | -                          | 467,831,831.64                   | 99,533,292.80                    | 23,164,822.95             | 122,698,115.75               | 358,687,627.07   | 331,314,484.50   |
| 13. Capital work-in-progress                                       | 26,174,000.00                    | 8,174,000.00             | 9,000,000.00               | 25,348,000.00                    | -                                | -                         | -                            | 25,348,000.00    | 26,174,000.00    |
| TRANSFER to assets                                                 | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| B) NETWORK -in- progress                                           | -                                | -                        | -                          | -                                | -                                | -                         | -                            | -                | -                |
| TOTAL (A+B)                                                        | 443,467,866.12                   | 58,711,965.52            | 9,000,000.00               | 493,179,831.64                   | 99,533,292.80                    | 23,164,822.95             | 122,698,115.75               | 384,035,627.07   | 357,488,484.50   |

**Schedule -8 INVESTMENTS****INVESTMENTS FROM EARMARKED/ENDOWMENT FUNDS**

|                                     | 2022-23               | 2021-22               |
|-------------------------------------|-----------------------|-----------------------|
| 1. In central government securities | -                     | -                     |
| 2. In state government securities   | -                     | -                     |
| 3. Other approved securities        | -                     | -                     |
| 4. Shares                           | -                     | -                     |
| 5. Debentures & bonds               | -                     | -                     |
| 6. Others (Fixed Deposit)           | 279,598,782.00        | 268,350,305.00        |
| <b>Total</b>                        | <b>279,598,782.00</b> | <b>268,350,305.00</b> |

**INVESTMENTS OTHERS**

|                                     | 2022-23  | 2021-22  |
|-------------------------------------|----------|----------|
| 1. In central government securities | -        | -        |
| 2. In state government securities   | -        | -        |
| 3. Other approved securities        | -        | -        |
| 4. Shares                           | -        | -        |
| 5. Debentures & bonds               | -        | -        |
| 6. Others                           | -        | -        |
| <b>Total</b>                        | <b>-</b> | <b>-</b> |

**Schedule -9 CURRENT ASSETS**

|                                                                                          | 2022-23  |                       | 2021-22  |                       |
|------------------------------------------------------------------------------------------|----------|-----------------------|----------|-----------------------|
| 1. Stocks                                                                                | -        | -                     | -        | -                     |
| a) Stores & Spares                                                                       | -        | -                     | -        | -                     |
| b) Loose tools                                                                           | -        | -                     | -        | -                     |
| c) Publications                                                                          | -        | -                     | -        | -                     |
| 2. Sundry debtors                                                                        | -        | -                     | -        | -                     |
| a) Debts Outstanding for a period exceeding six months                                   | -        | 12,230,996.00         | -        | 22,486,351.00         |
| b) Prepaid Exp.                                                                          | -        | -                     | -        | -                     |
| 3. Cash balances in hand (including cheques/drafts and imprest)                          | -        | 14,732.00             | -        | 6,023.00              |
| 4. Bank balances (to be further classified as pertaining to earmarked fund or otherwise) | -        | -                     | -        | -                     |
| a) With Scheduled Banks:                                                                 | -        | -                     | -        | -                     |
| -In Current Accounts                                                                     | -        | 128,670,138.57        | -        | 84,850,038.70         |
| -In term Deposit Accounts                                                                | -        | -                     | -        | -                     |
| -In Savings Accounts                                                                     | -        | -                     | -        | -                     |
| b) With Non-Scheduled Banks:                                                             | -        | -                     | -        | -                     |
| -In Current Accounts                                                                     | -        | -                     | -        | -                     |
| -In term Deposit Accounts                                                                | -        | -                     | -        | -                     |
| -In Savings Accounts                                                                     | -        | -                     | -        | -                     |
| 5. Post office-Savings Accounts                                                          | -        | -                     | -        | -                     |
| <b>Total</b>                                                                             | <b>-</b> | <b>140,915,866.57</b> | <b>-</b> | <b>107,342,412.70</b> |

**Schedule 10- LOANS, ADVANCES AND DEPOSITS**

|                                                                                           | 2022-23 |                      | 2021-22 |                      |
|-------------------------------------------------------------------------------------------|---------|----------------------|---------|----------------------|
| 1. Advances to employees: (Non- Interest bearing)                                         | -       |                      | -       | -                    |
| a) Salary                                                                                 | -       |                      | -       | -                    |
| b) Festival                                                                               | -       |                      | -       | -                    |
| c) LTC                                                                                    | -       |                      | -       | -                    |
| d) Medical advance                                                                        | -       |                      | -       | -                    |
| e) Other                                                                                  | -       | 5,000.00             | -       | 4,500.00             |
| 2. Long Term Advances to employees: (Interest bearing)                                    | -       |                      | -       |                      |
| a) Vehical loan                                                                           | -       |                      | -       |                      |
| b) Home loan                                                                              | -       |                      | -       |                      |
| c) Others (To be specified)                                                               | -       |                      | -       |                      |
| 3. Advances and other amounts recoverable in cash or in kind or for value to be received: | -       |                      | -       |                      |
| a) On capital account                                                                     | -       |                      | -       |                      |
| b) To Suppliers                                                                           | -       | -                    | -       | 81,176.00            |
| c) Others                                                                                 | -       | 7,268,103.00         | -       | 7,284,548.00         |
| 4. Prepaid expences:                                                                      | -       |                      | -       |                      |
| a) Insurance                                                                              | -       |                      | -       |                      |
| b) Other expences                                                                         | -       |                      | -       |                      |
| 5. Deposits:                                                                              | -       |                      | -       |                      |
| a) Telephone                                                                              | -       | 12,495.00            | -       | 12,495.00            |
| b) Lease Rent                                                                             | -       |                      | -       |                      |
| c) Electricity                                                                            | -       | 2,187,822.00         | -       | 977,060.00           |
| d) AICTE, If applicable                                                                   | -       |                      | -       | -                    |
| e) MCI, If applicable                                                                     | -       |                      | -       | -                    |
| f) Others (To be specified)                                                               | -       | 42,283,346.00        | -       | 27,102,022.00        |
| 6. Income Accured:                                                                        | -       |                      | -       | -                    |
| a) On Investments from Earmarked/Endowment Funds                                          | -       |                      | -       | -                    |
| b) On Investments-Others                                                                  | -       |                      | -       |                      |
| c) On Loans and Advances                                                                  | -       |                      | -       |                      |
| d) Others (includes income due unrealized-Rs....)                                         | -       |                      | -       |                      |
| 7. Other receivable                                                                       | -       |                      | -       |                      |
| a) Debit balances in sponsored projects                                                   | -       |                      | -       |                      |
| a) Debit balances in fellowship & scholarship                                             | -       |                      | -       |                      |
| c) Grants recoverable                                                                     | -       |                      | -       |                      |
| d) Other receivable( T.D.S.Receivable)                                                    | -       | 3,777,127.08         | -       | 5,800,841.45         |
| 8. Claims receivable                                                                      | -       |                      | -       | -                    |
| <b>Total</b>                                                                              | -       | <b>55,533,893.08</b> | -       | <b>41,262,642.45</b> |

**SCHEDULE 11 - ACADEMIC RECEIPTS**

|                                                 | 2022-23               | 2021-22               |
|-------------------------------------------------|-----------------------|-----------------------|
| <b>FEE FROM STUDENTS</b>                        |                       |                       |
| <b>Academic</b>                                 | -                     | -                     |
| 1. Tuition fee                                  | 126,796,780.91        | 121,438,324.18        |
| 2. Admission fee                                | -                     | -                     |
| 3. Enrolment Fee                                | -                     | -                     |
| 4. Library late fees                            | 35,908.00             | 7,768.00              |
| 5. Laboratory fee                               | -                     | -                     |
| 6. Art & Craft fee                              | -                     | -                     |
| 7. Registration fee                             | -                     | -                     |
| 8. Syllabus Fee                                 | -                     | -                     |
| <b>Total (A)</b>                                | -                     | -                     |
| <b>Examinations</b>                             | -                     | -                     |
| 1. Admission test fee                           | -                     | -                     |
| 2. Annual Examination fee                       | -                     | -                     |
| 3. Mark sheet, certificate fee                  | -                     | -                     |
| <b>Total (B)</b>                                | -                     | -                     |
| <b>Other fees</b>                               | -                     | -                     |
| 1. Library card reissuance fee                  | 514.00                | 50.00                 |
| 2. Fine/Miscellaneous fee                       | -                     | -                     |
| 3. Medical fee                                  | -                     | -                     |
| 4. Transportation fee                           | -                     | -                     |
| 5. Hostel fee                                   | -                     | -                     |
| 5. EMD Forfeited                                | -                     | -                     |
| 6. Student Welfare Receipt                      | -                     | -                     |
| 7. Others                                       | -                     | -                     |
| <b>Total ©</b>                                  | -                     | -                     |
| <b>Sale of publications</b>                     | -                     | -                     |
| 1. Sale of syllabus and Question Paper, etc.    | -                     | -                     |
| 2. Sale of prospectus including admission forms | -                     | -                     |
| <b>Total (D)</b>                                | -                     | -                     |
| <b>GRAND TOTAL (A+B+C+D)</b>                    | <b>126,833,202.91</b> | <b>121,446,142.18</b> |

**SCHEDULE 12 - GRANTS & DONATIONS (Irrevocable Grants & Subsidies Received)**

|                                | 2022-23               | 2021-22               |
|--------------------------------|-----------------------|-----------------------|
| 1) Central Government          | -                     | -                     |
| 2) State Government(s)         | 148,000,000.00        | 100,000,000.00        |
| 3) Government Agencies         | -                     | -                     |
| 4) Institutions/Welfare Bodies | -                     | -                     |
| 5) International Organisation  | -                     | -                     |
| 6) Others (Specify)            | -                     | -                     |
| <b>Total</b>                   | <b>148,000,000.00</b> | <b>100,000,000.00</b> |

**SCHEDULE 13- INCOME FROM INVESTMENTS**

| Investment from Earmarked/Endowment Fund | 2022-23  |          | 2021-22  |          |
|------------------------------------------|----------|----------|----------|----------|
| 1) Interest                              | -        | -        | -        | -        |
| a) On Govt. Securities                   | -        | -        | -        | -        |
| b) Other Bonds/Debentures                | -        | -        | -        | -        |
| 2) Income received                       | -        | -        | -        | -        |
| a) Restricted Fund                       | -        | -        | -        | -        |
| 3) Income accrued                        | -        | -        | -        | -        |
| a) Each Fund separately                  | -        | -        | -        | -        |
| 4) Others (Specify)                      | -        | -        | -        | -        |
| <b>TOTAL</b>                             | <b>-</b> | <b>-</b> | <b>-</b> | <b>-</b> |

**TRANSFERRED TO EARMARKED/ENDOWMENT FUNDS**

| Investment from Earmarked/Endowment Fund | 2022-23  | 2021-22  |
|------------------------------------------|----------|----------|
| 1) Interest                              | -        | -        |
| a) On Govt. Securities                   | -        | -        |
| b) Other Bonds/Debentures                | -        | -        |
| 2) Income received                       | -        | -        |
| a) Each Fund separately                  | -        | -        |
| 3) Income accrued                        | -        | -        |
| a) Each Fund separately                  | -        | -        |
| 4) Others (Specify)                      | -        | -        |
| <b>TOTAL</b>                             | <b>-</b> | <b>-</b> |

**SCHEDULE 14- OTHER INCOME**

|                                                                           | 2022-23             | 2021-22             |
|---------------------------------------------------------------------------|---------------------|---------------------|
| <b>A. Income from Land &amp; Building</b>                                 | -                   | -                   |
| 1. Hostel Room Rent                                                       | -                   | -                   |
| 2. License fee                                                            | 743,965.00          | 663,494.00          |
| 3. Hire Charges of Auditorium/Play ground/Convention Centre, etc          | -                   | -                   |
| 4. Electricity & water charges                                            | 374,163.00          | 139,477.00          |
| 5. Rental Income from guest house/canteen                                 | 428,806.00          | 56,122.00           |
| 6. Tender Cost                                                            | -                   | -                   |
| 7. Penalty Income                                                         | 265,942.00          | 198,090.00          |
| 8. Others                                                                 | 4,220,087.88        | 2,216,408.57        |
| <b>Total</b>                                                              | -                   | -                   |
| <b>B. Sale of Institute's publications</b>                                | -                   | -                   |
| <b>C. Income from holding events</b>                                      | -                   | -                   |
| 1. Gross Receipts from annual function/ sports carnival                   | -                   | -                   |
| Less: Direct expenditure incurred on the annual function/ sports carnival | -                   | -                   |
| 2. Gross Receipts from fetes                                              | -                   | -                   |
| Less: Direct expenditure incurred on the fetes                            | -                   | -                   |
| 3. Gross Receipts for educational tours                                   | -                   | -                   |
| Less: Direct expenditure incurred on the tours                            | -                   | -                   |
| 4. Others (to be specified and separately disclosed)                      | -                   | -                   |
| <b>Total</b>                                                              | -                   | -                   |
| <b>D. Interest on Term Deposits:</b>                                      | -                   | -                   |
| a) With Scheduled Banks                                                   | 2,793,749.00        | 2,433,180.00        |
| b) With Non-Scheduled Banks                                               | -                   | -                   |
| c) With Institutions                                                      | -                   | -                   |
| d) Others                                                                 | 668,780.00          | 52,204.00           |
| <b>E. Interest on Savings Accounts:</b>                                   | -                   | -                   |
| a) With Scheduled Banks                                                   | -                   | -                   |
| b) With Non-Scheduled Banks                                               | -                   | -                   |
| c) With Institutions                                                      | -                   | -                   |
| d) Others                                                                 | -                   | -                   |
| <b>F. Interest On Loans:</b>                                              | -                   | -                   |
| a) Employees/Staff                                                        | -                   | -                   |
| b) Others                                                                 | -                   | -                   |
| <b>G. Interest on Debtors and Other Receivables</b>                       | -                   | -                   |
| <b>H. Others</b>                                                          | -                   | -                   |
| 1. Income from consultancy                                                | -                   | -                   |
| 2. RTI fees                                                               | -                   | -                   |
| 3. Income from Royalty                                                    | -                   | -                   |
| 4. Sale of application form (recruitment)                                 | -                   | -                   |
| 5. Misc. receipts (Sale of tender form, waste paper, etc.)                | -                   | -                   |
| 6. Profit on Sale/disposal of Assets:                                     | -                   | -                   |
| a) Owned assets                                                           | -                   | -                   |
| b) Assets acquired out of grants, or received free of cost                | -                   | -                   |
| b) Discount Received                                                      | -                   | -                   |
| <b>Total</b>                                                              | -                   | -                   |
| <b>GRAND TOTAL (A+B+C+D+E+F+G+H)</b>                                      | <b>9,495,492.88</b> | <b>5,758,975.57</b> |

**SCHEDULE 15 - STAFF PAYMENTS & BENEFITS**

|                                             | 2022-23               | 2021-22              |
|---------------------------------------------|-----------------------|----------------------|
| a) Salaries and Wages                       | 91,809,630.00         | 75,282,818.00        |
| b) Allowances and Bonus                     | -                     | -                    |
| c) Contribution to Provident Fund           | 10,025,399.00         | 7,968,246.00         |
| d) Contribution to Other Fund (specify)     | -                     | -                    |
| e) Cumulative professional development fund | 1,516,196.00          | 826,319.00           |
| f) Retirement and Terminal Benefits         | -                     | -                    |
| g) LTC facility                             | 2,398,784.00          | -                    |
| h) Medical facility                         | 3,309,625.00          | 3,647,720.00         |
| i) Children Education Allowance             | 902,250.00            | 558,592.00           |
| j) Honorarium                               | 517,856.17            | 1,213,700.00         |
| k) Baggage allowance                        | 501,829.00            | 119,910.00           |
| l) Telephone expenses                       | 384,477.00            | 472,699.00           |
| M) Others                                   | 1,669,135.00          | 99,044.00            |
| <b>TOTAL</b>                                | <b>113,035,181.17</b> | <b>90,189,048.00</b> |

**SCHEDULE 16 - ACADEMIC EXPENSES**

|                                        | 2022-23       | 2021-22       |
|----------------------------------------|---------------|---------------|
| a) Laboratory expenses                 | 438,186.00    | 88,167.00     |
| b) Field work/Participation            |               |               |
| c) Seminar/Workshop                    | 907,484.00    | 620,493.00    |
| d) Payment to visiting faculty         |               |               |
| e) Examination                         |               |               |
| f) Student Welfare expenses            | 2,351,052.50  | 1,585,535.00  |
| g) Admission expenses                  |               |               |
| h) Training and Placement Expenses     | 1,839,492.00  |               |
| i) Convocation expenses                |               | 2,375,089.00  |
| j) Publications                        |               |               |
| k) Stipend/means-cum-merit scholarship |               |               |
| l) Subscription Exp.                   | 5,898,467.59  | 8,340,631.00  |
| m) Canteen Exp.                        | 1,299,753.00  | 895,198.00    |
| n) Others (Village adoption)           | 42,326.00     | 25,480.00     |
|                                        | 12,776,761.09 | 13,930,593.00 |

**SCHEDULE 17 - ADMINISTRATIVE AND GENERAL EXPENSES**

|                                                   | 2022-23              | 2021-22              |
|---------------------------------------------------|----------------------|----------------------|
| a) Electricity and power                          | 20,248,993.37        | 6,423,286.55         |
| b) Water charges                                  | 1,760,160.00         | 966,806.00           |
| c) Insurance                                      |                      |                      |
| d) Rent, Rates and Taxes (including property tax) |                      |                      |
| e) Postage & telegram                             | 16,427.00            | 18,074.00            |
| f) Telephone and Internet Charges                 | 1,777,053.00         | 1,609,915.00         |
| g) Printing and Stationary                        | 840,362.00           | 653,244.00           |
| h) Traveling and Conveyance Expenses              |                      |                      |
| i) Expenses on seminar/workshops                  |                      |                      |
| j) House Keeping Exp.                             | 12,058,164.00        | 11,182,063.00        |
| k) Auditors Remuneration                          |                      | 133,360.00           |
| l) Professional Charges                           | 4,884,510.00         | 763,934.00           |
| m) Advertisement and Publicity                    | 3,459,852.00         | 2,419,773.00         |
| n) IEEE Exp.                                      |                      |                      |
| o) Board Meeting Exp.                             | 23,900.00            |                      |
| p) Magazine & Journals                            | 355,375.00           |                      |
| q) Others (IT peripheral, software renewal)       | 872,704.00           | 1,060,785.00         |
| r) Security Exp.                                  | 7,204,614.00         | 5,018,139.00         |
| s) office Exp.                                    | -                    |                      |
| t) Sitting fees                                   | 330,000.00           |                      |
| u) inauguration Exp.                              |                      |                      |
| v) Other Office & Pooja Exp.                      | 1,377,066.00         | 273,547.00           |
| w) Travelling Expenses                            | -                    | 59,201.00            |
| x) Dispensary expenses                            | 83,596.00            |                      |
| <b>TOTAL</b>                                      | <b>55,292,776.37</b> | <b>30,582,127.55</b> |

**SCHEDULE 18 - TRANSPORTATION EXPENSES**

|                                                | 2022-23             | 2021-22             |
|------------------------------------------------|---------------------|---------------------|
| 1. Vehicles (owned by educational institution) |                     |                     |
| a) Running expenses                            | 613,900.00          | 534,075.00          |
| b) Repairs & maintenance                       |                     |                     |
| c) Insurance expenses                          |                     |                     |
| 2. Vehicles taken on rent/lease                |                     |                     |
| a) Rent/lease expenses                         | 622,942.00          | 757,810.00          |
| <b>TOTAL</b>                                   | <b>1,236,842.00</b> | <b>1,291,885.00</b> |

**SCHEDULE 19 - REPAIRS & MAINTENANCE**

|                                          | 2022-23             | 2021-22              |
|------------------------------------------|---------------------|----------------------|
| a) Building (Manpower)                   |                     |                      |
| b) Annual Maintenance charges            | 3,149,870.43        | 3,056,316.00         |
| c) Plant & Machinery                     |                     |                      |
| d) Office Equipments                     | 3,146,536.00        | 1,489,068.00         |
| e) Cleaning material & services          |                     |                      |
| f) Others Civil & Electrical maintenance | -                   | 4,049,484.00         |
| g) Garden Exp.                           | 73,820.00           | 4,784,242.00         |
| <b>TOTAL</b>                             | <b>6,370,226.43</b> | <b>13,379,110.00</b> |

**SCHEDULE 20 - FINANCE COSTS**

|                            | 2022-23         | 2021-22          |
|----------------------------|-----------------|------------------|
| a) Interest on fixed loans | -               | -                |
| b) Interest on other loans | -               | -                |
| c) Bank charges            | 1,376.71        | 10,737.28        |
| d) Others (suspense)       | -               | -                |
| <b>TOTAL</b>               | <b>1,376.71</b> | <b>10,737.28</b> |

**SCHEDULE 21 - OTHER EXPENSES**

|                                                   | 2022-23             | 2021-22             |
|---------------------------------------------------|---------------------|---------------------|
| a) Provision for Bad and Doubtful Debts/ Advances | -                   | -                   |
| b) Irrecoverable Balances Written-off             | -                   | -                   |
| c) Others (specify)                               | 3,533,629.00        | 1,192,929.00        |
| <b>TOTAL</b>                                      | <b>3,533,629.00</b> | <b>1,192,929.00</b> |



**SCHEDULE 22 - NOTES TO ACCOUNT**  
**SIGNIFICANT ACCOUNTING POLICIES**

**1 BASIS OF PREPARATION OF FINANCIAL STATEMENTS**

The financial statements are prepared under the historical cost convention, on accrual basis, in accordance with the generally accepted accounting principles in India, the Accounting Standards issued by the Institute of Chartered Accountants of India and the provisions of the Companies Act, 2013.

**2 FIXED ASSETS**

Fixed Assets are stated at cost of acquisition with any subsequent improvement thereto and less accumulated depreciation. The cost includes taxes, duties, freight and other incidental expenses relating to acquisition and installation. Land is stated at cost.

**3 DEPRECIATION**

Depreciation on fixed assets has been provided on straight line method as per MHRD guidelines and rate of depreciation has also been used as per MHRD guidelines:

**INVESTMENTS**

Investments are stated at cost. Current Investment are valued at the lower of cost or net realizable value, computed script wise. Provision for diminution in the value of long term investments is made only if such a decline is other than temporary in the opinion of the management

**5 REVENUE RECOGNIZATION :**

Receipts & Expenditure is accounted for on accrual basis.

**6 TAXES ON INCOME**

Current tax is accounted for after taking into account deductions that are expected to be available to the company. In view of the uncertainty of the profits in the future, no Deferred Tax Asset/ Liability has been recognised.

**7 IMPAIRMENT OF ASSETS**

The carrying amounts of assets are reviewed at each Balance Sheet date, if there is any indication of impairment based on internal/ external factors. An asset is treated as impaired when the carrying cost of assets exceeds its recoverable value. An impairment loss is charged to the Profit & Loss Account in the year in which an asset is identified as impaired. The impairment loss recognized in prior accounting periods is reversed if there has been a change in the estimate of recoverable amount.

**8 PROVISIONS, CONTINGENT LIABILITIES AND CONTINGENT ASSETS**

The company recognizes a provision when there is a present obligation as a result of a past even that probably requires an outflow of resources and a reliable estimate can be made of the amount of the obligation. A disclosure for a contingent liability is made when there is a possible obligation or a present obligation that may, but probably will not, require an outflow of resources. Where there is a possible obligation or a present obligation that the likelihood of outflow of resources is remote, no provision or disclosure is made. Contingent Assets are neither recognized nor disclosed in the financial statements.

**9 USE OF ESTIMATES**

The preparation of financial statements requires estimates and assumptions to be made that affect the reported amount of assets and liabilities on the date of financial statements and the reported amount of revenues and expenses during the reporting period. Difference between the actual results and estimates are recognized in the period in which the results are known / materialized

**10 MAJOR FUND OPENING BALANCE AND DEDUCTION DURING THE YEAR**

**Corpus Fund**

|                                                       | 2022-23 |       | 2021-22 |       |
|-------------------------------------------------------|---------|-------|---------|-------|
|                                                       | (Rs).   | (Rs). | (Rs).   | (Rs). |
| Balance as at the beginning of the year               | -       | 46.00 | -       | 46.00 |
| Add.: Contribution Towards Corpus                     | -       | -     | -       | -     |
| Less: Asset written off during the year out of corpus | -       | -     | -       | -     |
| <b>BALANCE AT THE YEAR END</b>                        | -       | 46.00 | -       | 46.00 |

**Schedule -2 General fund**

|                                                                                  | 2022-23 |                | 2021-22 |                |
|----------------------------------------------------------------------------------|---------|----------------|---------|----------------|
|                                                                                  | (Rs).   | (Rs).          | (Rs).   | (Rs).          |
| Balance as at the beginning of the year                                          | -       | 468,190,017.60 | -       | 422,977,852.21 |
| Add.: Contribution Towards General fund                                          | -       | -              | -       | -              |
| Less:                                                                            |         |                |         |                |
| Balance of net Income(Expenditure) Transferred From Income & Expenditure Account | -       | 68,917,080.07  | -       | 45,212,165.39  |
| <b>BALANCE AT THE YEAR END</b>                                                   | -       | 537,107,097.67 | -       | 468,190,017.60 |

**12 RELATED PARTY DISCLOSURE**

According to As 24 no transaction exist between the educational institution and the trust or society of the author of the trust or founder of the institution.



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### **Editorial Team**

#### **Editor-in-Chief**

Pradeep K. Sinha  
VC and Director, IIIT-NR

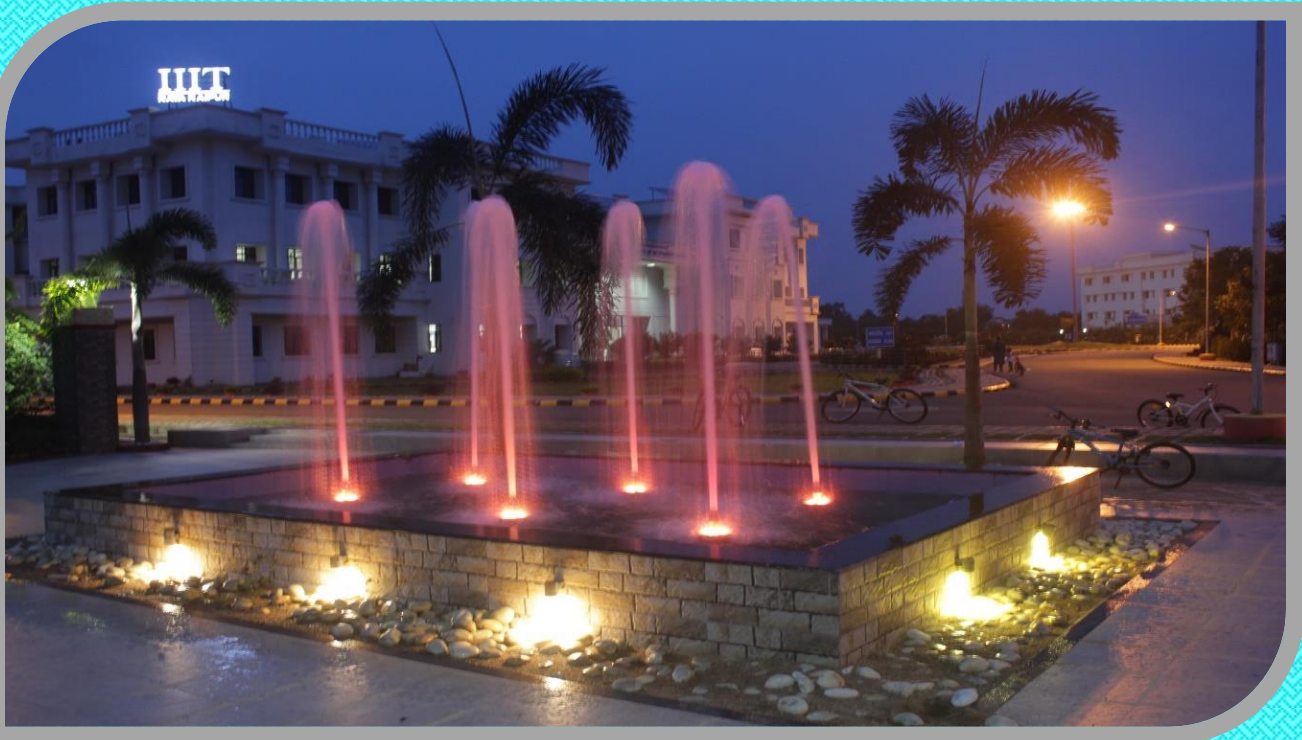
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